

Operator's Manual
The Coherent Infinity™
Nd:YAG Laser System

 **COHERENT®**
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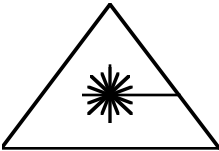
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Operation of any Coherent laser with any of its interlocks defeated is always at the operator's own risk.

Operating Information

Operating the laser without understanding the following information may cause extensive damage to the system that is not covered by the warranty. Read the Operator's Manual for more detailed information. Keep this document with the laser at all times.



Direct eye contact with the output beam from the laser may cause permanent blindness. Laser safety glasses must be worn at all times when operating the laser.



Lasing is immediately terminated and all systems in the laser are sequentially switched off if an interlock is tripped. To restart the laser, close the interlock, clear the fault message and press "On" in the main control window.



By manually increasing the flash lamp voltage the Infinity system can deliver output energies significantly higher than its specifications. Doing this will, however, permanently damage optical components in the amplifier. Do not run the system at higher energies or powers than what it is rated for.



Modifying the parameters in the Lamp Timer window will affect laser performance. Deviation from factory settings may result in lower output power and unstable DPSS laser performance.



Operate the laser at maximum specified output energy for minimum intensity noise and optimum mode quality.



It is important that the depolarized energy, displayed in the Monitor window, is kept at a minimum. If it is above 1.5 %, of the IR pulse energy, see Procedure 4 in Chapter Four.



All optical components must be cleaned in their mounts. Removing and reinserting any of the optical components in the laser will severely affect the mode quality and may require a complete realignment to be performed. This can only be done by a Coherent Field Service Engineer. Do not attempt to optimize the output power by adjusting the alignment of the laser.



Set the Lamp max voltage in the main Control window to be only 5–10 V higher than what is needed to produce the required output power. This prevents the energy servo from applying excessive voltage to the flash lamp in case the beam is partially blocked.



The resistivity of the internal cooling water, displayed in the Monitor window, should be greater than 2 MΩ/cm. If necessary adjust the bypass valve in the power supply. Replace the deionization cartridge when a resistivity higher than 2 MΩ/cm can not be obtained even with the bypass valve fully open.



Use only steam distilled water in the internal water cooling system. Do not use deionized tap water.



If the internal water cooling system is drained, remove the deionization cartridge and seal it so that it does not dry out. If the cartridge is allowed to dry it will lose its capacity to maintain a high enough water resistivity. Do not insert a new cartridge until you are ready to fill the system with distilled water.



Make sure that the beam is never retro-reflected back into the laser. Even a very weak return reflex can be amplified by residual gain in the Nd:YAG rods and destroy parts of the laser. If a reflected pulse is too intense a fault message stating “Depolarized energy too high” appears and the laser is automatically turned off. The same pulse that causes the laser to turn off itself might, however, also damage the system.



The material in the Infinity beam dump can safely absorb 42 W of average power of IR. It can thus be used for the IR power produced by any combination of pulse energy and repetition rate. It is, however, not designed to absorb any more of the harmonic radiation than what goes through the respective dichroics. It is therefore important to observe the following:

1. The beam dump can only handle the residual IR + the harmonic light that leaks through the dichroics.
 2. When installing a harmonic option, the harmonic crystal must not be left detuned for any time longer than necessary to complete the alignment procedure. Damage caused by directly dumping harmonic radiation (produced at 532 nm, 355 nm and 266 nm) into the beam dump is not covered by the warranty. Always use the SHG, THG and FHG crystals with their respective dichroics.
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Table 1. Glossary of Units

UNIT	QUANTITY
AC	Alternating Current
Amp	Amperes
°C	Degrees Centigrade
DC	Direct Current
°F	Degrees Fahrenheit
Hz	Hertz
J	Joule
lbs	Pounds
mrاد	milliradians = 10^{-3} radians
nm	nanometers = 10^{-9} m
psi	Pounds per square inch
V	Volts
W	Watts

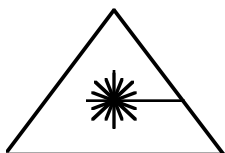
Table 2. Glossary of Abbreviations

ABBREVIATION	MEANING
AOM	Acousto-Optical Modulator
BBO	Beta Barium Borate
DPMO	Diode Pumped Master Oscillator
DPSS	Diode Pumped Solid State Laser
FHG	Fourth Harmonic Generation
IR	Infrared
LED	Light Emitting Diode
PCM	Phase Conjugate Mirror
SBS	Stimulated Brillouin Scattering
SHG	Second Harmonic Generation
SLM	Single Longitudinal Mode
THG	Third Harmonic Generation

Preface

Congratulations on your purchase of the Infinity Nd:YAG laser. We at Coherent hope that this laser will become a valuable tool for you. The Infinity laser system represents a major breakthrough in pulsed high power solid state laser technology. Its design is based on a diode pumped master oscillator, followed by a multi-pass power amplifier stage that incorporates a phase conjugate mirror, serving as a corrective element for phase distortions. As a result the Infinity offers true single longitudinal mode operation, diffraction-limited brightness, near top-hat intensity distribution and a repetition rate that is continuously variable from single-shot up to 100 Hz for 40-100 systems, 50 Hz for 25-50 systems, and 30 Hz for 15-30 systems.

As with all sophisticated equipment, proper maintenance can significantly extend the life span of components and also ensure that the Infinity laser continuously will perform at the specified level. It is therefore strongly recommended that you allocate some time to study this manual before operating the system. By studying this manual you will gain a better understanding of how the laser works. This translates into less down time, a more reliable system, and better performance. Problems will be detected earlier and the range of applications extended. If you have any questions about the laser, its operation or the content of this manual, please do not hesitate to contact Coherents Service Department at 800-367-7890 (within the US), 408-764-4557 (outside the US) or your local Coherent representative.



Use of controls, adjustments, or performance of procedures other than those specified herein, may result in hazardous radiation exposure.



Read this manual carefully before operating the laser for the first time. Special attention should be given to the material in Chapter One, Laser Safety.

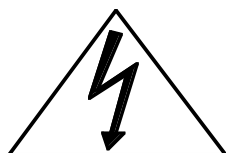
U.S. Export Control Laws Compliance

It is the policy of Coherent to comply strictly with U.S. export control laws.

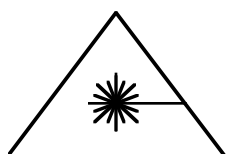
Export and re-export of lasers manufactured by Coherent are subject to U.S. Export Administration Regulations, which are administered by the Commerce Department. In addition, shipments of certain components are regulated by the State Department under the International Traffic in Arms Regulations.

The applicable restrictions vary depending on the specific product involved and its destination. In some cases, U.S. law requires that U.S. Government approval be obtained prior to resale, export or re-export of certain articles. When there is uncertainty about the obligations imposed by U.S. law, clarification should be obtained from Coherent or an appropriate U.S. Government agency.

Symbols Used In This Document



This symbol is intended to alert you to the presence of dangerous voltage within the product's enclosure that may be of sufficient magnitude to constitute a risk of electric shock and to indicate possible risk of equipment damage.



This symbol is intended to alert you to the danger of exposure to hazardous visible and invisible laser radiation.



This symbol is intended to emphasize the presence of important operating and maintenance instructions.

OPERATOR'S MANUAL

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CHAPTER ONE

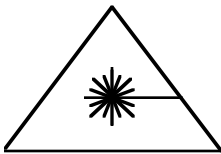
LASER SAFETY

Introduction

The Infinity laser is a high performance laser and has undergone extensive testing to ensure that, with proper usage, it is a safe and reliable system. Due to the nature of laser light and the performance level of the Infinity laser there are some important safety considerations that the you must be aware of. These relate to optical safety, electrical safety, and for the Infinity system, chemical safety. The chemical safety issues are due to CFC 113 used in the cell in which the phase conjugate return beam is generated. In order to use the Infinity system in a safe way, it is important that all users, and everyone near the laser system, are aware of the dangers involved.

Optical Safety

Laser light, because of its special properties, poses safety hazards not associated with light from conventional sources. The safe use of lasers requires that all laser users, and everyone near the laser system, are aware of the dangers involved. The safe use of the laser depends upon the user being familiar with the instrument and the properties of coherent, intense beams of light.



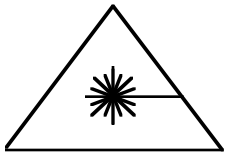
Direct eye contact with the output beam from the laser will cause serious damage and possible blindness.

The greatest concern when using a laser is eye safety. In addition to the main beam, there are often many smaller beams present at various angles near the laser system. These beams are formed by specular reflections of the main beam at polished surfaces such as lenses or beamsplitters. While weaker than the main beam, such beams may still be sufficiently intense to cause eye damage.

Laser beams are powerful enough to burn skin, clothing or paint. They can ignite volatile substances such as alcohol, gasoline, ether and other solvents, and can damage light-sensitive elements in video cameras, photomultipliers and photodiodes. The laser beam can ignite substances in its path, even at some distance. The beam may also cause damage if contacted indirectly from reflective surfaces. For these reasons, and others, the user is advised to follow the precautions below.

1. Observe all safety precautions in the preinstallation and operator's manual.

2. Extreme caution should be exercised when using solvents in the area of the laser.
3. Limit access to the laser to qualified users who are familiar with laser safety practices and who are aware of the dangers involved.
4. Never look directly into the laser light source or at scattered laser light from any reflective surface. Never sight down the beam into the source.
5. Maintain experimental setups at low heights to prevent inadvertent beam-eye encounter at eye level.

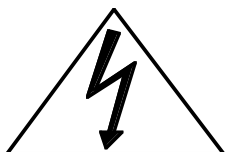


Laser safety glasses can present a hazard as well as a benefit; while they protect the eye from potentially damaging exposure, they block light at the laser wavelengths, which prevents the operator from seeing the beam. Therefore, use extreme caution even when using safety glasses.

6. As a precaution against accidental exposure to the output beam or its reflection, those using the system should wear laser safety glasses as required by the wavelength being generated.
7. Avoid direct exposure to the laser light. The intensity of the beam can easily cause flesh burns or ignite clothing.
8. Use the laser in an enclosed room. Laser light will remain collimated over long distances and therefore presents a potential hazard if not confined.
9. Post warning signs in the area of the laser beam to alert those present.
10. Advise all those using the laser of these precautions. It is good practice to operate the laser in a room with controlled and restricted access.

Electrical Safety

Lethal voltages and currents are present in both the Infinity power supply and laser head. For every laser pulse, up to 50 Joules of energy is deposited into the flash lamp, and the voltage across the capacitor bank can be as high as 570 V. High voltages are also present at various points on the circuit boards. These units are therefore designed to be operated with the protective covers in place. Certain procedures such as changing the flash lamp, water filter, or cleaning optical components, require removal of the protective covers. It is important that all safety precautions contained in this manual are observed by anyone using the laser. The most important rule when performing work on any part of the laser is first to switch it off completely!



There are no user serviceable parts in the electrical side of the power supply. Service procedures on system electronics must be carried out by a Coherent certified field service engineer. Always switch off the laser when working on any part of it.

Troubleshooting and adjustments on the circuit boards are strictly service procedures and should only be done by a Coherent field service technician. Unauthorized attempts by the user to modify or repair the circuit boards will void any warranty!

Chemical Safety

The Infinity laser incorporates a cell filled with CFC 113, which is used to generate the backward traveling return beam going through the power amplifiers. The choice of CFC 113 is motivated by its superior physical properties with respect to reflectivity, chemical stability and threshold for the phase conjugation process. Under normal operating conditions there are no situations in which the user will be exposed to the liquid. In the event of a cell leak or breakage, avoid direct skin and eye contact with the liquid. Ensure good ventilation until it has evaporated completely and the environment is odor free (CFC 113 has an ethereal and a faint sweetish odor). Call Coherent technical support for information about replacement cells and handling of damaged cells.

Safety Features and Compliance to Government Requirements

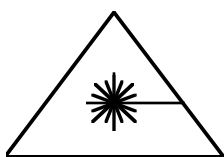
The following features are incorporated into the instrument to conform to several government requirements. The applicable United States Government requirements are contained in 21 CFR, subchapter J, part II administered by the Center for Devices and Radiological Health (CDRH). The European Community requirements for product safety are specified in the "Low Voltage Directive" (published in 73/23/EEC and ammended in 93/68/EEC). The "Low Voltage Directive" requires that lasers comply with the standard EN 61010-1 "Safety Requirements For Electrical Equipment For Measurement, Control and Laboratory Use" and EN60825-1 "Radiation Safety of Laser Products". Compliance of this laser with the requirements is certified by the CE mark.

Classification

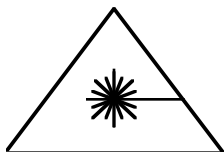
The governmental standards and requirements specify that the laser must be classified according to the output power or energy and the laser wavelength. The Infinity is classified as Class IV based on 21 CFR, subchapter J, part II, section 1040-10 (d). According to the European Community standards, the Infinity classified as Class 4 based on EN 60825-1, clause 9. In this manual and other documentation of the Infinity, the classification will be referred to as Class 4.

Protective Housing

The Infinity head is enclosed in a protective housing that prevents human access to radiation in excess of the limits of Class I radiation as specified in the Federal Register, July 31, 1975, Part II, Section 1040.10 (f) (1) and Table 1-A/EN 60825-1, clause 4.2 except for the output beam, which is Class IV.



Operation of laser with cover removed will allow access to hazardous visible and invisible radiation. The laser housing should only be opened for the purposes of maintenance and service by trained personnel cognizant of the hazards involved.



Use of controls or adjustments or performance of procedures other than those specified in the manual may result in hazardous radiation exposure.



Use of the system in a manner other than that described herein may impair the protection provided by the system.

Safety Interlocks

Electrical safety interlock breakers are positioned at four places in the Infinity system to prevent accidental exposure to dangerous levels of electricity or laser radiation. (21 CFR 1040.10 (f)(2)). The four interlocks are: head cover (1), flash lamp terminal housing (1) and amplifier housing latches (2).

External Interlock Connector

A RCA plug on the rear of the power supply is provided for use as an external interlock connection. This connector must be short circuited for the laser to operate. The laser is shipped with a plug installed. If this is removed then an external interlock circuit can be connected in its place (21 CFR 1040.10 (f)(3)).

Key Control

The instrument cannot be turned on or operated until the system key switch, located at the bottom of the power supply beneath the BNC connectors, is turned on (21 CFR 1040.10 (f)(4)).

Laser Radiation Emission Indicator

The appropriately labeled indicator lights on the laser head are turned on at least five seconds before laser emission can occur. It is visible without exposing the operator to laser emission. White light is used so that it may be seen regardless of the type of safety glasses employed (21 CFR 1040.10 (f)(5)).

Beam Attenuators

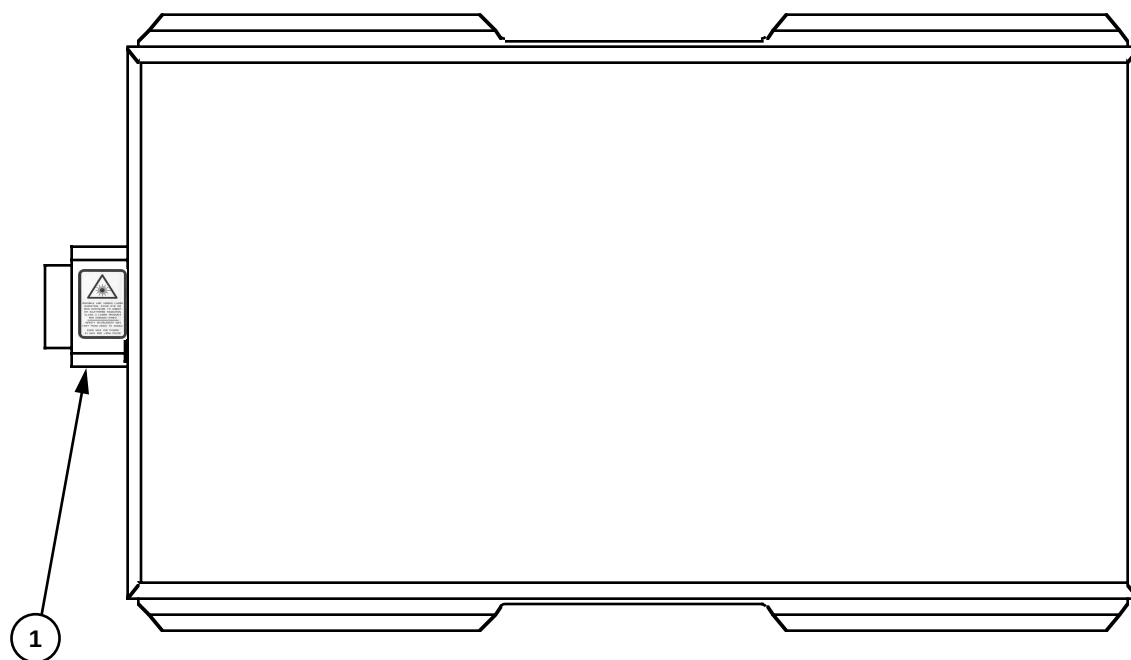
A mechanical shutter is provided with the Infinity. The shutter is actuated through the computer interface (21 CFR 1040.10 (f)(6)).

Manual Reset

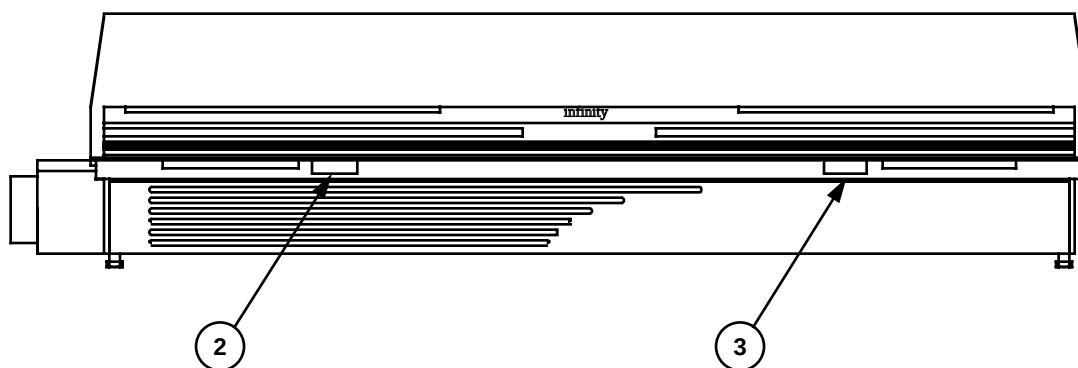
After an interruption in laser activity because of remote interlock activation, or from an interruption due to loss of electrical power, the laser will not restart until the operator presses the On button in the main control window in the computer interface (21 CFR 1040.10 (f)(10)).

Location of Safety Labels

Refer to Figures 1-1 through 1-11 for a description and location of all safety labels. These include warning labels indicating removable or displaceable protective housings, apertures through which laser radiation is emitted and labels of certification and identification [CFR 1040.10(g), CFR 1040.2, and CFR 1010.3/EN61010-1 EN60825-1, Clause 5].

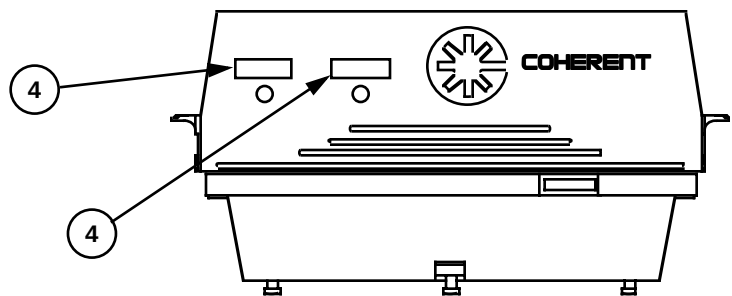


Laser Head Exterior, Top View

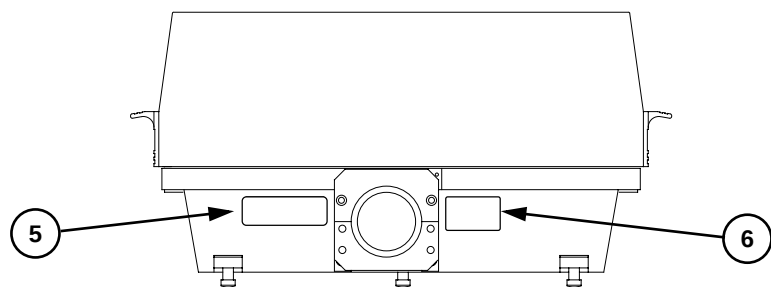


Laser Head Exterior, Side View

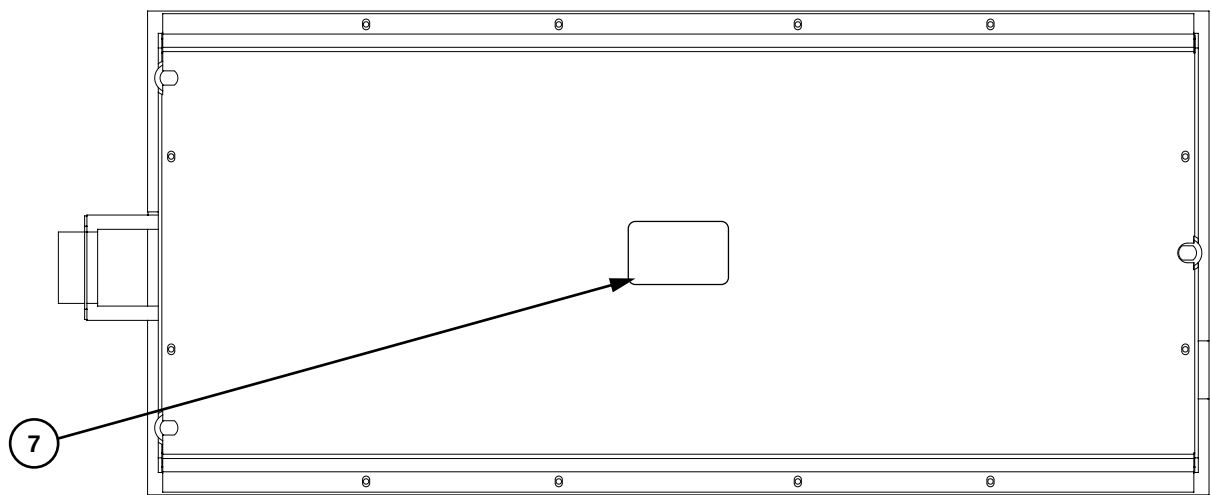
Note: Key is located on sheets 7 through 11.
Figure 1-1. Location of Safety Labels (Sheet 1 of 11)



Laser Head Exterior, Front View

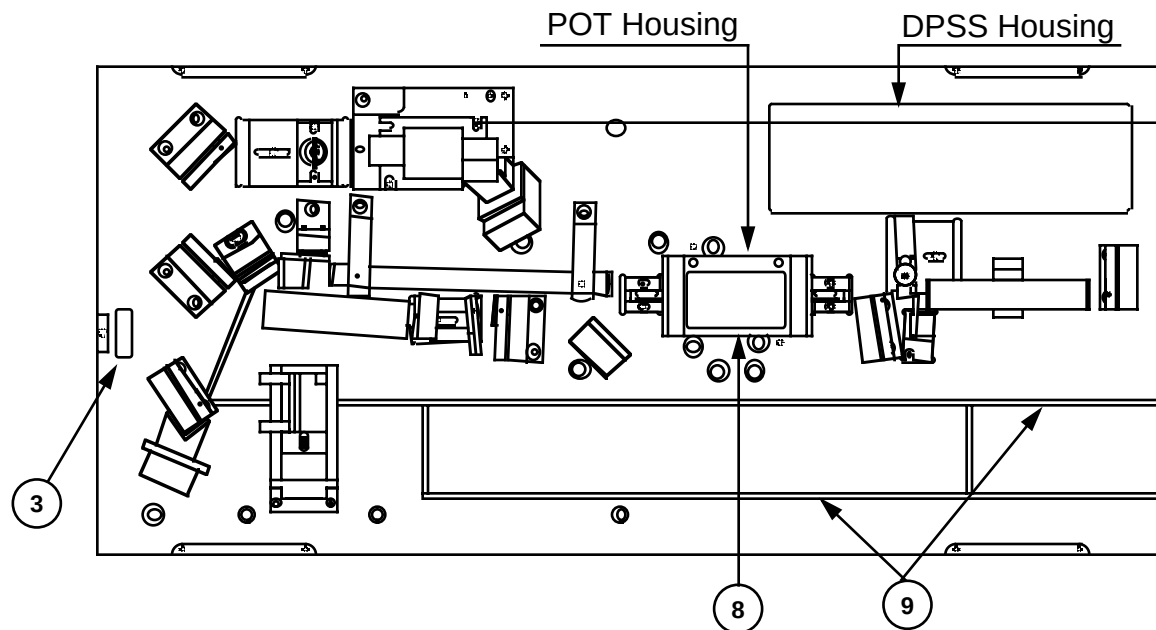


Laser Head Exterior, Rear View

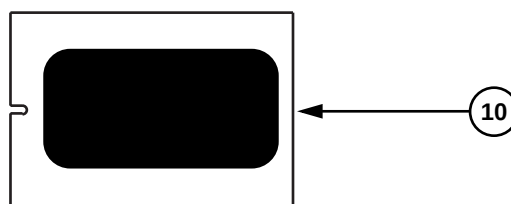


Laser Head Exterior, Bottom View

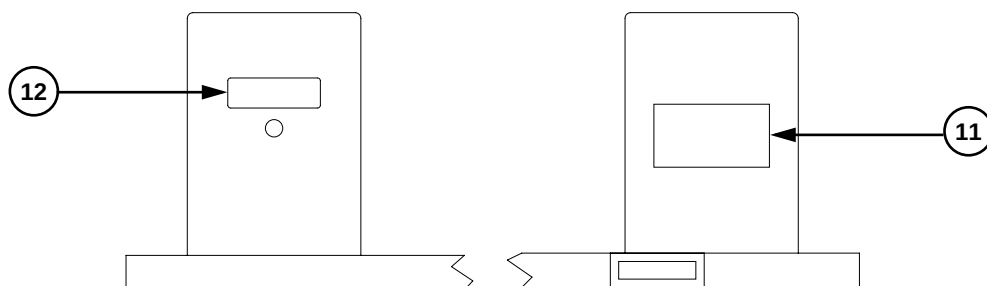
Note: Key is located on sheets 7 through 11.
Figure 1-2. Location of Safety Labels (Sheet 2 of 11)



Laser Head Interior Without Cover, Top View



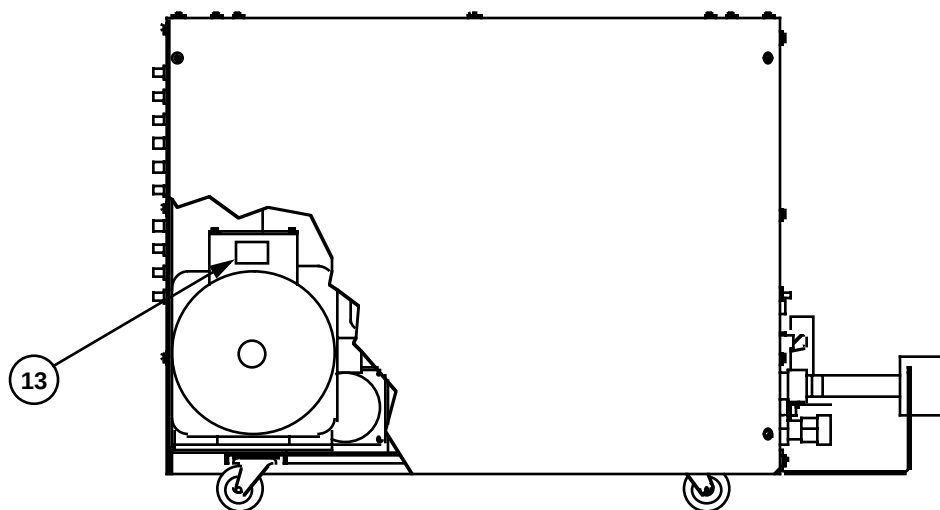
Laser Head Interior, POT Terminal Housing Cover



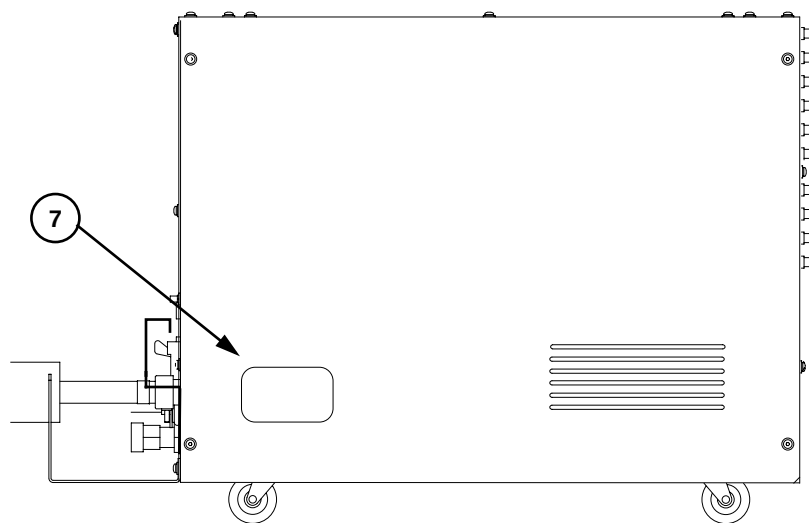
Laser Head Interior, DPSS Housing, Exit Aperture (12) and Rear (11)

Note: Key is located on sheets 7 through 11.

Figure 1-3. Location of Safety Labels (Sheet 3 of 11)



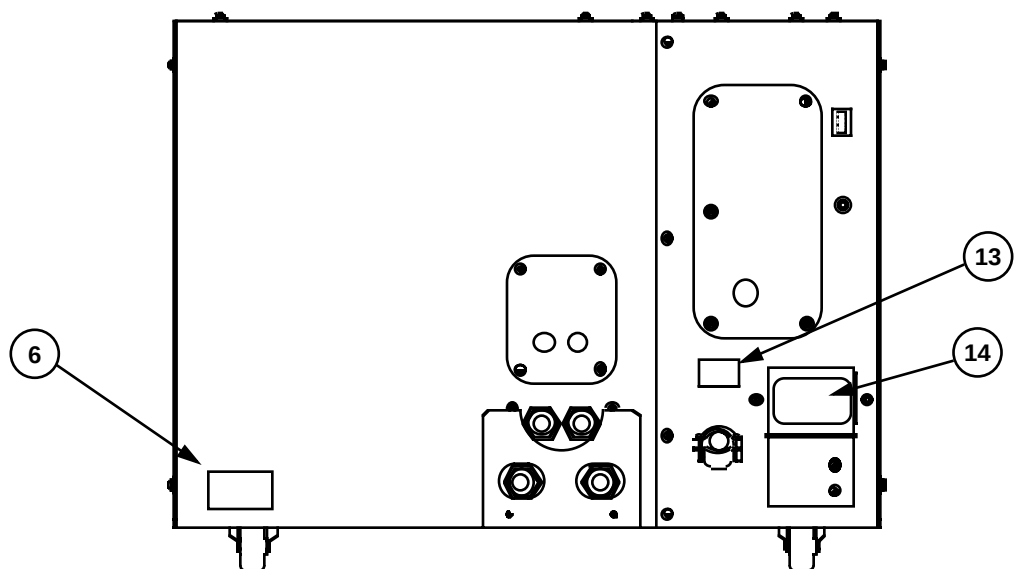
Power Supply, Cut-away, Right Side View, Water Pump



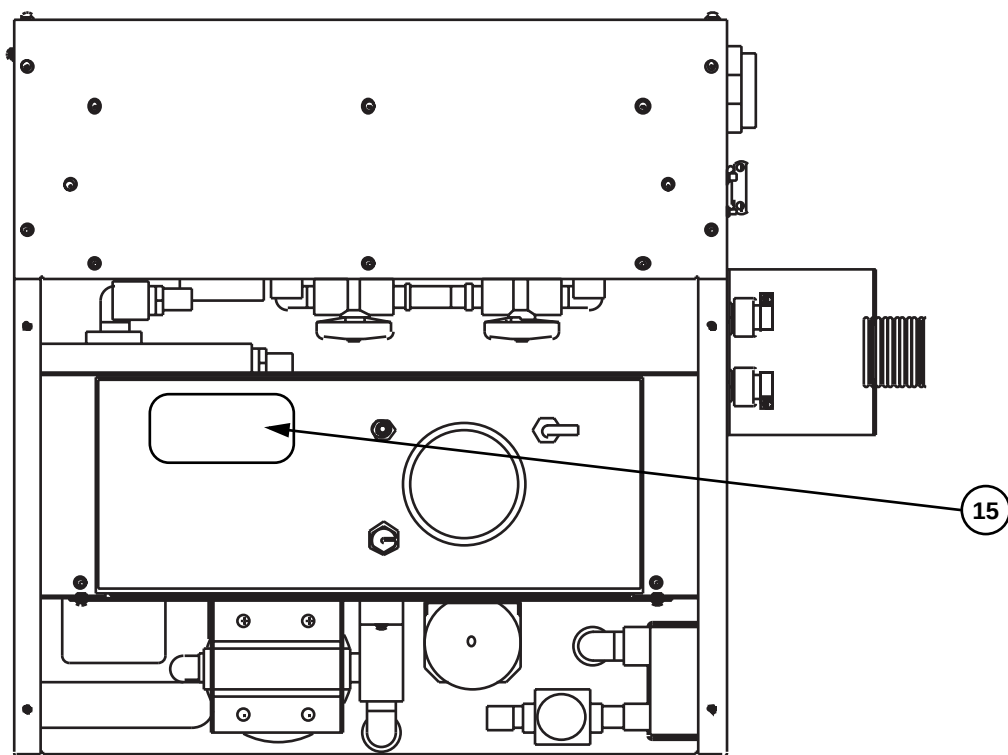
Power Supply, Left Side View

Note: Key is located on sheets 7 through 11.

Figure 1-4. Location of Safety Labels (Sheet 4 of 11)



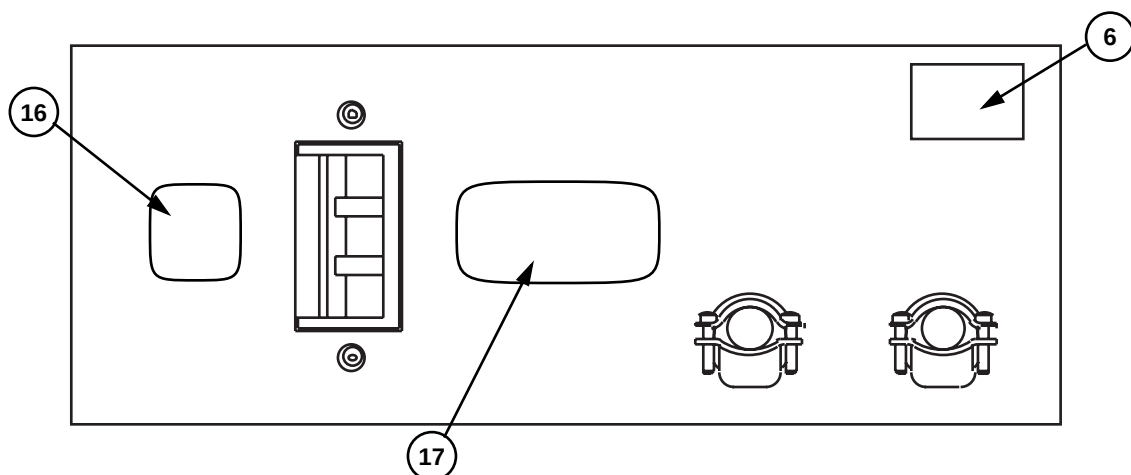
Power Supply, Rear View



Power Supply, Top View

Note: Key is located on sheets 7 through 11.

Figure 1-5. Location of Safety Labels (Sheet 5 of 11)



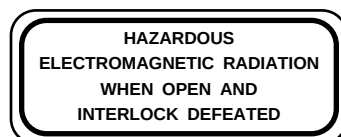
External EMI Filter (European Systems Only), Back View

Note: Key is located on sheets 7 through 11.

Figure 1-6. Location of Safety Labels (Sheet 6 of 11)



Label 1



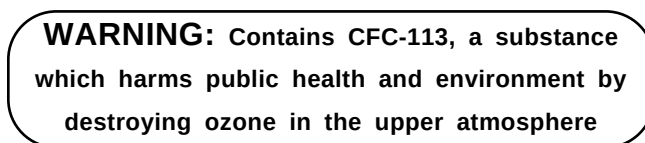
Label 2



Label 3



Label 4

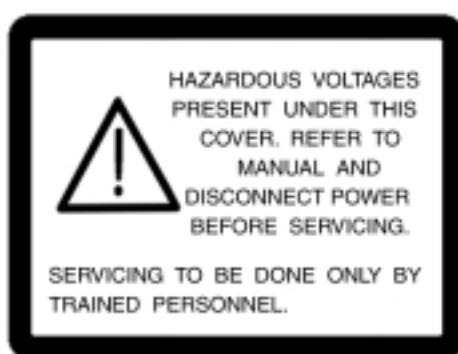


Label 5

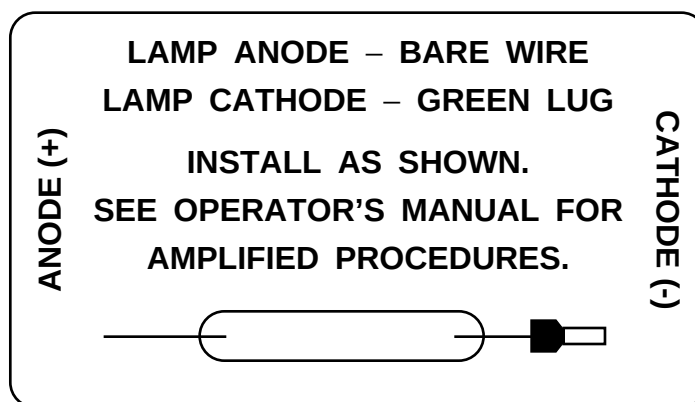
Figure 1-7. Location of Safety Labels (Sheet 7 of 11)



Label 6



Label 7



Label 8

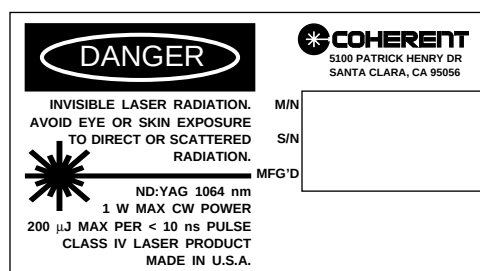
Figure 1-8. Location of Safety Labels (Sheet 8 of 11)



Label 9



Label 10



Label 11

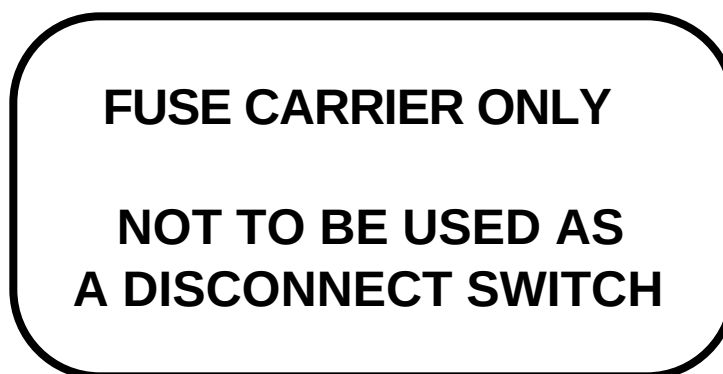


Label 12

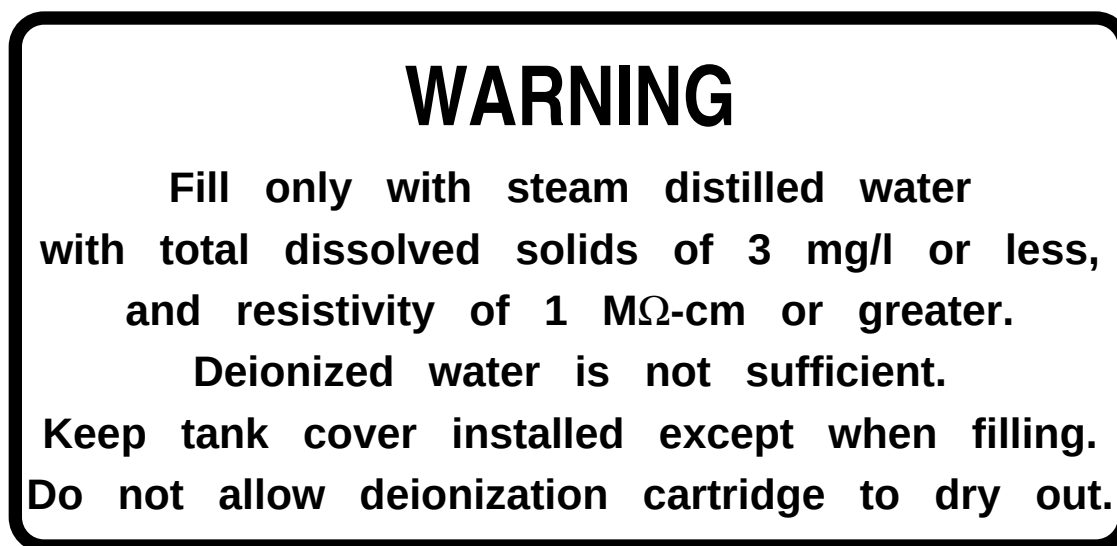
Figure 1-9. Location of Safety Labels (Sheet 9 of 11)



Label 13

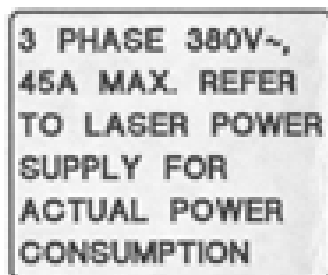


Label 14

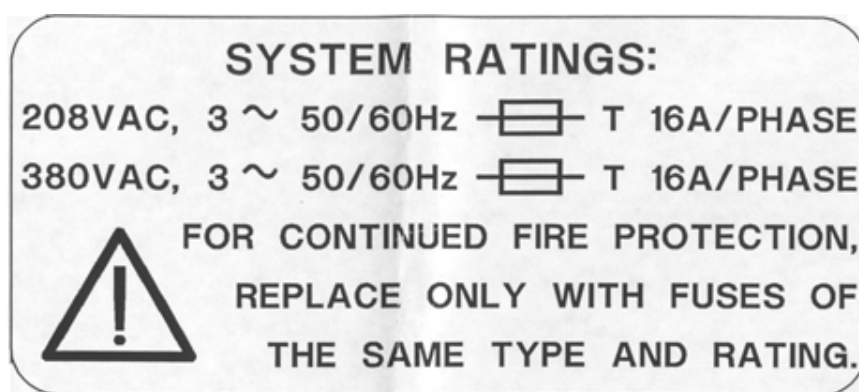


Label 15

Figure 1-10. Location of Safety Labels (Sheet 10 of 11)



Label 16



Label 17

Figure 1-11. Location of Safety Labels (Sheet 11 of 11)

References

More detailed information on laser safety can be found in the following publications:

"LIA Laser Safety Guide"

Laser Institute of America, 12424 Research Parkway, Suite 125, Orlando, FL 32826-3274, Telephone: (800)-345-2737 or (407)-380-1553

"American National Standard for the Safe Use of Lasers"

American National Standards Institute, 1430 Broadway, New York, NY 10018

"Safety with Lasers and Other Optical Sources"

D. Sliney and M. Wolbarsht, Plenum Publishing Company, New York, N.Y. 1980

Additional safety information and material can also be found through the following sources:

Laser Institute of America

Books, Safety standards, Training courses, Warning signs.

12424 Research parkway, Suite 125

Orlando, FL 32826-3274

Telephone: (800)-345-2737 or (407)-380-5588

Coherent, Inc.

"Laser Safety Comes to Light", Video

5100 Patrick Henry Drive

P.O. Box 54980

Santa Clara, CA 95056

Telephone: (408)-764-4800 or (800)-527-3786

Laser Focus World Buyer's Guide

List of suppliers of safety materials, e.g. signs, goggles etc.

One Technology Park Drive

P.O. Box 989

Westford, MA 01886

Telephone: (508)-692-0700

OPERATOR'S MANUAL

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CHAPTER TWO

DESCRIPTION AND SPECIFICATIONS

Description

The Infinity™ Nd:YAG laser system consists of an internal diode pump, Q-switched, master oscillator which provides the seed pulse for a dual rod, single lamp, amplifier stage. An overview of the Infinity's three primary assemblies is given in Figure 2-1. Also illustrated in Figure 2-1 is the suggested setup orientation for each of the components.

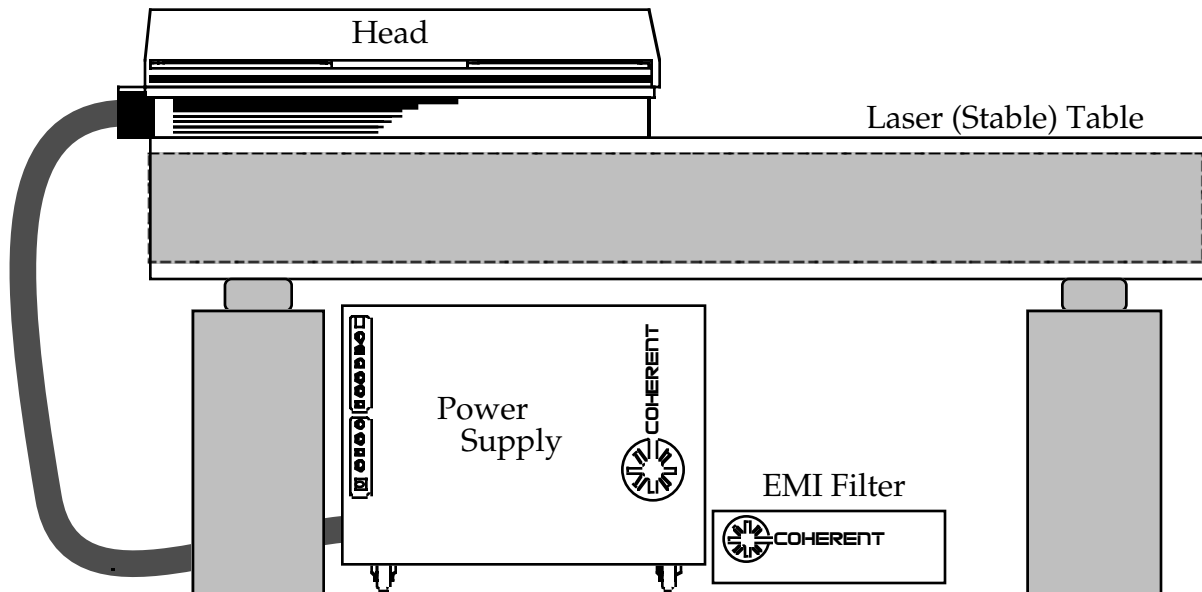


Figure 2-1. Infinity Main Assemblies.

Note that the external EMI filter (under table, lower right of figure) is only part of systems which are shipped to the European Community. This component is wired between the facility power the Infinity power supply.

An optical schematic of the Infinity laser head is presented in Figure 2-2. Refer to Chapters Three and Eight for complete descriptions of each component in the Infinity Laser System.

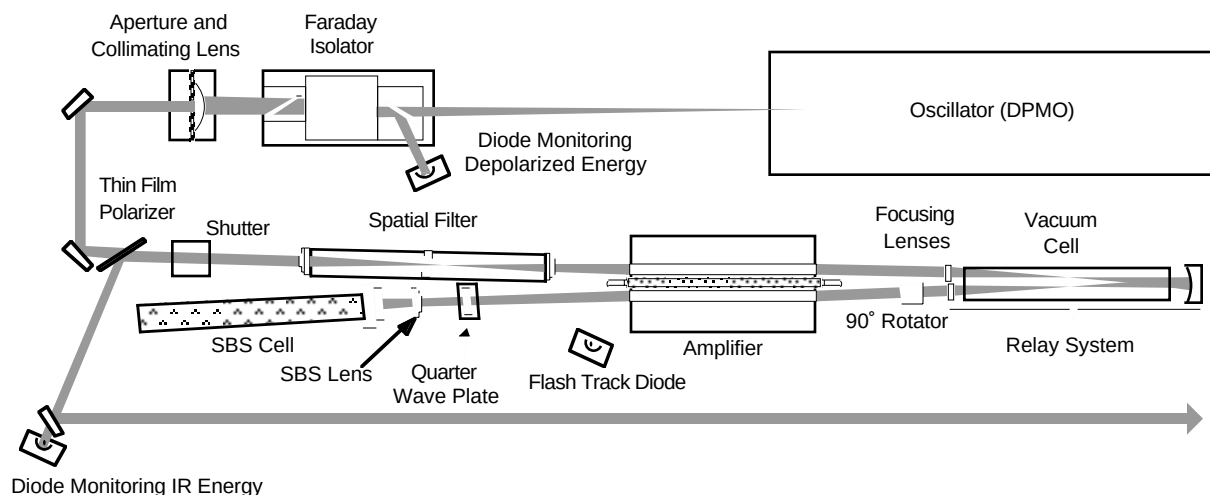


Figure 2-2. Infinity Optical Schematic.

Pulse Energy Servo

The Infinity is equipped with a feedback system that improves long term stability of the laser output. This system utilizes the output from a photo-diode measuring the IR energy to adjust the voltage across the capacitor bank. This feature is enabled or disabled in the computer interface. When the output power is set by choosing the voltage applied to the flash lamp, the light regulation system is disabled and when it is set by choosing mJ/pulse it is enabled. Operating the system using the lamp voltage control is normally a service procedure. See Chapter Seven, System Software, for complete details.

Safety Interlocks

In order to protect the user(s), four mechanical safety interlocks and a mechanical shutter have been built into the system. The interlocks can be tripped by opening the laser head cover, opening the lid covering the flash lamp leads, or removing one of dowel pins adjacent to the amplifier assembly latches. In addition there is an external interlock connector at the rear of the power supply. This provides the user with the option of installing a safety interlock switch connected to an external device such as a lab door.

If any of the interlocks are tripped while the laser is running, all systems except the water pump will be switched off immediately and the fault message: "Interlock open" displayed (obviously interlocks inside the laser head can only be tripped if the head cover interlock first has been defeated). Before the laser can be restarted, the off button needs to be pressed and the closed position of the interlocks restored. Thereafter the normal start-up procedure, as described in Chapter Four, can be used. If the fault message is cleared but

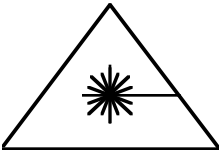
the fault condition still exists, the message will reappear within a few seconds.

If the laser is operated with the laser head cover interlock defeated, and the cover is either opened or closed while the laser is running, the beam will strike the inside of the cover while it is being translated through the beam path. This may give rise to sputtering which can deposit onto the optics resulting in irreversible damage.

The electromagnetically activated shutter is positioned in the beam path between the thin-film polarizer and the spatial filter. It provides a way of preventing a laser pulse from leaving the laser head without changing the thermal load on the Nd:YAG rods.



Never open or close the laser head cover while operating the laser with the interlock defeated. Irreversible damage may occur to optical components.



Operating the laser with the interlocks defeated is a service procedure. It should only be carried out by a qualified technician who is well aware of the potential hazards. Damage that occurs to the system while operated with defeated interlocks is not covered by Coherent warranty.

PulseGuard™

PulseGuard is a safety system designed to protect the laser by continuously monitoring the depolarized light intensity. Depolarized light is laser energy that after amplification, instead of being reflected out of the laser by the thin film polarizer, is transmitted through it and directed back into the Faraday Isolator. It is desirable to keep this energy component less than or equal to 1.5% of the output energy at the systems maximum repetition rate and pulse energy. If the energy going through the Faraday Isolator, as measured by the photodiode attached to it, exceeds 8 mJ/pulse the on board microprocessor will shut off the system. This is done in order to protect the master oscillator from being damaged. A bar graph display showing the depolarized energy is available in the Infinity system "Monitor" window. This window can be accessed by clicking on the "Monitors..." button found in the main Infinity software window. See Chapter Seven, System Software, for additional information.

FlashTrack Servo Controls

FlashTrack is a control system that reduces shot-to-shot intensity fluctuations. A photo-diode monitors the emission intensity of the flash lamp during the first half of the lamp pulse. A feed-forward circuit then uses this information to make a small correction to the lamp current during the second half of the lamp pulse such that the total amplification is constant over time. The principle of operation is illustrated in Figure 2-3. A photo-diode connected to an integrator monitors the flash lamp emission. After half of the lamp width time has elapsed a sample and hold circuit reads out the integrator and converts it to a voltage. This voltage is summed with V_{Coarse} and V_{Vernier} (both inverted) and then amplified. The amplified signal is converted to a pulse that has fixed height and a length proportional to the voltage. This pulse is used to drive a transistor switch regulating the current going through the lamp. If the first half of a lamp pulse is brighter than average, the amplifier output goes down (signal applied to negative input of amplifier). This yields a shorter drive pulse for the transistor switch which decreases the amount of current going through the lamp during the second half of the lamp pulse. See Chapter Seven, System Software, for additional information.

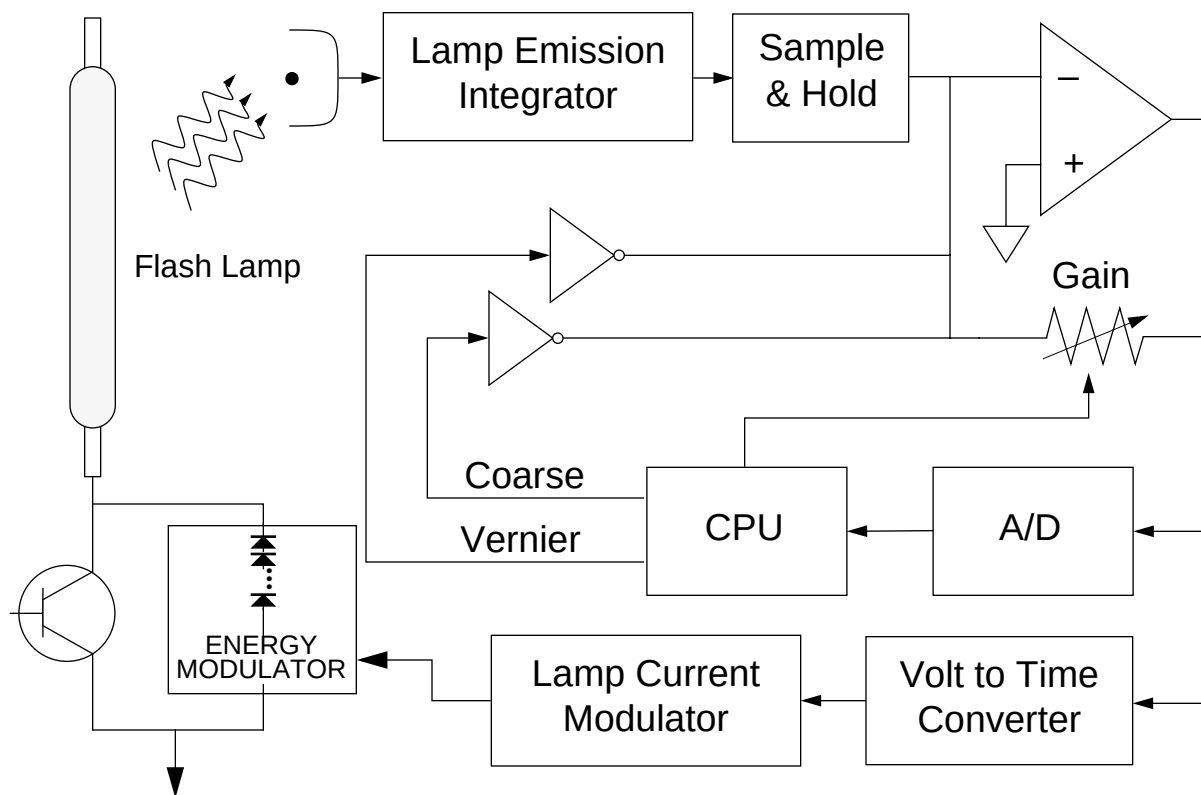


Figure 2-3. Block Diagram for FlashTrack Servo.

Water Cooling System

For every laser pulse of a 100 Hz system, about 40 Joules of energy is deposited into the flash lamp. This means that 4000 W are deposited into the amplifier head. Only a small fraction of this energy is converted into laser light. Because of the large heat load, the small dimensions, and the stringent requirements, an internal water cooling loop is used. This water is cooled using externally supplied water in a heat exchanger located inside the power supply. The layout of the water cooling system is illustrated in Figure 2-4.

Tap water enters the power supply and goes through a heat exchanger and then back out to the drain. The internal water goes (after being cooled in the heat exchanger) up to the laser head, where it cools off the flash lamp, the laser rods, and some of the electronics located under the base plate. After this the water flows back to the power supply where it goes through a water filter with 10 μm pore size, followed by an instrument that monitors the water resistivity. If the conductivity of the water rises above or below preset values, the computer will display a fault message. In order to keep the water deionized, a fraction of it is sent through a deionization filter while the main flow is directed back to the main reservoir. The internal cooling water is also used to regulate the temperature of some of the electronics in the power supply, see Figure 2-4.

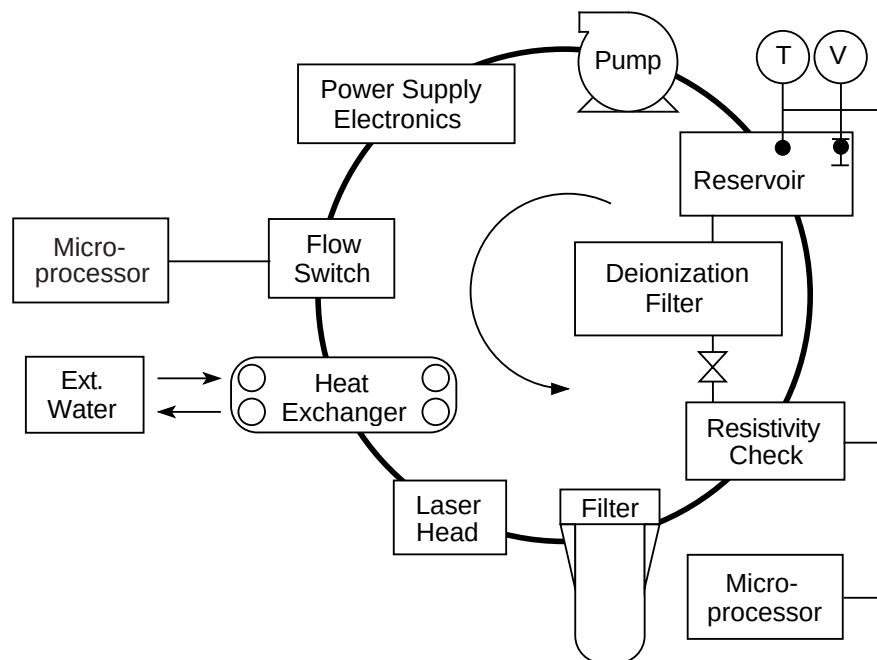
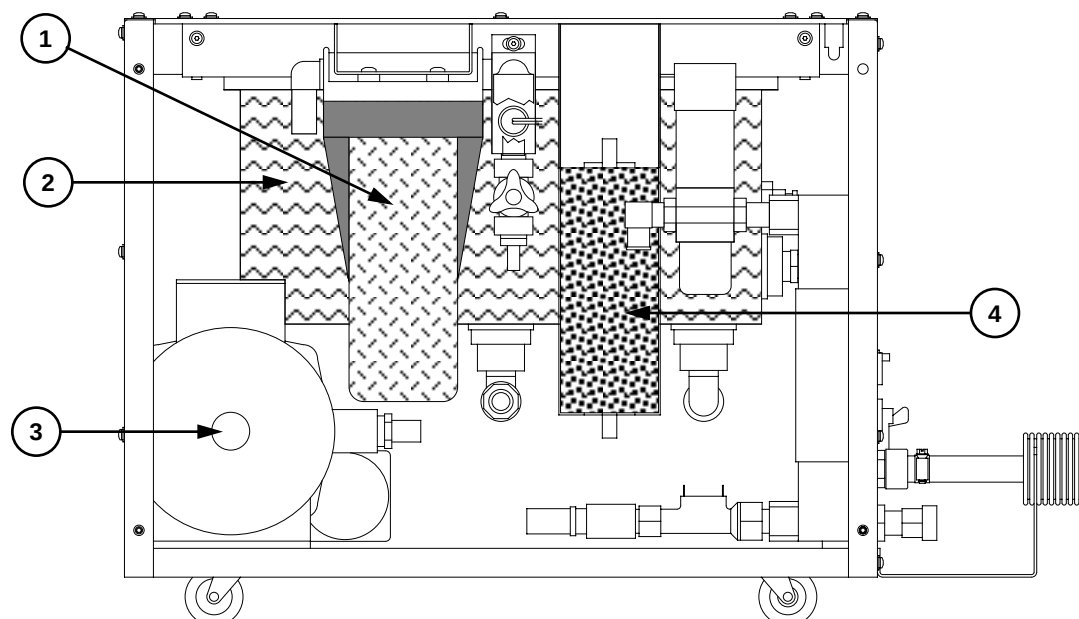


Figure 2-4. Schematic of Infinity Water Cooling Loop.

Heat Exchanger

It is recommended that the internal cooling water, the deionization filter, and the particle filter be changed once every year. Major heat exchanger components, the right hand side of the Infinity power supply, are illustrated in Figure 2-5.



- | | |
|--------------------|---------------------------|
| 1. Particle Filter | 3. Three Phase Water Pump |
| 2. Water Reservoir | 4. Deionization Cartridge |

Figure 2-5. Major Heat Exchanger Components.

Infinity Specifications

Table 2-1. Infinity Pulse Energy Specifications

	PULSE ENERGY AT [mJ]/pulse]			
REP. RATE	1064 nm	532 nm	355 nm	266 nm
0.1 - 20 Hz	600	300	240	50 ¹
20 - 50 Hz	500	250	200	40
50 - 100 Hz	400	200	160	25

¹ Pulse energy at 266 nm is 50mJ/pulse at 10 Hz.

Table 2-2. Infinity Noise Specifications

	NOISE AT [% std. dev]			
REP. RATE	1064 nm	532 nm	355 nm	266 nm
0.1 - 20 Hz	1.0	1.8	2.0	<2.5
20 - 50 Hz	1.0	1.8	2.0	<2.5
50 - 100 Hz	1.7	2.5	2.5	<4.0

Table 2-3. Infinity Typical Pulse Widths

WAVELENGTH [nm]	PULSE WIDTH [ns]
1064	3.5
532	3.0
355	3.0
266	2.5

Table 2-4. Infinity System Specifications

PARAMETER	SPECIFICATIONS
IR mode quality	< 1.5x Diffraction limit
Beam diameter, near field ¹	> 95% of pulse energy through 5.5 mm at SHG crystal
Beam diameter, far field	> 84% of pulse energy into < 0.7 mrad divergence
Divergence, full angle	< 0.7 mrad
Linewidth Single-longitudinal mode Single shot (typical)	< 2x transform limit < 250 MHz
Timing jitter (rms) rel. Q-switch synch	< 500 psec

¹ Near-field image is formed at the SHG crystal position.

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CHAPTER THREE

CONTROLS AND INDICATORS

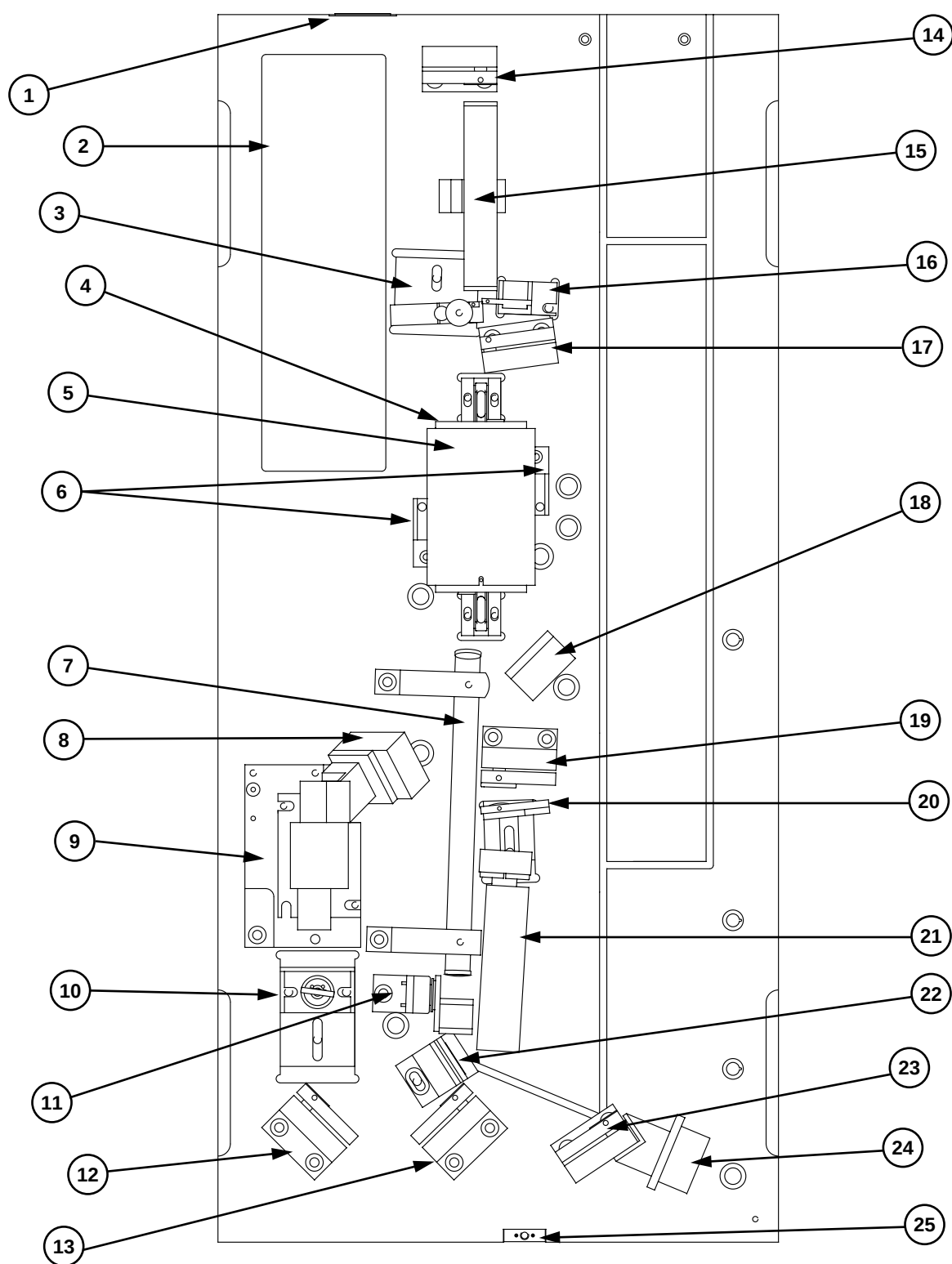


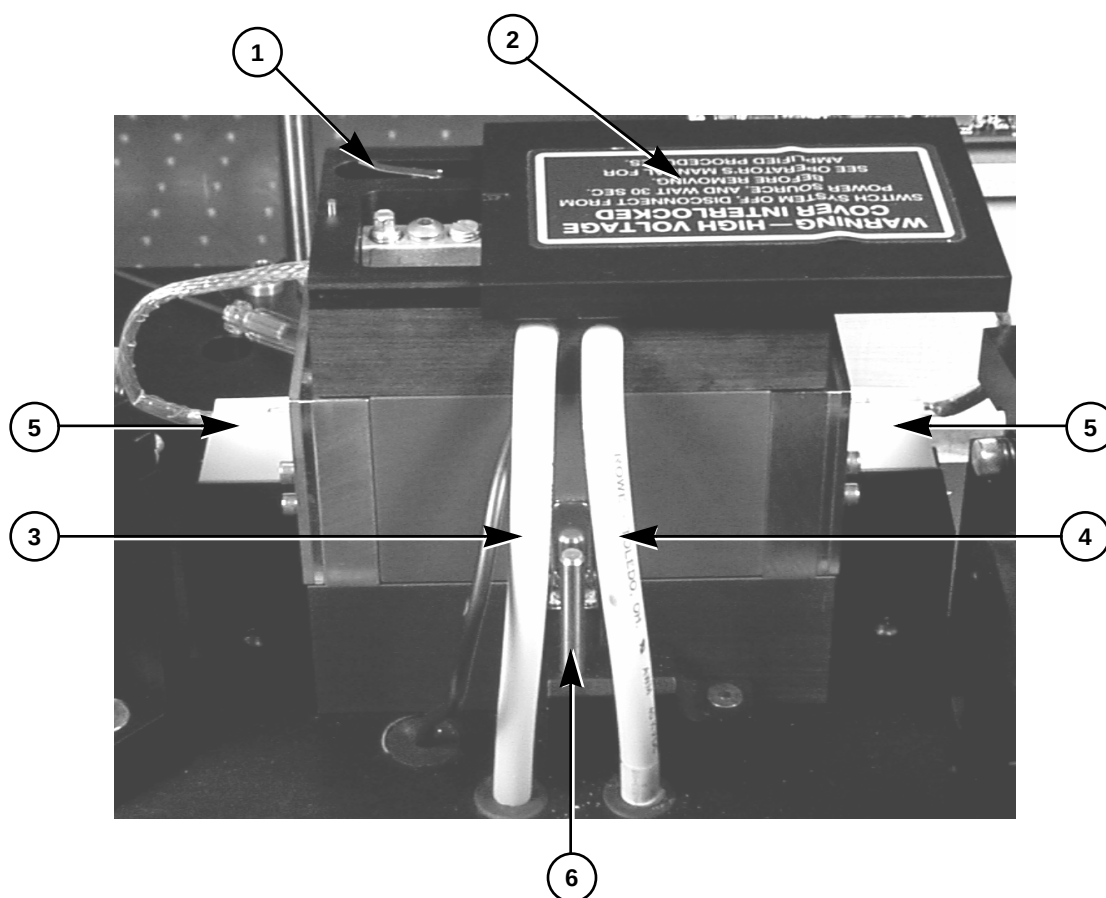
Figure 3-1. Laser Head Controls and Indicator Locations.

Table 3-1. Laser Head Controls and Indicators

CALLOUT	DESIGNATION	REFERENCE FIGURE
1	Laser emission indicator	3-1
2	DPMO, Diode Pumped Master Oscillator	3-1
3	Relay system lens #1	3-4
4	Amplifier assembly, POT	3-2
5	Flash lamp terminal box (above amplifier assembly)	3-2
6	Safety interlock dowel pin	3-2
7	Vacuum spatial filter	3-1
8	Depolarized energy detector	3-3
9	Faraday isolator	3-3
10	Collimating lens	3-1
11	Mechanical shutter	3-1
12	Folding mirror, M1	3-1
13	Reversed folding mirror, M2	3-1
14	Concave mirror, relay system	3-1
15	Relay system vacuum cell	3-1
16	Relay system lens 2	3-4
17	90° Polarization rotation prism	3-4
18	Flashtrack photodetector	3-5
19	Quarter wave plate	3-5
20	SBS lens	3-1
21	SBS cell	3-1
22	Thin film polarizer	3-1
23	Folding mirror, M3	3-6
24	Photo-diode monitoring IR energy	3-6
25	Head cover interlock	3-1



If a back reflection, from an optic external to the laser head, is sent into the laser it can be amplified by residual gain in the POT assembly. After amplification the back reflection, even after being attenuated by the Faraday isolator, may have enough energy to damage the master oscillator permanently. Always be aware of the location of back reflections from optical components when moving them in, or through, the Infinity output beam path.

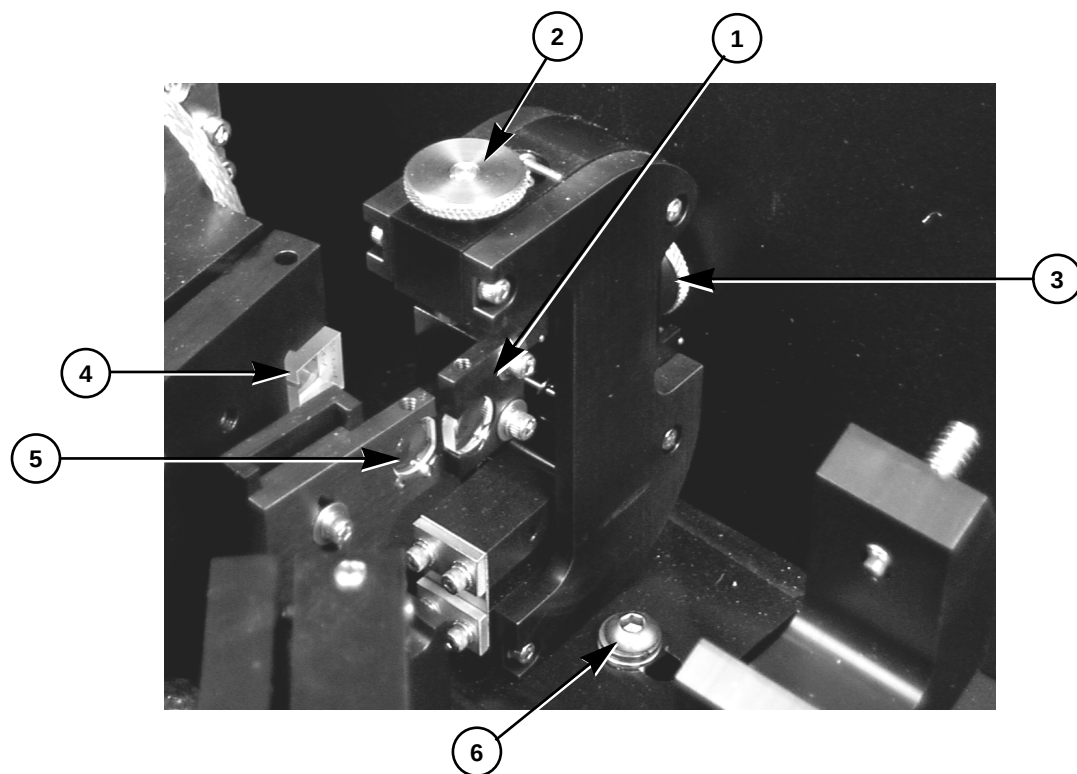


- | | |
|--|------------------------------------|
| 1. Flash lamp terminal box slide cover | 4. Lamp cathode wire |
| 2. Terminal box cover interlock | 5. Teflon lamp end shields (2x) |
| 3. Lamp anode wire | 6. Safety interlock dowel pin (2x) |

Figure 3-2. Infinity POT Assembly.

Table 3-2. Infinity POT Assembly Components

CONTROL	FUNCTION
Terminal box cover interlock	One of three interlocks on the Infinity POT assembly. Opening the terminal box cover with the laser on will result in the laser shutting down and an interlock fault being displayed by the Infinity computer system.
Anode and Cathode wires	Complete the current path, through the lamp, for the capacitor bank discharge to B ⁻ (ground). When verifying or setting the lamp pulse width the cathode wire (closest to relay lens #1) should be used. See Chapter Six, Maintenance, for the complete system timing procedure.
Safety interlock dowel pin	There is an interlock dowel pin on each side of the Infinity POT assembly. Their purpose is to prevent the POT clips from opening during normal operation of the laser system. If the laser is moved these pins should be removed from the head and placed in the "Infinity Maintenance Kit". If a dowel pin is removed while the laser system is running, the system will shut down and an interlock fault will be displayed by the Infinity computer system.

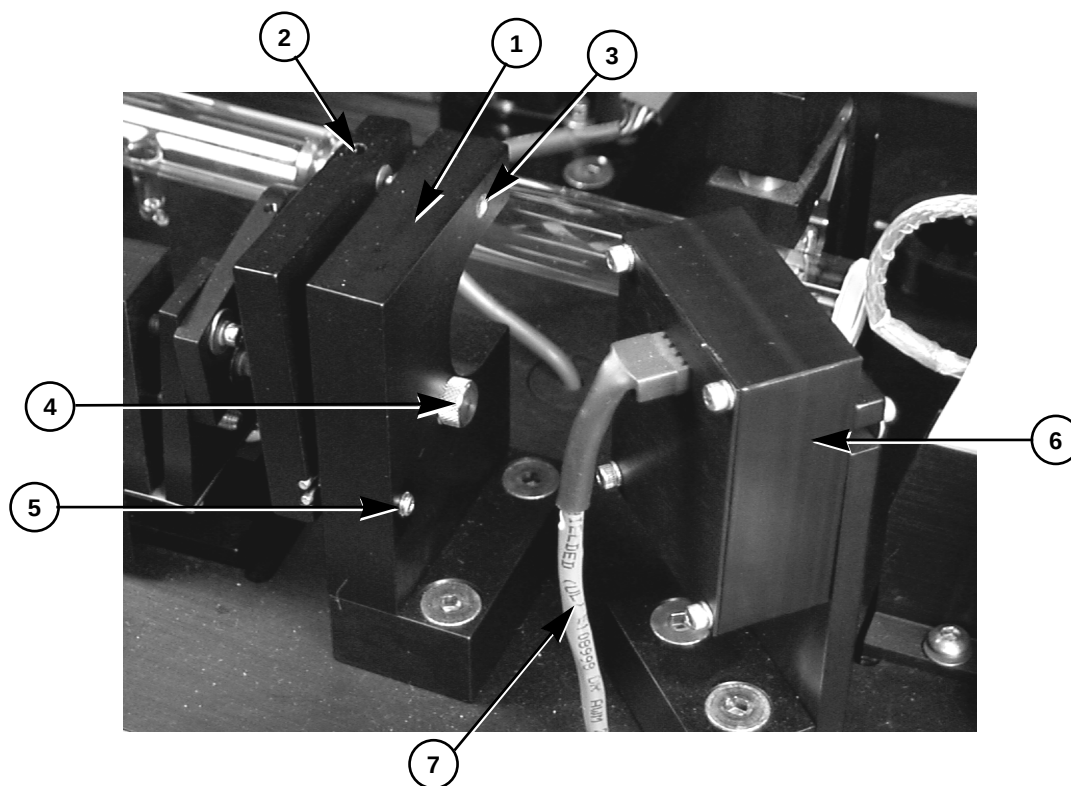


- | | |
|-------------------------------------|--|
| 1. Relay lens #1 assembly | 4. 90 degree rotator assembly |
| 2. Relay lens #1 vertical control | 5. Relay lens #2 assembly |
| 3. Relay lens #1 horizontal control | 6. Relay lens #1 translation set screw |

Figure 3-3. Relay Lens #1, #2, and 90 Degree Rotator.

Table 3-3. Relay Lens #1, #2, and 90 Degree Rotator Components

CONTROL	FUNCTION
Relay lens #1 assembly	First lens in the Infinity relay imaging system. This lens can be adjusted to minimize the system depolarized energy at high repetition rates. See Chapter Four, Procedure 4, for additional information.
Relay lens #1 vertical and horizontal adjustments	Allows the vertical and horizontal positioning of the relay lens. See Chapter Four, Procedure 4, for additional information.
Relay lens #1 translation set screw	Locks the translational position of relay lens #1. Under normal operating conditions the factory determined position should not be changed.
Depolarized energy detector bracket retaining screw	Removing this screw allows the depolarized energy detector, including the bracket, to be detached from the Faraday isolator. To prevent damage to the electronics the photodetector electronics plug should be unplugged with the system key switch in the "off" position, before the bracket retaining screw is removed.
90 degree rotator assembly	This optical component can be adjusted to minimize the system depolarized energy. See Chapter Four, Procedure 4, for additional information. See Table 3-4, quarterwave plate, for a description of the Infinity fold assembly vertical and horizontal tilt adjustments.

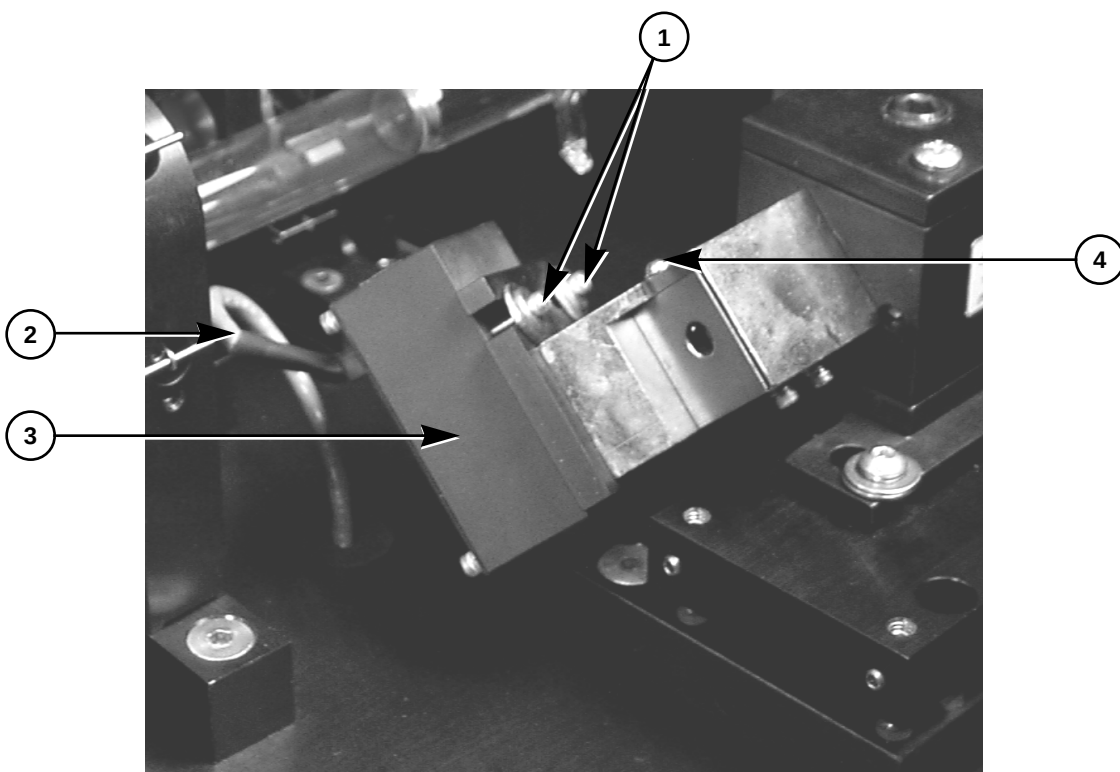


- | | |
|---|--|
| 1. Quarter waveplate assembly | 5. Fold assembly horizontal tilt adjustment |
| 2. Quarter waveplate setscrew | 6. Flashtrack photodetector assembly |
| 3. Fold assembly vertical tilt adjustment | 7. Flashtrack photodiode electronics connector |
| 4. Fold assembly tensioning screw | |

Figure 3-4. Quarter Waveplate and Flashtrack Photodetector Assemblies.

Table 3-4. Quarter Waveplate and Flashtrack Photodetector Assembly Components

CONTROL	FUNCTION
Quarter waveplate assembly	Rotates the polarization of the Q-switched DPMO (DPSS) output from horizontal to vertical. This optical component can be adjusted to minimize the system depolarized energy. See Chapter Four, Procedure 4, for additional information.
Quarter waveplate set screw	Secures the quarter waveplate in the fold assembly. Loosening the setscrew will allow the waveplate to be rotated. See Chapter Four, Procedure 4, for additional information.
Fold assembly vertical and horizontal tilt adjustments	Allow for the horizontal and vertical adjustment of the optical component be held in the fold assembly. In this case, that optical component is the quarter waveplate.
Fold assembly tensioning screw	Determines the amount of pressure required to adjust the vertical and horizontal tilt controls of the fold assembly.
Flashtrack photodetector assembly	Provides real time corrections to the lamp flash energy. See Chapter Eight, System Theory, for additional information on this feature.

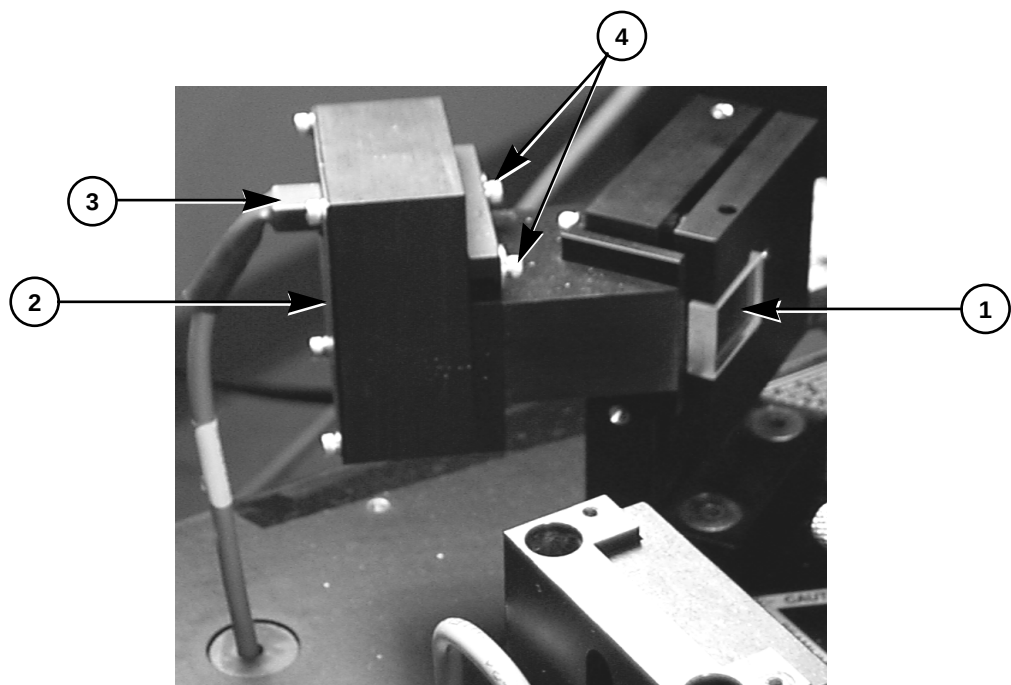


- | | |
|---------------------------------|--|
| 1. Diode housing securing bolts | 3. Photodiode housing |
| 2. Photodiode electronics plug | 4. Depolarized energy detector bracket retaining screw |

Figure 3-5. Depolarized Energy Photodetector.

Table 3-5. Depolarized Energy Detector Components

CONTROL	FUNCTION
Diode housing securing bolts	Loosening these bolts will allow the photodetector signal to be maximized, a step in the calibration procedure for this detector. Under normal operating conditions repositioning of the depolarized energy detector is not required.
Depolarized energy detector bracket retaining screw	Removing this screw allows the depolarized energy detector, including the bracket, to be detached from the Faraday isolator. To prevent damage to the electronics the photodetector electronics plug should be unplugged with the system key switch in the "off" position, before the bracket retaining screw is removed.



1. IR detector fold assembly

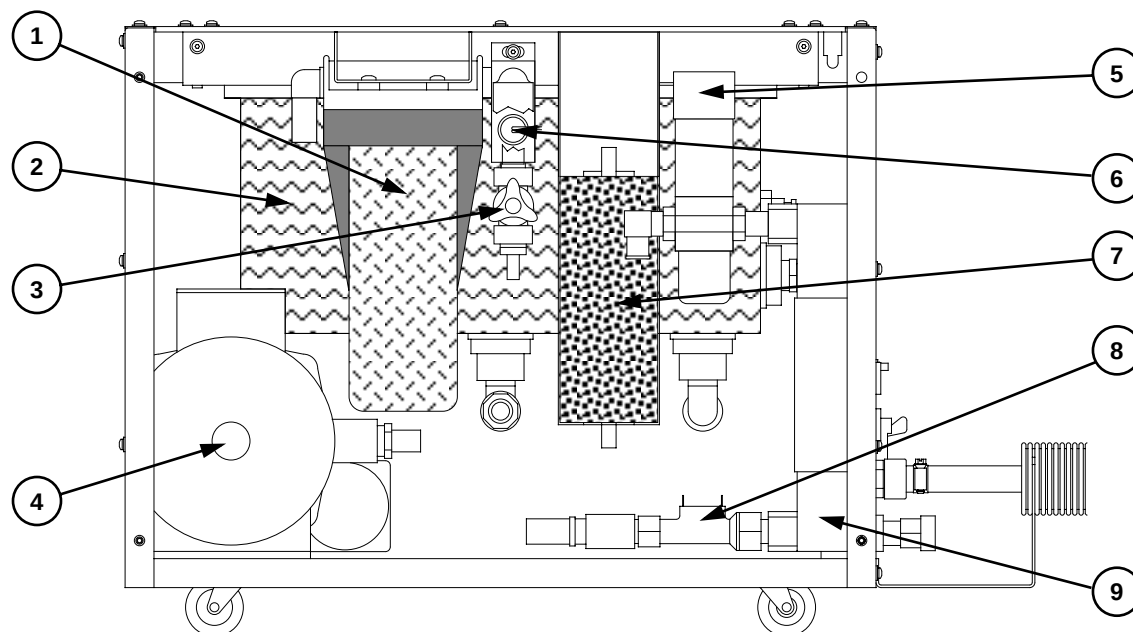
2. Photodiode housing
3. Photodiode electronics plug

6. Diode housing securing bolts

Figure 3-6. IR Photodetector Assembly

Table 3-6. IR Photodetector Assembly Components

CONTROL	FUNCTION
IR fold assembly	Final fold assembly after amplification stage. See Table 3-4, quarter waveplate, for a description of the horizontal and vertical tilt adjustments.
Diode housing securing bolts	Loosening these bolts will allow the photodetector signal to be maximized, a step in the calibration procedure for this detector. Under normal operating conditions repositioning of the depolarized energy detector is not required.

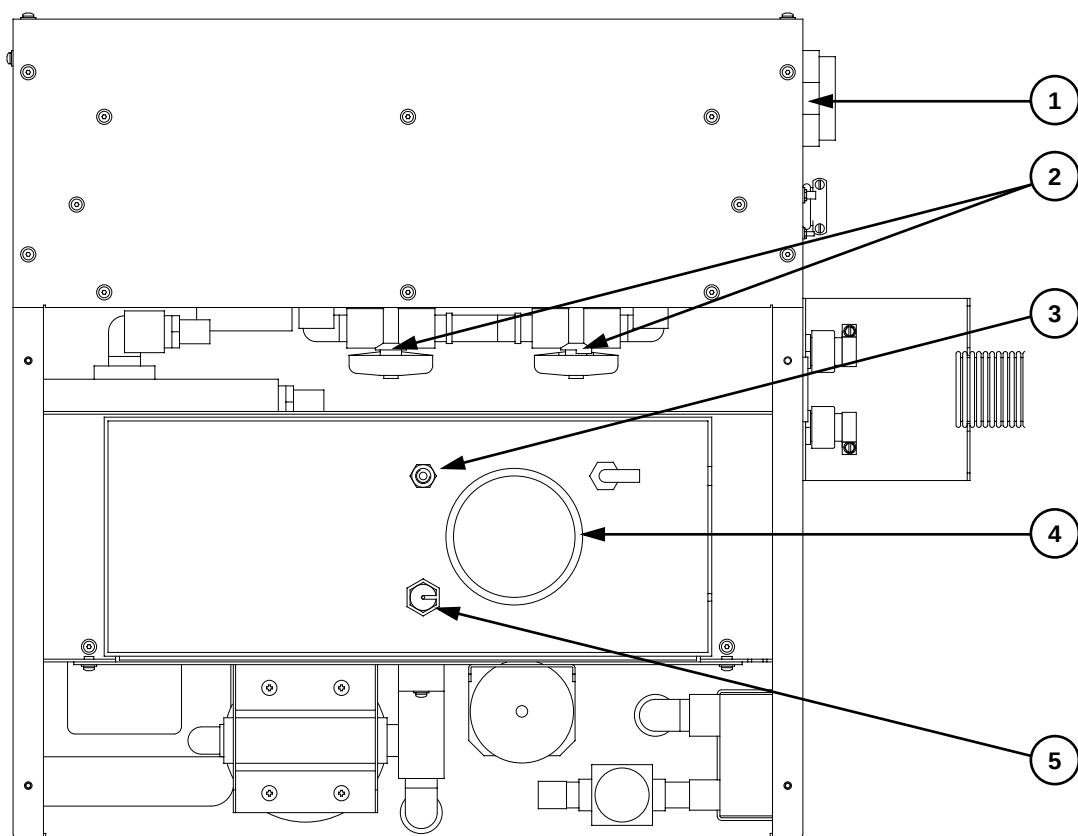


- | | |
|---------------------------------|---------------------------|
| 1. Particle filter | 6. Resistivity sensor |
| 2. Water reservoir | 7. Deionization Cartridge |
| 3. Resistivity bypass valve | 8. Water flow switch |
| 4. Water pump | 9. Heat exchanger |
| 5. Temperature regulation valve | |

Figure 3-7. Power Supply Side View, Water Side Components Locations.

Table 3-7. Power Supply Side View, Water Side Components

CONTROL	FUNCTION
Particle filter	Only the particle filter supplied by Coherent should be used in the Infinity laser system. Standard filters, such as those used in ion lasers, can contain manufacturing residues which can greatly affect the performance of the Infinity laser system. See Appendix B, Part Numbers, for the Infinity particle filter order number.
Water reservoir	Volume ~4.5 gallons. Only steam distilled water should be used in the Infinity heat exchanger.
Resistivity bypass valve	Valve controlling the proportion of water going through the deionization cartridge. Turning it clockwise decreases flow which decreases the resistivity.
Temperature regulation valve	Valve controlling flow rate of primary cooling water. Turning it clockwise increases flow, i.e. lowers the temperature of the internal water.



1. Fuse block

2. Booster board water valves

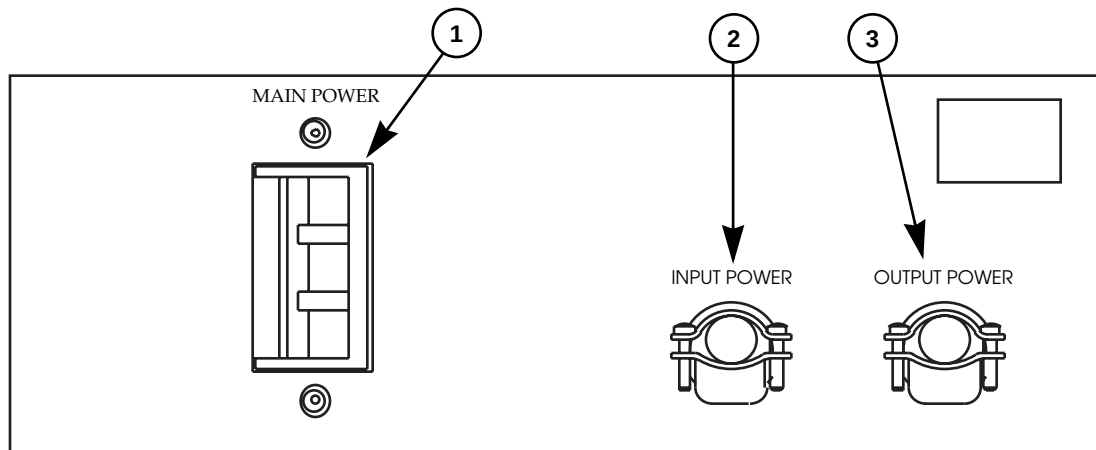
3. Water level sensor
4. Water reservoir fill lid

5. Water temperature sensor

Figure 3-8. Power Supply Top View, Water Side Components Locations.

Table 3-8. Power Supply Top View, Water Side Components

CONTROL	FUNCTION
Fuse block	Contains the fuses for the facility power to the Infinity laser system.
Booster board valves	Under normal operating conditions these values should be in the open position (parallel to the cooling lines). Operating the Infinity laser system with one or both of the valves closed could result in the catastrophic failure of the booster boards.



1. Fuse block
2. 3-phase input line from facility
3. 3-phase output line to Infinity power supply

Figure 3-9. External EMI Filter, Back View.

Table 3-9. External EMI Filter, Back View.

CONTROL	FUNCTION
External EMI filter	To meet CE-MARK requirements an external EMI filter is supplied with systems shipped to Europe. The purpose of this filter is to prevent the Infinity laser system from feeding noise onto the facility power lines. The filter has no effect on the performance of the laser system, and has no internal components which are user serviceable.

OPERATOR'S MANUAL

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CHAPTER FOUR

DAILY OPERATION

Introduction

This chapter contains daily turnon, after temporary or long term shutdowns, troubleshooting procedures and an over view of the system triggering/synchronization features.

Refer to Chapter Seven, System Software, for a discussion of the Infinity control software. Refer to Chapter Three, Controls and Indicators, for a discussion of the Infinity optical mounts and their adjustment.



Lasing is immediately terminated, and all systems in the laser are sequentially switched off, if an interlock is tripped. To restart the laser close the interlock, clear the Fault Message box, and press the “On” button in the main Infinity control window.



By manually increasing the flash lamp voltage the Infinity system can deliver output energies significantly higher than it is specified to provide. Doing this will, however, permanently damage optical components. Do not run the system at higher energies than what the system is rated for.

Daily Start-Up Procedure

The following procedure should be employed to start the Infinity laser system.

1. Turn the key switch located on the power supply, below the BNC connectors, to **On**.
2. Start the computer and run the Infinity GUI by double clicking on the Infinity icon. After the Infinity library files are bound, the “main” Infinity window should be displayed and the text in the system status panel should read “AC power on”.
3. Select the desired **Repetition rate [Hz]**. This can be done in three different ways.
 - a. Position the mouse cursor over the repetition rate indicator (e.g., downward pointing arrow head in side bar) and depress the left mouse button. This will allow the indicator to be dragged along the slide bar with a resolution of 1 Hz.
 - b. Position the mouse cursor over the side bar to the left or right of the indicator and click the left mouse button. This action will increase or decrease the repetition rate in 10 Hz increments.

- c. Positioning the cursor over the left or right rectangular boxes at the end of the slide bar. Left clicking the mouse at this location will increase or decrease the repetition rate by 0.1 Hz.

Note that as the repetition rate is changed the pulse energy panel is rescaled at 20 and 50 Hz (40-100 systems). Even though it is possible to start the laser at any repetition rate, it is advisable to start at a lower setting in order to minimize thermal stress on components in the experiment (e.g., power meters and optics). The repetition rate can be changed while the laser is running.

4. Select the desired **Pulse energy [mJ]**. Again this can be done in three different ways.
 - a. Position the mouse cursor over the repetition rate indicator (e.g., downward pointing arrow head in side bar) and depress the left mouse button. This will allow the indicator to be dragged along the slide bar with a resolution of 5 mJ/pulse.
 - b. Position the mouse cursor over the side bar to the left or right of the indicator and click the left mouse button. This action will increase or decrease the output power in 25 mJ/pulse increments.
 - c. Positioning the cursor over the left or right rectangular boxes at the end of the slide bar. Left clicking the mouse at this location will increase or decrease the pulse energy by 1 mJ.

The pulse energy can be changed while the laser is running.

5. Press either the **On** or the **Standby** button.

Pressing the **Standby** button will initiate the following sequence of events:

- a. The water pump starts and the water heater is turned on.
- b. The mechanical shutter opens, the master oscillator (DPMO/DPSS) is turned on and the flash lamp is simmered after a 5 second delay.
- c. The "Standby" button is highlighted and Standby is displayed in the status window.

Pressing the **On** button, in addition to the steps listed above, also:

- d. Enables triggering of the flash lamp, and after a ~3 second delay, the Q-switch.

- e. Displays “Laser on” in the status window.

The laser should now be running. If the on-board microprocessor has detected a problem in the system, a new window will appear and an error message will be displayed. Refer to the troubleshooting section of this chapter for more information about error messages.

Lamp Voltage Clamp

As a safety feature you can set the maximum voltage that will be applied to the flash lamp in the main control panel. It is strongly recommended that you set this voltage to only 10 V to 15 V higher than what is needed in order to achieve the desired power. As the lamp ages it can gradually be increased. Be aware that the voltage required to produce 600 mJ/pulse on an old lamp, will when applied to a new lamp, will produce enough energy to destroy optical components.

Under normal operation of the Infinity system the output energy should be set in the pulse energy panel, located in the main control window. The option of manually setting the lamp voltage should only be used for special procedures such as calibration of the internal energy meter.

The LightLoop servo, maintaining long term stability of the laser, is enabled by setting the output energy in the Control window (mJ/pulse). It is disabled by setting the lamp voltage in the Service window rather than the output energy.

Shut-down Procedure

The lasing action can be terminated by pressing either the **Standby** or the **Off** button in the main user interface.

Pressing the **Standby** button immediately terminates lasing by inhibiting the flash lamp trigger and the Q-switch.

Pressing the **Off** button also switches off the simmer current, the oscillator, the water pump and closes the mechanical shutter. If the computer locks up (i.e., does not respond to mouse or keyboard commands) the laser can be turned off by tripping the external interlock on the power supply. The status of the sub-systems are listed in Table 4-1.

Table 3-1. System Status in Different Modes

	OFF KEY SWITCH ON	STANDBY	ON
Lamp	Not Triggered	Not Triggered	Triggered
Lamp Simmer	Not Simmered	Simmered	Simmered
Mechanical Shutter	Closed	Open	Open
Oscillator	Not Lasing	Lasing	Lasing
Q-Switch	Not Triggered	Not Triggered	Triggered
Water Pump&Heater	Off	On	On

Troubleshooting

This section consists of two parts:

1. A listing of System faults, Table 4-2. System faults are conditions that may exist which make the laser inoperable or cause its performance to be lower than the specifications. How to deal with a specific System fault is described in the respective procedures.
2. A listing of all Fault messages with their accompanying explanations and possible causes and remedies, Table 4-3.

To use the troubleshooting section, find the system fault or fault message that best matches the problem encountered and read the accompanying explanation or remedy to clarify or correct the situation.

For some of the problems listed, the remedy or final step in a procedure may require scheduling a service call. A description of the symptoms and the steps taken attempting to resolve the problem will facilitate the planning of a course of action. The troubleshooting guides may request recording of certain system parameters. Have these at hand when calling Coherent technical support.

Table 3-2. System Faults

SYSTEM FAULTS	PROCEDURES
System does not turn on	Refer to Procedure 1
Low output power	Refer to Procedure 2
Unstable laser power	Refer to Procedure 3
Depolarized energy high	Refer to Procedure 4
Poor spatial mode quality	Refer to Procedure 5
Low SHG efficiency	Refer to Procedure 6
Low THG efficiency	Refer to Procedure 7
Optical breakdown in the SBS liquid	Refer to Procedure 8
Erroneous IR energy meter reading	Refer to Procedure 9
Erroneous depolarized energy meter reading	Refer to Procedure 10

Table 3-3. Fault Messages

MESSAGE	EXPLANATION
±12 VA, ±12 VB, +2 VB, +4 VB, +5 V out of range	The listed supply voltage has fallen outside its normal operating range. This message can sometimes appear as transients disturb the system. If the system otherwise behaves normal clear the message and continue to operate it. If the message reappears check that the line voltages on all three phases are within their specifications. If they are, call Coherent's technical support.
AC line voltage low	The voltage on at least one of the incoming lines is too low to ensure the proper and safe operation of the system. Check all incoming phases against neutral.
Boost board fan not turning	The fan cooling the booster boards does not work. If you continue to run the system, the booster boards will overheat and eventually fail catastrophically. If this message appears call Coherent's technical support.
Cap bank not charged	At the end of the charge cycle, the voltage across the capacitor bank does not equal the value set by the computer. If this message appears call Coherent's technical support.
Contactor fan not turning	The fan cooling the two boards supplying low voltages (±12 VA, ±12 VB, +2 VB, +4 VB, +5 V) to the system electronics does not work. If you continue to run the system the boards will overheat and eventually fail catastrophically. If this message appears call Coherent's technical support.
Control system failure	The system status as read by the computer is not the same as that set by the computer. Consequently the computer may not have full control over the system. When this fault occurs, the CPU attempts to shut down the entire system. However, owing to the nature of this fault there is no guarantee that this will be successful. If this message appears turn off the system if it is still running. Disconnect any test equipment that may be connected to it. Remove possible sources of EMI from the lasers vicinity. Disconnect all BNC cables from it. Make sure that the PC computer is working. If this message continues to appear call Coherent's technical support.
Depolarization too high	The depolarized energy returning from the amplifier is too high. Since this energy goes back into the DPSS laser it may jeopardize the safe and optimum performance of the master oscillator. See Procedure 4 for the depolarized energy minimization procedure.
DPSS electronic fault	The diode pumped master oscillator (DPSS) is stabilized by several servo loops. If one or more of these fail to keep the system parameters within their operating ranges, this message is displayed. Possible causes are overheating and electromagnetic interference. There are no serviceable parts inside the DPSS housing. If this message appears repeatedly call Coherent's technical support.

MESSAGE	EXPLANATION
EPROM hardware fault	The EEPROM in the power supply does not respond to commands from the PC computer. If this message appears call Coherent's technical support.
External (CDRH) interlock open	The Infinity system is equipped with an external interlock circuit that can be used to monitor the status of external devices. It is located at the rear of the power supply. If this circuit is open the system will not start. The system can be restarted by clicking on ON in the user interface as soon as the interlock circuit has been short circuited. See Chapter 5 for a description of how to install an external safety switch.
FlashTrack at limit	The FlashTrack servo system has, at least temporarily, been unable to perform its task. Verify that the cable is connected to the photo diode housing.
High water temperature	The temperature of the internal cooling water exceeds 45°C. All subsystems in the laser are switched off. Make sure that the water pump is working and that the incoming external cooling water is colder than 30°C. Check that the incoming water can flow freely through the hoses and the umbilical. If the problem persists, remove the water side cover on the power supply. Turn the regulating valve, located between the heat exchanger and the deionization filter, see Chapter 3, half a turn clockwise. Restart the water pump (from the service window) when the internal water has cooled down.
Interlock open	There are five interlocks in the system. They are: 1. Laser head cover interlock, 2. Flash lamp terminal box cover, 3. External interlock and 4&5, the two interlock dowel pins next to the latches holding the POT. If any of these interlocks are tripped all systems in the laser are switched off. In order to restart the system close the interlock circuit and click on the ON button in the main control window.
Lamp not simmering	The system has failed in igniting the lamp. When the lamp is simmered a steady whitish glow can be seen from the rods. Check that the lamp leads are connected and that the lamp itself is undamaged. Try to simmer the lamp again by first turning the laser off and then on again. If the lamp does not simmer after a couple of attempts, call Coherent's technical support.
Low capacitor bank voltage	If the voltage across the capacitor bank falls below the line voltage this error message is displayed and the system is turned off. The voltage regulation circuits can only function above the line voltage. If the voltage across the capacitors is lower at the beginning of the charge cycle it will behave as a short circuit until the voltage is brought up to the line voltage. In order to avoid the sudden rush of current into the system that would occur, the capacitor bank voltage is monitored. If this message cannot be defeated by turning the laser off and restarting it, call Coherent's technical support.

MESSAGE	EXPLANATION
Low water flow	The flow rate in the internal cooling loop is too low to ensure safe operation of the system. Remove the water side cover on the power supply. Check that there are no leaks and that the particle filter is in good shape. Remove the cap on the end of the pump and by switching the pump on and off establish that it is spinning clockwise.
Power module A over current Power module A over temperature Power module A voltage fault Power module B over current Power module B over temperature Power module B voltage fault	These messages indicate that one or both of the booster boards have failed. If these messages cannot be defeated by turning the laser off and restarting it, call Coherent's technical support.
Remote buffer overflow	The PC sent a message to the power supply before the previous message was acknowledged by the power supply. Since the message exchange is under software control, this error would normally indicate a software defect, although it could possibly be caused by a noisy serial line or defective serial port. Check that the serial port and the cable between the computer and power supply is OK. Eliminate any possible sources of EMI. If possible try another computer.
Remote framing error	The UART on the CPU reported a framing error, which typically results from a baud rate mismatch between the serial sender and receiver. Since the baud rate settings are under software control, this error would normally indicate a software defect, although it could possibly be caused by a noisy serial line or defective serial port. Check that the serial port and the cable between the computer and power supply is OK. Eliminate any possible sources of EMI. If possible try another computer.
Remote parity error	The UART on the CPU board reported a parity error indicating noise on the serial line. Check that the serial port and the cable between the computer and power supply is OK. Eliminate any possible sources of EMI. If possible try another computer.
Remote UART overrun	The UART on the CPU reported a character overrun, a condition in which a character was received while the previous character was still read by the CPU. This indicates a software defect. Reinstall the software.
SCR failure	The Infinity system has two solid state rectifiers. If one or both of them fails, this message is displayed. Call Coherent's technical support.
Servo error: no lasing	If the internal power meter reads less than 10 mJ/pulse and the lamp voltage is above 460 V this message is displayed. Check that there are no obstructions in the beam path and that the cable to the IR photodiode housing is connected, see Chapter Three.

MESSAGE	EXPLANATION
Transformer fan not turning	The fan cooling the transformer does not work. If the system does not have a built in transformer the system can be safely operated without the fan working (only 380 V systems have this transformer). Call Coherent's technical support for further assistance.
Water flow low	See low water flow
Water level low	The water level in the reservoir, located in the power supply, is to low. Refill with deionized water and check for leaks.
Water resistivity low	The resistivity of the internal cooling water is to low. Remove the cover on the water side on the power supply. Turn the bypass valve, see Chapter 3 for location, half a turn counter clockwise. Wait until the resistivity, value available in the Monitor window, stabilizes. Repeat if necessary. The recommended value is 3 M Ω /cm and the recommended range 2 – 6 M Ω /cm. The laser can not be operated when the resistivity is lower than 1.5 M Ω /cm. Replace the deionization cartridge when the resistivity, even with the valve fully open, stays below 2 M Ω /cm.
Water resistivity high	The resistivity of the internal cooling water is to high. Remove the cover on the water side on the power supply. Turn the bypass valve, see Chapter 3 for location, half a turn clockwise. Wait until the resistivity, value available in the Monitor window, stabilizes. Repeat if necessary. The recommended value is 3 M Ω /cm and the recommended range 2 – 6 M Ω /cm.
Wrong AC phase rotation	This means that the relative order of the three electrical phases are wrong. Interchange any two of the incoming AC phases and try again.

Procedure 1. System fault: Laser does not start

When the power supply is turned on, a 0 should be displayed on the readout above the BNC connectors and the fans inside the power supply and laser head should come on. If these signs are absent check the following:

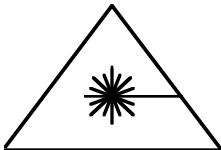
1. That the facility main power circuit breaker is turned on.
2. That the line voltage is correct and that all three phases are active.
3. That none of the three line fuses, located at the rear of the power supply, are blown.
4. That the key switch on the power supply is turned all the way to the ON position.

If the power supply does not start, schedule a service call.

Procedure 2. System fault: Low output power

Note that the internal power meter only monitors the IR energy. If the energy harmonic energy is low see Procedures 6 through 8. The procedure below assumes that the energy at 1064 nm is low. Low output power can be due to a number of factors. Check the following:

1. Confirm that the laser power is low using an external calibrated power meter. If necessary recalibrate the built-in energy meter, see Procedure 9.
2. That there are no obstructions in the beam path and that the entire beam goes into the power meter. Verify that the beam is not being clipped anywhere by taking a burn pattern of the mode just after the last folding mirror. It should appear round and symmetrical. If the burn mark exhibits any irregularities or asymmetries, the laser might be misaligned or one or more of its optical components damaged. See Chapter Six, Maintenance, for optics inspection procedures.
3. That the maximum lamp voltage in the Infinity Control window is at least 10 V higher than what is needed.
4. The number of shots the flash lamp has fired. A shot counter is available in the Infinity Control panel. If this number exceeds 75 million shots, change the flash lamp. The flash lamp replacement procedure is described in Chapter Six, Maintenance.
5. That the flash lamp and Q-switch timing sequence is optimized as well as the emission time of the flash lamp. In the Infinity Control panel click on the Service button and then chose first Lamp from the menu bar and then Timing from the menu. The Lamp pulse width should be close to 220 μ s and the QS trigger position about 250 μ s. These delays are factory optimized and there is, under normal operating conditions, no reason for the user to change them.



The procedure described below requires the laser to be operated with the laser head cover open and the interlock defeated. It is strongly recommended that everyone present in the room wears appropriate safety glasses and that access to the room is restricted.



Even though positioning the Q-switch trigger position later with respect to the lamp pulse than indicated above, may yield higher pulse energies it will also lead to instabilities in the laser mode and intensity. It is imperative that the Q-switch trigger is positioned before the turn off time of the flash lamp.

6. That the output power from the oscillator is at least 20 μJ /pulse. While performing this measurement be aware of the cw prelasing radiation emitted from the oscillator. This radiation amounts to ~15 mW of power while the integrated Q-switched output at 100 Hz repetition rate equals 3 mW (100 Hz*30 μJ /pulse). It is therefore necessary to measure the pulsed output with a calibrated photo-diode, a Gentec energy meter or similar pulse sensitive meter. If the output energy is below 20 μJ /pulse schedule a service call.

Procedure 3. System fault: Unstable laser power

The shot-to-shot intensity fluctuations are specified to have a standard deviation of less than 1.0 % for repetition rates up to 50 Hz at maximum system pulse energy (600 mJ/pulse up to 20 Hz, and 500 mJ/pulse from 20.1 up to 50 Hz repetition rate). At repetition rates above 50 Hz the specified standard deviation is 1.7 % for an output of 400 mJ. If the output at 1064 nm exhibits larger intensity variations, check the following

1. Verify that the laser is being operated at full power. Running the laser at lower pulse energies will inevitably lead to increased intensity fluctuations. This is especially true at energies close to the SBS threshold energy.



Attenuating the master oscillator output or moving the Q-switch trigger with respect to the flash lamp emission will reduce the output power, but it will also increase the standard deviation of the shot to shot noise!

To achieve a combination of low noise and low power the beam should be attenuated outside the laser using neutral density filters or glass wedges.

2. Verify that FlashTrack is active by opening the “Servo” window. FlashTrack is an electronic control servo that, based on the integrated lamp emission during the first half of the pulse, makes a small correction to the current going into it during the second half to keep the total amplification constant.
3. The number of shots the flash lamp has fired. A shot counter is available in the Infinity Control panel. If this number exceeds 75 million shots, change the flash lamp. The flash lamp replacement procedure is described in Chapter Six, Maintenance.
4. That the flash lamp and Q-switch timing sequence is optimized as well as the emission time of the flash lamp. These parameters are set in the Infinity lamp timing window that is accessed from the Lamp menu in the Service window. Verify that the parameter values, Lamp pulse width and Q-switch delay, match those on the performance sheet delivered with the system. These delays are factory optimized and there are, under normal operating conditions, no reasons for the user to change them.



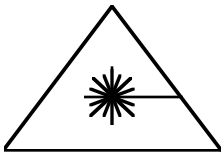
Even though positioning the Q-switch trigger position later with respect to the lamp pulse than indicated above, may yield higher pulse energies it will also lead to instabilities in the laser mode and intensity. It is imperative that the Q-switch trigger is positioned before the turn off time of the flash lamp.

5. The DPSS temperature. If the oscillator runs in a cavity mode that is close to an adjacent mode, occasional mode-jumps might occur. Multi-mode output will lead to breakdown in the SBS cell, which can be heard as a distinct clicking sound. If this is present a small change in the oscillator temperature set-point is usually enough to remedy the problem. To do this, first in the Infinity Control panel click on the "Service" button, secondly in the Infinity Diagnostic panel move the DPSS heater slide bar, located in the lower left corner, 5 counts in either direction. Allow a few minutes for the oscillator to reach its new equilibrium temperature. If necessary repeat a couple of times until the sound disappears.
6. Verify that there are no intense emitters of electromagnetic interference close to the Infinity system. Examples of such emitters are Excimer lasers and radio frequency devices.

Procedure 4. System fault: Depolarized Energy High

Depolarized energy is defined as all radiation which is not vertically polarized after transversing the Infinity amplifier stage two times. Since this radiation is not vertically polarized, it is not coupled out of the amplifier by the thin film polarizer. It is instead transmitted through the thin film polarizer and directed back into the Faraday isolator.

It is desirable to keep this energy component to no more than 1.5% of the output energy, when operating the laser at its maximum energy and repetition rate. If the energy going through the Faraday isolator, as measured by the photo diode attached to it, exceeds 2 % of the output energy, the on-board microprocessor will shut off the system. This is done in order to protect the master oscillator from being damaged. A bar graph display showing the depolarized energy is available in the Infinity software Monitor window which can be accessed by clicking on the "Monitor..." button in the main Infinity control window.



The procedure described below requires the laser to be operated with the laser head cover open and the interlock defeated. It is strongly recommended that everyone present in the room wears appropriate safety glasses and that access to the room is restricted.

Follow the procedure described below to minimize the amount of depolarized energy.

1. Increase the repetition rate to the maximum value for the system, and thereafter the IR energy until the depolarized energy is 1% of the maximum output power. If the depolarized energy falls below 1%, as a result of the adjustments described below, increase the IR pulse energy until 1% is again reached. Continue to increase the IR energy until the maximum pulse energy of the system is reached.
2. Slowly adjust the horizontal and vertical translation controls of the first relay lens, mount 1 in Figure 4-1, while monitoring the amount of depolarized light. Iterate between the two controls until the minimum is reached.
3. Adjust the vertical and horizontal tilt angle of the 90° polarization rotator, mount 2 in Figure 4-1. Use the long 5/32" ball driver supplied with the laser. Slowly tilt

the holder in the vertical and horizontal direction back and forth until a minimum in the depolarized energy is found. Iterate between the two directions.

4. Make small vertical and horizontal tilt adjustments of the quarter wave plate holder, mount 3 in Figure 4-1, until the amount of depolarized energy has been minimized.

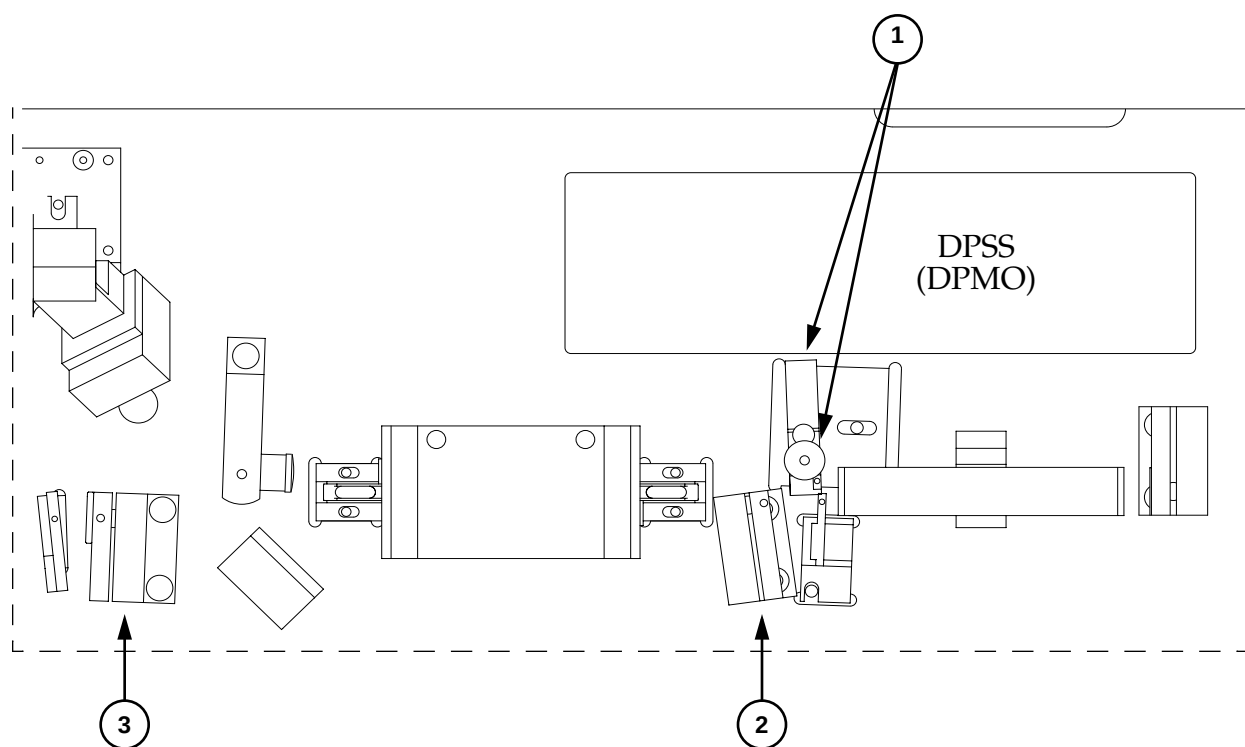


Figure 3-1. Order of Adjustments for Minimization of Depolarized Energy

5. Reduce the power to its minimum value and then reduce the repetition rate to 10 Hz. Slowly increase the power until the depolarized energy reaches 5 mJ or until the maximum pulse energy of the system is reached, 600 mJ/pulse.
6. While holding the quarter wave plate holder in place with a finger, loosen the setscrew holding the quarter wave plate as shown in Figure 4-2. Carefully, while monitoring the intensity of the depolarized light, make small rotational adjustments until the amount of depolarized energy has been minimized.
7. Repeat steps 5 through 6 until no further decrease in the depolarized energy can be obtained or the maximum pulse energy of the system is reached.

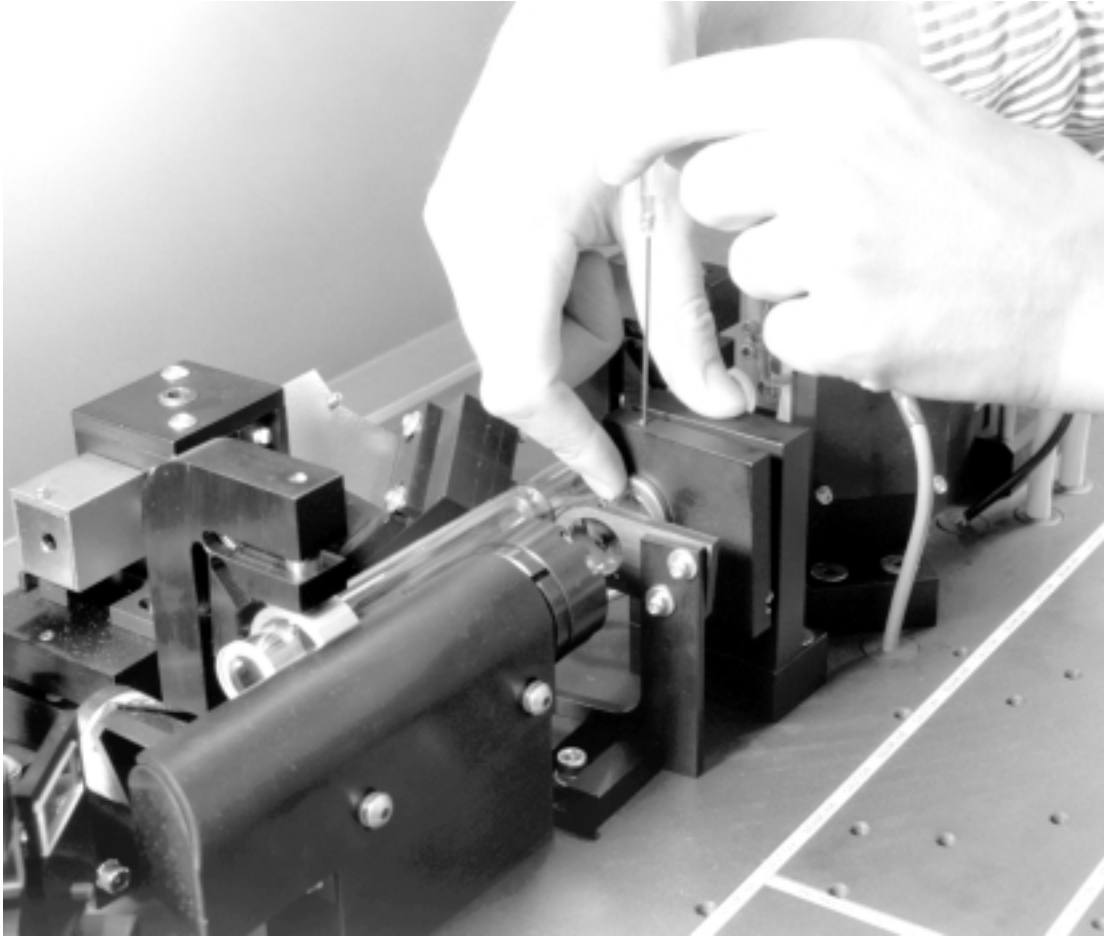


Figure 3-2. Alignment of the Quarter Wave Plate

8. Repeat steps 1-7 until it is possible to run the system at its maximum average output power (i.e., maximum energy and pulse rate) with less than 1.5% depolarized energy.

If the depolarized energy can't be brought below 1.5% call Coherent's technical support group.

Procedure 5. System fault: Poor Mode and Beam Quality

Poor mode quality means that the beam exhibits an asymmetrical intensity distribution, for example ellipticity or hot spots, while poor beam quality means that the beam is more divergent than what is specified for the system. They are related in so far that a poor mode usually implies poor beam quality. The reverse is, however, not necessarily true. A beams intensity distribution can appear symmetrical and it can still be highly diverging. In order to achieve the very high beam quality that the Infinity system is capable of delivering it is important that the beam is round and symmetrical.

The mode quality can be checked using burn paper or, preferably, a CCD camera based beam analyzing system, such as the Coherent BeamView Analyzer.

The Infinity system has the following two specifications for its beam quality. In the near field 95% of the energy can be transmitted through an aperture with a diameter of 5.5 mm. In the far field the beam will have a divergence no greater than 0.7 mrad full angle, measured between points defining an area containing 84% of the total energy.

If the IR beam appears asymmetrical check the following:

1. That the laser is operated at full power
2. That the mode is diagnosed in the near field. The near field is located about 5 cm after the thin film polarizer.
3. That there are no obstructions in the beam path and that the entire beam strikes the burn paper or the CCD camera. In particular make sure that the white teflon shields protecting the flash lamp leads are pushed all the way down into their mounts.
4. All optical components including the rod surfaces (see Chapter Six for optics inspection procedures and cleaning). If any of the optics are found to be damaged schedule a service call.



Do not try to improve the mode quality by modifying the alignment. Doing this may damage the optics in the laser.

If the IR beam exhibits a larger diameter in the near field or greater divergence in the far field than specified, verify the following:

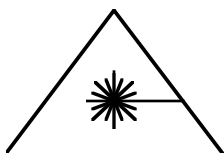
1. That the collimating lens, positioned after the Faraday isolator, has not been moved.
2. That the beam is not striking the pinhole in the vacuum spatial filter. This can be checked with an IR viewer. A consequence of the amplified return beam striking the pinhole is the development of a dark discoloration on the inside of the glass envelope. In extreme cases it might be possible to see a bluish plasma on the pinhole on the POT side. If this is the case, turn off the system and call Coherent's technical support group.



It should be noted that a smoky discoloration of the spatial filter vacuum cell will occur during normal operation. An indication that the filter is misaligned would be a deep blackening of the glass vacuum cell.

Procedure 6. System fault: Low Second Harmonic Generation Efficiency

Second Harmonic Generation, or SHG for short, is an optical non-linear process in which two incoming photons interact with a medium to form a third new photon. Since energy is conserved, the new photon has an energy equal to the two incoming ones. This means that the efficiency of the process depends mainly on the intensity of the incoming radiation field. However, there are two other important factors which must be considered when determining the efficiency of the SHG process. These are beam quality (its brightness) and the quality of the crystal used. It is therefore only meaningful to talk about efficiency for a given beam/crystal configuration.



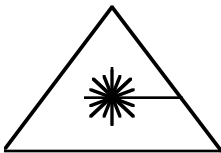
Some of the steps in the procedure below require the head cover to be opened and the head interlock defeated. Use appropriate safety glasses and restrict access to the room in which the laser is operated to qualified personnel.

When the Infinity system is operated at 600 mJ/pulse the SHG efficiency should be higher than 50%. If this is not the case check the following:

1. That the crystal is optimally phase-matched. While monitoring the laser power at 532 nm slowly turn, using a 9/64" allen wrench, the horizontal adjustment screw on the SHG mount until maximum power is achieved.
2. That the IR energy is 500 mJ/pulse using an external power meter.
3. That the mode appears round and symmetrical. Use the burn paper supplied with the laser.
4. That the BBO crystal and the cell windows are undamaged. Remove the cell by first unscrewing the two lock nuts, then remove the plate and finally pull the crystal housing out of the mount. Inspect the cell for damages, burn marks, discolorations etc. If any damage is found on the windows or the crystal call your Coherent representative.
5. That the harmonic mount temperature remains stabilized (i.e., remains in the hold configuration) during operation. See Chapter Seven, System Software, for more information on this window.

Procedure 7. System fault: Low Third Harmonic Generation Efficiency

Third Harmonic Generation, or THG for short, in a Nd:YAG laser, is an optical non-linear process in which the radiation fields at 1064 nm and 532 nm interact in a non-linear medium to form a third field radiating at 355 nm. The efficiency of the process depends on the intensity of the beams, their spatial and temporal mode and, the quality of the crystal.



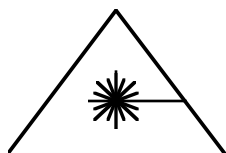
Some of the steps in the procedure below require the head cover to be opened and the head interlock defeated. Use appropriate safety glasses and restrict access to the room in which the laser is operated to qualified personnel.

If the IR energy is 600 mJ/pulse and the SHG energy is greater than 250 mJ/pulse, the energy at 355 nm should be higher than 200 mJ/pulse. If this is not the case check the following:

1. That both crystals are optimally phase-matched. While monitoring the laser power at 355 nm slowly turn, using a 9/64" allen wrench, the vertical adjustment screw on first the THG mount until maximum power is achieved and then the horizontal adjustment screw on the SHG mount. Note, the optimum phase-matching angle for generation of 532 nm is different when generating only 532 nm and when generating it as an intermediate step for 355 nm generation.
2. That the IR energy is 600 mJ/pulse and that the energy at 532 nm is above 250 mJ/pulse using an external calibrated power meter.
3. That the waveplate between the SHG and THG crystals is rotated to its optimum position.
4. That the mode appears round and symmetrical. Use the burn paper supplied with the laser.
5. That the BBO crystals and the cell windows are undamaged. If any damage is found on the windows or the crystals call Coherent.
6. That the harmonic mount temperature remains stabilized (i.e., remains in the hold configuration) during operation. See Chapter Seven, System Software, for more information on this window.

Procedure 8. System fault: Low Fourth Harmonic Generation Efficiency

Fourth Harmonic Generation, or FHG for short, is an optical non-linear process in which two incoming photons interact with a medium to form a third new photon. Since energy is conserved, the new photon has an energy equal to the two incoming ones. This means that the efficiency of the process depends mainly on the intensity of the incoming radiation field. However, there are two other important factors which must be considered when determining the efficiency of the SHG process. These are beam quality (its brightness) and the quality of the crystal used. It is therefore only meaningful to talk about efficiency for a given beam/crystal configuration.



Some of the steps in the procedure below require the head cover to be opened and the head interlock defeated. Use appropriate safety glasses and restrict access to the room in which the laser is operated to qualified personnel.

With the repetition rate set to 10 Hz and a SHG energy equal to or greater than 250 mJ/pulse, the energy at 266 nm should be greater than or equal to 50 mJ/pulse. If this is not the case check the following:

1. That both crystals are optimally phase-matched. While monitoring the laser power at 266 nm slowly turn, using a 9/64" allen wrench, the vertical adjustment screw on FHG mount until maximum power is achieved and then the horizontal adjustment screw on the SHG mount.
2. That the IR energy is 600 mJ/pulse and that the energy at 532 nm is above 250 mJ/pulse using an external calibrated power meter.
3. That the SHG beam is cleanly passing through the strip out dichroic and that the back reflection from the FHG cell is also cleanly passing through the strip out dichroic and striking the beam dump (white ceramic diffuser) mounted on the SHG stage.
4. That the mode appears round and symmetrical (A white business card works well for this).
5. That the BBO crystals and the cell windows are undamaged. If any damage is found on the windows or the crystals call Coherent.

6. That the harmonic mount temperature remains stabilized (i.e., remains in the hold configuration) during operation. See Chapter Seven, System Software, for more information on this window.



Due to the large nonlinear coefficient of BBO at 266 nm, 133 nm radiation can also be generated by the same crystal employed to generate the 266 nm radiation. BBO is not transparent to 133 nm radiation resulting in a significant temperature gradient as the pump energy and repetition rate of the laser system are changed. It is therefore necessary, and normal, to have to adjust the phase matching angle of the FHG crystal if the laser system operational parameters are changed by more than a few percent.

Procedure 9. System fault: Breakdown in SBS Cell

If the oscillator runs in a cavity mode that is close to an adjacent mode, occasional mode-jumps may occur. Multi-mode output will lead to laser induced breakdown in the SBS cell, which can be heard as a distinct clicking sound. If this is present, a small change in the oscillator temperature is usually enough to remedy the problem. To do this, first in the Infinity Control panel click on the "Service" button, secondly in the Infinity Diagnostic panel move the DPSS heater slide bar, located in the lower left corner, 5 counts in either direction. Allow a few minutes for the oscillator to reach its new equilibrium temperature. If necessary repeat a couple of times until the sound disappears.

If no improvement can be achieved call Coherent Technical Support.

Procedure 10. System fault: Erroneous IR Energy Meter Reading

It is possible to recalibrate the built in energy meter through the system software. Since some parts of the energy meter system are temperature sensitive allow the laser to reach thermal equilibrium before evaluating and calibrating it. Run the laser at its maximum repetition rate and pulse energy for one hour with the cover on (the average power should be measured on an external power meter!). If the energy meter still reads incorrectly, follow the procedure below. Numbers in the text are given with reference to Figure 4-3.

1. In the main Infinity control window click on "Service..." and then chose "IR..." from the Service window Calibration pull down menu.
2. The window shown below, Figure 4-3, should now be displayed on the screen. Move the Calibration and the Service windows on the screen so they do not totally obscure each other.
3. Optimize the signal on the detector. Note that the "Lamp volts" slide bar in the Service window should be used to set the IR pulse energy of the laser system during this procedure.

For 40-100 systems, set the repetition rate to 100 Hz and the output energy to 200 mJ using an external power meter.

For 25-50 systems, set the repetition rate to 50 Hz and the output energy to 250 mJ using an external power meter.

For 15-30 systems, set the repetition rate to 30 Hz and the output energy to 250 mJ using an external power meter.

Loosen the screws which hold the photodiode housing to the fold optic assembly. While watching the "Detector counts" bar graph, move the photodiode housing until a maximum is found, see Figure 4-3.

4. Set the lamp voltage in the service window so that the external energy meter reads the maximum specified average power of the system.
5. Set the power measured on the energy meter on the slide bar in the calibration window, item 1.
6. When the detector counts value, item 3, has stabilized, click on Accept, item 2. The entered value will appear in the window, item 4.
7. In the Service window decrease the lamp voltage by approximately 15 volts.

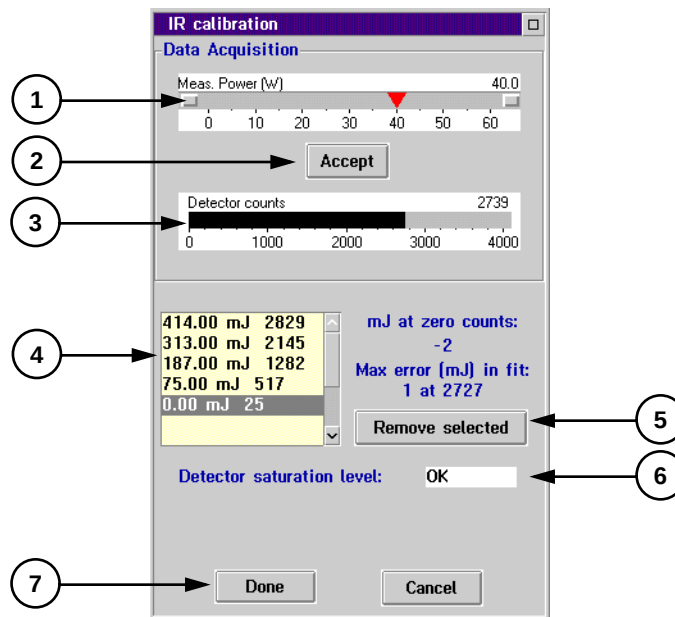


Figure 3-3. IR Energy Calibration Window

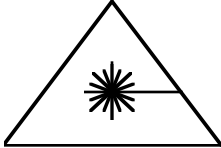
8. Repeat steps 5 to 7 at least 3 times. The exact value of the entered powers does not matter but it is important that 0 mJ be the last data point entered. Record this data point with the lamp voltage set lower than the lasing threshold. (i.e., 360 V).
9. Compute a new set of interpolation constants by clicking on "Done", item 7.
10. The IR energy meter is now calibrated. Go back to the main control window.

If an incorrect value is entered you can removed it by first highlighting it and then click on "Remove selected", item 5.

If there is not enough, or too much light, on the photodetector a message will be displayed stating in which direction the light level needs to be adjusted, item 6. The detector adjustment should be made at the current pulse energy (i.e., lamp voltage). For example; if after entering the second data point the system indicates that the detector counts need to be increase by 10%, and the current number of counts is 2000, then the detector counts should be increased to 2200 with the system operating at the lamp voltage setting of data point number two. After adjusting the detector counts, all data points should be removed and the calibration procedure restarted from step 4.

Procedure 11. System fault: Erroneous Depolarized Energy Meter Reading

The photodiode monitoring the depolarized energy can be recalibrated through the system software. Follow the procedure below. It is assumed that the internal IR energy meter is calibrated. If not, the IR detector should be calibrated first, see Procedure 10.



The procedure described below requires the laser to be operated with the laser head cover open and the interlock defeated. It is strongly recommended that everyone present in the room wears appropriate safety glasses and that access to the room is restricted.

In order to access the depolarized beam, the photodiode attached to the Faraday isolator will need to be removed. Numbers in the text are given with reference to Figure 4-4.

1. Open the head cover and install the head interlock defeat. Disconnect the electrical plug on the back of the photodiode assembly, item 2. Loosen the retaining screw, item 1, and pull the diode assembly (grey in figure) up and out of the laser. Be careful not to touch the spatial filter.
2. Position the Infinity beam dump, or an external power meter, to intercept the main IR beam.
3. In the main Infinity control window set the output energy to 0 mJ and the repetition rate to 10 Hz. Start the laser by clicking on "On".
4. In the main Infinity control window click on "Service..." and then chose "Depolarization..." from the Service window Calibration pull down menu.
5. The window shown below, Figure 4-5, should now be displayed on the screen. Move the Calibration and the Service windows on the screen so they do not totally obscure each other.
6. Gradually increase the lamp voltage until depolarized light can be detected using an IR fluorescent card. This light will be directed towards the Infinity baseplate in the direction of the Spatial Filter.

7. Position a IR folding mirror so that the beam reflected off the exposed polarizer is directed into an external power meter. Make sure, using the IR viewing card, that the beam goes into the power meter unobstructed.



It is imperative that a reliable and calibrated power meter is used for this procedure and that it is carried out in a meticulous way. Erroneous calibration of the depolarized energy monitor may lead to irreparable damage.

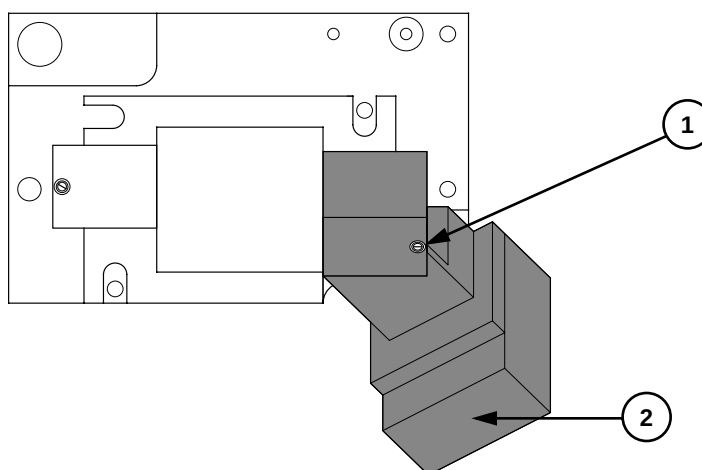


Figure 3-4. Faraday Isolator and Photo-Diode Assembly Measuring Depolarized Energy

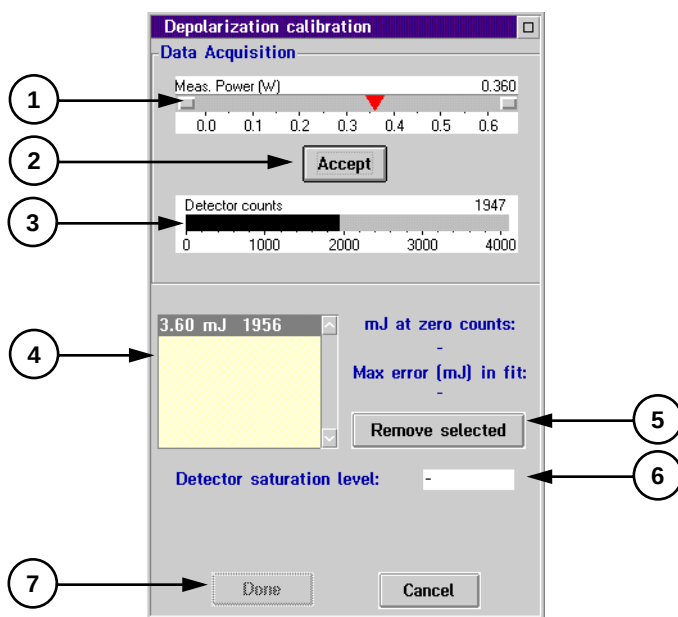


Figure 3-5. Depolarized Energy Calibration Window

8. Set the lamp voltage, in the Service window, to 360 V and then close the Service Window. Increase the repetition rate to system maximum. Let the system warm up by running it for >15 minutes.
9. Collect data points, depolarized energy vs. output energy, according to the following procedure. Note that the software will want the depolarized energy readings to be in “watts”.
 - a. Record the internal energy meter value, available in the main Infinity control window, and the corresponding depolarized energy on the external power meter. Take the first reading at 360 V (i.e., 0 mJ output).
 - b. Set the IR energy to 100 mJ.
 - c. Take a reading of the depolarized energy.
 - d. Increase the IR energy by 50 mJ.
 - e. Repeat steps “c” and “d” until the maximum output energy of the system is reached.
10. Set the IR energy to 1 mJ, and place the laser system in “Standby”. Reinstall the photodiode assembly.
11. Turn the laser system back on and set the IR energy to the system maximum. Loosen the screws which hold the photodiode housing to the fold optic assembly. While watching the “Detector counts” bar graph, move the photodiode housing until a maximum is found, see Figure 4-5.
12. Set the “Meas. Power” slide bar, item 1, to the measured depolarized energy value.
13. When the detector counts value, item 3, has stabilized, click on Accept, item 2. The entered value will appear in the window, item 4.
14. Set the IR energy to the value used for the next depolarized energy data point.
15. Repeat steps 12 through 14 for each of the depolarized energy values recorded. Note that Service window “Lamp volts” slide bar will need to be used for the last data point (i.e., the 0 mJ data point at a lamp voltage of 360 V).
16. Compute a new set of interpolation constants by clicking on “Done”, item 7.

17. The depolarized energy meter is now calibrated. Go back to the main control window.

If an incorrect value is entered you can remove it by first highlighting it and then click on "Remove selected", item 5.

If there is not enough, or too much light, on the photodetector a message will be displayed stating in which direction the light level needs to be adjusted, item 6. The detector adjustment should be made at the current IR pulse energy. For example; if after entering the second data point the system indicates that the detector counts need to be increased by 10%, and the current number of counts is 200, then the detector counts should be increased to 220 with the system operating at the IR pulse energy of data point number two. After adjusting the detector counts, all data points should be removed and the calibration procedure restarted from step 11.

Timing and Synchronization

Infinity's on-board, digital timing and triggering system provides a powerful set of tools for synchronizing Infinity with external laboratory equipment. Several input triggers and output sync signals are available, allowing Infinity to operate as either a master or a slave. All interface signals use TTL logic, the most common and convenient standard for laboratory instruments. All timing parameters are under control of the Infinity CPU and are therefore accessible through any of Infinity's user interfaces. The system includes advanced features designed to facilitate laboratory automation, such as programmable delay generators and sync signals. In some applications, Infinity's built-in capabilities will eliminate the need for external delay generators.

The material in this section can be divided into two topics: triggering Infinity from external instrumentation, and synchronizing external instrumentation to Infinity. These two topics are concerned with Infinity's inputs and outputs, respectively.

Inputs - Infinity can respond to two sets of input signals: (1) external triggers that allow direct control of repetition rate and pulse timing; and (2) blanking inputs that allow laser emission to be inhibited by an external logic level. These are described in the section on triggering.

Outputs - Infinity provides a set of output sync signals for triggering external devices. Both pre-triggers (before the laser output pulse) and post-triggers (after the laser pulse) are available. These are described in the section on synchronization.

Triggering

Firing the Infinity is a two-step process, involving two separate triggers: the first one for the lamp, the second for the Q-switch. Each trigger has two modes, internal and external, making a total of four possible triggering modes for the system. In the internal modes, Infinity's on-board electronics produce the necessary signals in accordance with the operational parameters set by the user. Internal triggering is the simplest way to run the system, since the user does not have to concern himself with the details of the triggering process.

In some applications, however, it is necessary to fire the laser on demand. Infinity offers three ways to do this:

1. Trigger the lamp with a TTL pulse and let the Infinity fire the Q-switch automatically (Lamp external, Q-switch internal).

2. Let Infinity fire the lamp automatically, but trigger the Q-switch with a TTL pulse (Lamp internal, Q-switch external).
3. Trigger both the lamp and the Q-switch with external TTL pulses (Lamp external, Q-switch external).

The choice among these three modes will depend on the application requirements. In selecting a mode, bear in mind that the time interval between the two triggers is an important parameter in system performance. When triggering the Q-switch externally, care must be taken to insure that the trigger is correctly timed with respect to the lamp pulse.

Regardless of the mode, Infinity's internal electronics have control over the *duration* of the lamp flash. The external lamp trigger simply causes the flash to begin and has no effect on its duration. The user interface allows control of the duration within certain limits, but normally it is not advisable to change it from the factory-optimized setting.

Considerations in External Lamp Triggering

External triggering of the lamp can be used to set a custom (possibly irregular) repetition rate or to synchronize the firing of Infinity with an external device. The lamp trigger does not directly control the instant of laser emission, rather, it begins the sequence of events that ultimately results in laser output. Because the inherent jitter in these events is about 100 ns, the external lamp trigger offers relatively coarse control over laser timing.

It is not possible to exceed a repetition rate of 100 Hz in this mode. This is enforced by an on-board protection circuit that inhibits the external lamp trigger for 10 ms after each lamp flash. Any trigger pulse that arrives during this interval will be ignored. The system provides an output signal, LAMP READY, that indicates the state of this inhibit logic; this line can be monitored by external circuitry to determine when the laser is ready to fire.

Although Infinity's optical design offers uniform system performance over a wide range of repetition rates, abrupt changes in repetition rate are not recommended. Each time the repetition rate changes significantly, the temperature of critical laser components shifts to a new operating point. Until these thermal effects stabilize (in a few seconds), system performance will not be optimum; in extreme cases an excessive amount of depolarized light may result, activating the systems protection feature and shutting off the laser temporarily. Applications that encounter this problem can avoid it by firing the lamps at an approximately constant rate

(maintaining the laser in a state of thermal equilibrium), enabling and disabling the Q-switch as desired. A Q-SWITCH INHIBIT input line is supplied for this purpose.

All of Infinity's advanced control features, such as the pulse energy monitor, the depolarization monitor, and pulse energy regulation, operate independently of the lamp trigger mode. All of the TTL synchronization pulses act as described in the section on synchronization.

Considerations in External Q-Switch Triggering

External triggering of the Q-switch allows precise control over the instant of laser emission: the time jitter between an external Q-switch trigger and the output laser pulse is approximately 1 ns.

In order for the system to work properly when the Q-switch is externally triggered, two timing constraints must be satisfied:

1. The lamp can be triggered either externally or internally, but it must be triggered before the Q-switch;
2. The delay between the lamp trigger and the Q-switch trigger must be close to its optimum value, which can vary slightly from one laser system to another. The wrong delay will cause low output power; jitter in the delay will cause output amplitude noise. Deviations of 1 μ s from the optimum should be considered significant.

Infinity supplies a TTL output signal that makes it simple to satisfy these timing constraints. This signal (QS READY), which is active in both internal and external lamp trigger modes, indicates that the proper interval has elapsed since the lamp trigger. The application can monitor this signal and respond by producing the Q-switch trigger, preferably within 1 μ s. Proper timing of the Q-switch is then assured without the need for external delay generators.

Within this general external triggering scheme are the two following modes:

1. External Q-switch Delayed. This mode is designed to support an application that requires overlapping the Infinity pulse with another light source, such as a sync-pumped tunable laser. Such setups typically involve arbitrary delays, both electronic and optical, that must be compensated by an adjustable delay somewhere in the system. In Infinity's External Delayed mode, the Q-switch trigger is passed through a high-resolution delay generator, programmable through the user interface in steps of approximately 0.1 ns. Furthermore,

in this mode the Q-switch trigger input is disabled until the lamps are ready (as indicated by the QS READY signal). Together these features make it quite simple to synchronize Infinity to a high repetition rate external source: trigger pulses from the external source are fed continuously to Infinity's Q-switch trigger input, and all the necessary timing and gating is handled by the on-board electronics.

2. External Q-switch Immediate. In this mode, the Q-switch fires upon receipt of the trigger. This is a high-precision, minimum-jitter trigger that allows the application to have direct control over the laser timing. This trigger is always enabled, and will cause the master laser oscillator to emit a pulse into the amplifier regardless of whether the lamps have been fired or whether the time delay between lamp and Q-switch triggers is optimal.

The same input BNC is used for both External Q-switch delay and External Q-switch immediate.

All of Infinity's advanced control features, including pulse energy regulation, operate independently of Q-switch triggering modes. All of the TTL Synchronization pulses act as described in the section on Synchronization, with one exception: the Variable Q-switch Delay cannot be used as a pre-trigger (i.e., cannot be programmed to produce a synch pulse that is *before* the Q-switch trigger).

Set-Up of External Q-Switch Triggering

When the current into the flash lamp is turned off a strong electromagnetic pulse is sent through the system electronics. To avoid any degradation of system performance due to interference from this pulse it is very important that the Q-switch is timed relative the lamp current turn off with a precision greater than 0.5 μ s. In order to do this both the Q-switched laser pulse and the lamp current needs to be monitored. A convenient way of doing this is described in the following procedure.

1. Turn the key switch on the power supply to off.
2. Position a fast photodiode ($t_{\text{rise}} < 1$ ns) in front of the DPSS laser so that it monitors the output from it. It is often easier to let the DPSS output first strike a diffuser and let the photo-diode detect scattered light. Connect the photodiode to an oscilloscope.

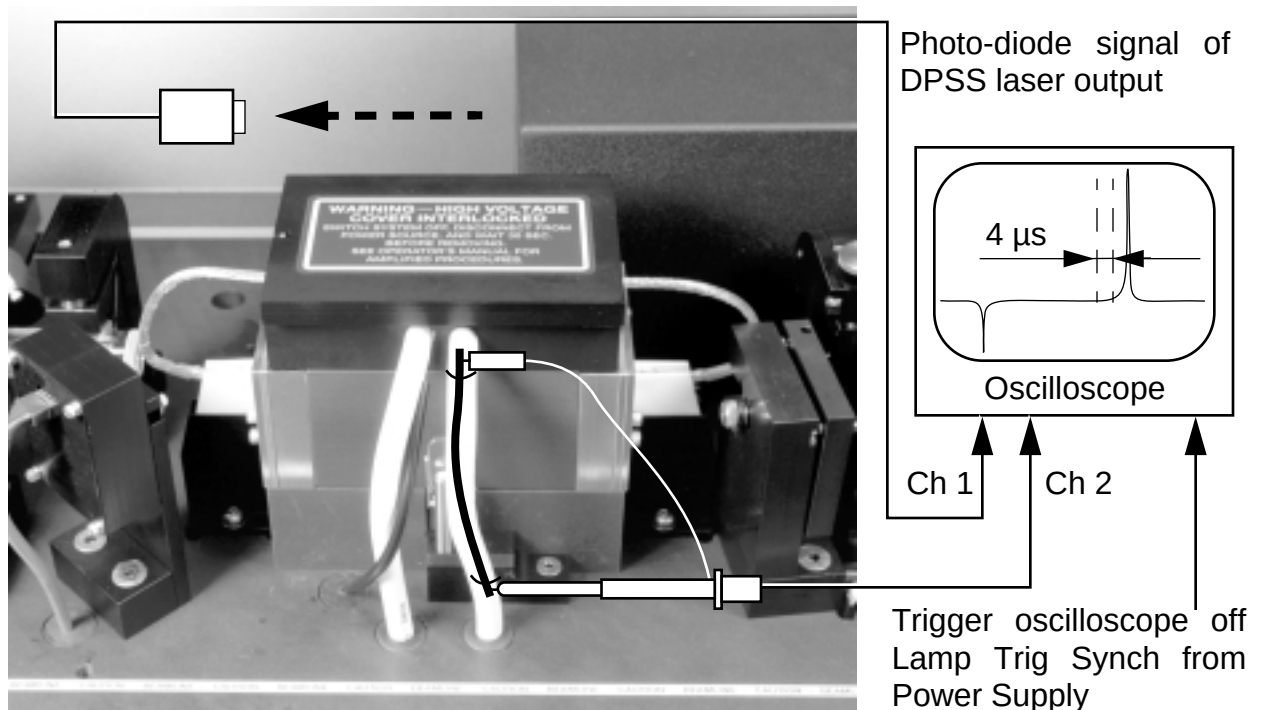


Figure 3-6. Set-up to Determine Timer Settings for Lamp Pulse and Q-switch.

3. Tape a piece of electrical wire, ~ 5 cm long, to the white flash lamp cathode lead according to Figure 4-6. Connect an oscilloscope probe to the ends of the lead as shown in Figure 4-6. Connect the probe to the oscilloscope.
4. Turn on the power supply and the computer. Run the Infinity program. Use Internal lamp and Q-switch triggering. Go to Standby mode.
5. Open the Service window. Set the lamp voltage to 360 V. Close the Shutter, click on lamp and Q-switch triggers. The lamp should now be flashing but the no laser light should exit from the system.
6. Trigger the oscilloscope of the LAMP TRIG SYNCH on the power supply.
7. Locate the signal from the probe on the oscilloscope. Use 1 M Ω termination, 50 μ s time division and 5 mV sensitivity. The signal appearing on the oscilloscope shows the time derivative of the current pulse through the flash lamp, see insert in Figure 4-6. It is created through induction in the wire taped to flash lamp lead. The inductive voltage along the wire is given by the expression: $V = -L \frac{di}{dt}$, where L is the inductance of the

system and di/dt the time derivative of the current through the lamp lead.

8. Locate the photo-diode signal on the oscilloscope. It should appear close to 4 μ s before the lamp turn off.
9. Note the timing relationship between the lamp turn off and the Q-switch.
10. Disable the lamp trigger in the Service window. Open the Application window.

Chose external Q-switch and if desired external lamp trigger.

Note that the duration of the lamp current pulse is always under computer control and can not be changed through the BNC connectors.

11. Connect the external trigger signals to their respective BNCs on the power supply, LAMP TRIG and QS TRIG. Note that triggering occurs on the falling edge of the applied signals.
12. Enable the lamp and Q-switch trigger signals in the Service window.
13. Position the Q-switch relative the experiment.
14. Using the signals on the oscilloscope position the lamp trigger relative the Q-switch so that the lamp turn off occurs 4 μ s after the Q-switch. Reproduce the timing sequence recorded when triggering in internal mode.
15. Go to Standby. Remove the probe and the photodiode.

Synchronization

There are six TTL output signals from Infinity, divided into three functional groups:

Sync signals. Three signals are available for triggering external equipment, offering different timing and jitter characteristics. These are described in detail below.

Ready signals. Two signals indicate that Infinity is ready to accept external triggers, one each for the lamp and Q-switch. Named LAMP READY and QS READY, their use was described in the section on Triggering.

Output 10 MHz clock reference. All of Infinity's timing circuits are driven by an on-board 10 MHz clock. This clock is provided as an output allowing external counters or delay generators to Synchronize their counting rate to Infinity.

Output Synch Signals

The three output triggers from Infinity serve different purposes. In their simplest terms, they provide a trigger that precedes the laser pulse by a programmable amount (LAMP TRIG SYNCH, Pre trigger in software), one that follows the laser pulse by a programmable amount (VAR QS SYNCH, High precision trigger in software), and one that is coincident with the laser pulse (QS SYNCH). The QS SYNCH pulse is a minimum jitter (< 0.5 ns) trigger that permits the most precise possible synchronization of external equipment.

Note that the BNCs labeled LAMP TRIG SYNCH and VAR QS SYNCH are controlled from panels, in the Timing and triggering window, labeled Lamp pre-trigger and Q-switch sync trigger, respectively. These trigger signals and their relationship are illustrated in Figure 4-7.

The LAMP TRIG SYNCH is a pre-trigger that can be programmed to precede the Q-switch opening by between $240\text{ }\mu\text{s}$ and 6.77 ms . The resolution is one tick of the 10 MHz clock (100 ns). Equipment such as streak cameras, which typically require set-up times greater than $100\text{ }\mu\text{s}$, can be conveniently triggered by this signal.

The VAR QS SYNCH can be programmed as either a pre-trigger or post-trigger. As a pre-trigger, it can lead the laser pulse by approximately $150\text{ }\mu\text{s}$. As a post-trigger it can follow the laser pulse by up to 6.15 ms . It is a high-precision trigger with a coarse and fine stage, the former having a resolution of 100 ns (one tick of the 10 MHz clock) and the latter 0.5 ns . Each stage of the delay is independently programmable, effectively subdividing the available 6.2 ms range into approximately 10^8 steps. This signal can be used to trigger data acquisition equipment, and is particularly useful in experiments that measure the time evolution of a sample in response to a light pulse. Such experiments can be easily automated through Infinity's programmable interfaces.

Unlike the other SYNCH signals the QS SYNCH output is not programmable, but it has the lowest timing jitter with respect to the output laser pulse. The QS SYNCH signal that appears on the power supply BNC also goes directly to the master oscillator trigger. Although laser light does not appear immediately after this trigger (there are electronic and optical delays in the laser head), the interval between the trigger and the appearance of a laser pulse is nearly constant, with a jitter of $< 0.5\text{ ns}$. QS SYNCH is the most precise trigger available from Infinity, and is useful for applications that demand the minimum triggering jitter.

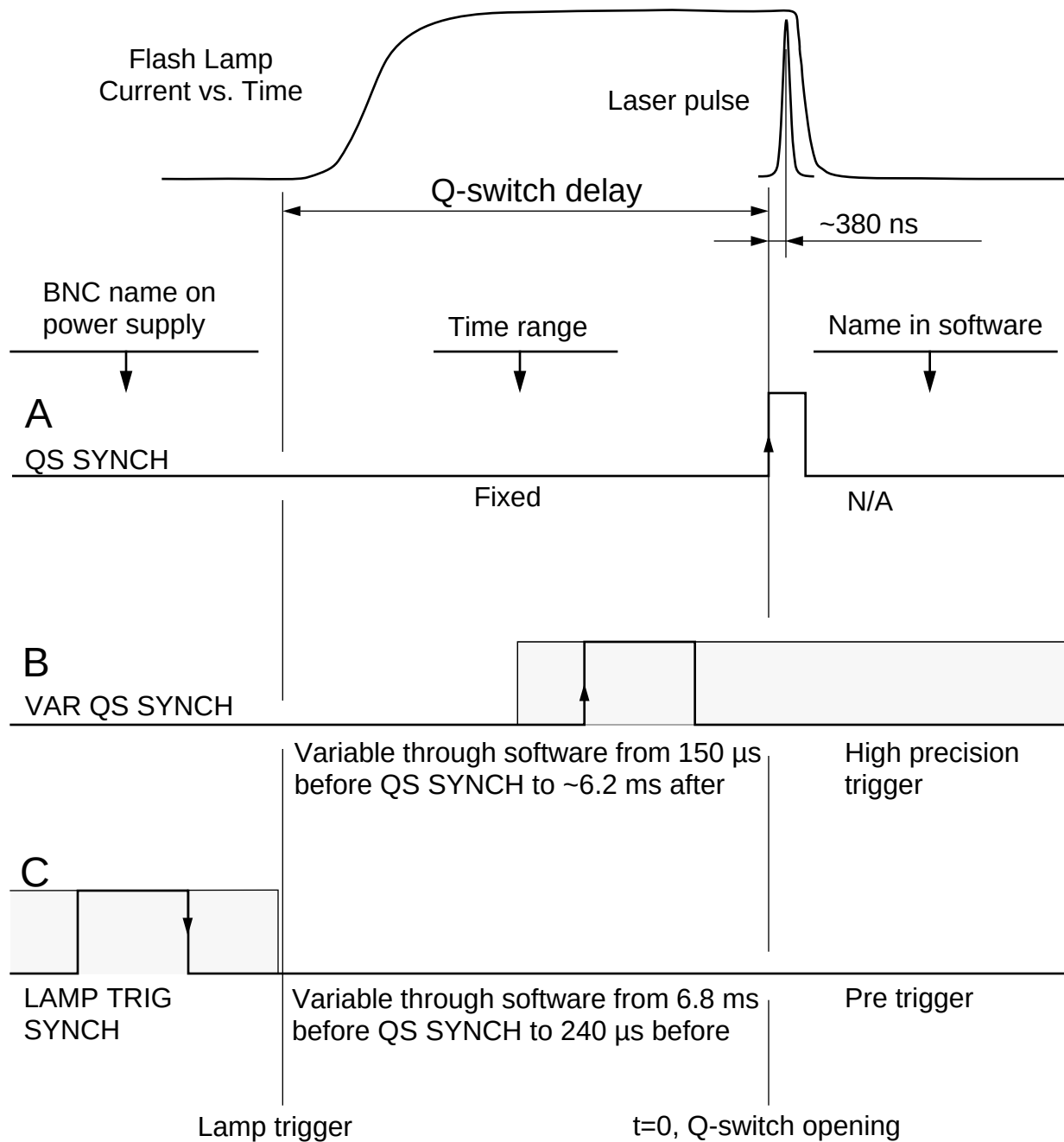


Figure 3-7. Relationship between Available Synchronization Signals.

Table 3-4. Output Signals.

BNC	FUNCTION
10 MHz	Master clock for Infinity timing circuits. Provided as a reference signal.
LAMP TRIG SYNCH	Programmable trigger. Precedes the laser pulse by between 6.77 ms and 200 μ s. It is controlled from the lamp pre-trigger panel in the Timing and Triggering window, Figure 4-6.
LAMP READY	Indicates by its level whether the lamp can be triggered; active only in external lamp triggering modes. Lamp triggers are disabled for 10 ms after each laser shot, enforcing a maximum repetition rate of 100 Hz.
QS READY	Indicates that the Q-switch is ready to be triggered. In external Q-switch modes, the Q-switch should be fired as soon as possible after this line changes state.
VAR QS SYNCH	Programmable trigger. Can precede the laser pulse by 150 μ s and follow it by 6.15 ms. It is controlled from the Q-switch sync trigger panel in the Timing and Trigger window, Figure 4-6.
QS SYNCH	A trigger having low jitter (<0.5 ns) with respect to the laser pulse

Table 3-5. Input Signals.

BNC	FUNCTION
QS TRIG	Fires the Q-switch on the falling edge
QS BLANK	Inhibits Q-switch firing. Active high.
LAMP TRIG	Fires the flash lamp on the falling edge
LAMP BLANK	Inhibits the lamp from flashing. Active high.

Table 3-6. Timers and Triggering Interface.

PANEL	RANGE	UNIT	FUNCTION
Lamp pre-trigger	-0.2 to - 6.77	ms	Sets the time for the pre-trigger (negative time values are <i>before</i> the laser pulse)
High precision trigger, coarse	-0.15 to 6.15	ms	Sets the time for the coarse stage of the programmable high-resolution trigger (negative time values are <i>before</i> the laser pulse)
High precision trigger, vernier	0 to 255	DAC counts	Sets the time for the vernier stage of the programmable high-resolution trigger. Each count represents approximately 0.1 ns; the exact calibration may vary slightly from one laser system to another.
Lamp trigger mode	One of: Internal External		Selects the lamp triggering mode
QS Trigger mode	One of: Internal External External delayed		Selects the Q-switch triggering mode
QS Trigger External delay	0 to 255	DAC counts	Sets the delay that is applied to an input Q-switch trigger in the Q-switch External Delayed mode. Each count represents approximately 0.5 ns; the exact calibration may vary slightly from one laser system to another.

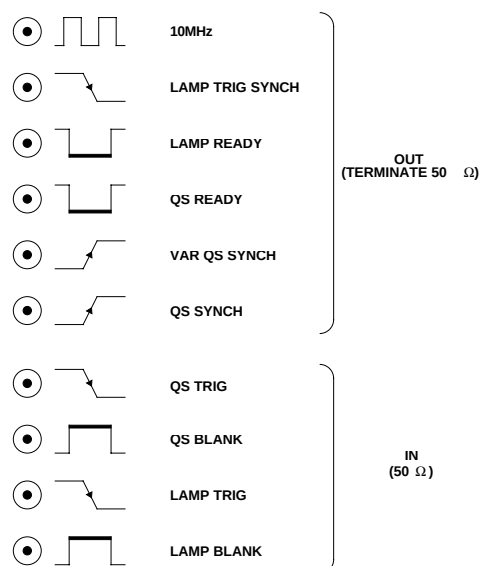


Figure 3-8. Layout of BNC Connectors on the Power Supply.

OPERATOR'S MANUAL

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CHAPTER FIVE

INSTALLATION AND ALIGNMENT

Introduction

In this chapter system installation requirements, and option alignment procedures, for the Infinity laser system are presented. The user should consult these sections if the laser system has or will be moved. When the system is delivered from Coherent, and set up for the first time, these procedures will be performed by the field service engineer. The user does, however, need to make some preparations before this installation. These are outlined below and in the Infinity Preinstallation Manual. Refer to Chapter Four, Daily Operation, for start-up and system troubleshooting procedures.

Setting up the Work Space

The Infinity laser system is a precision instrument whose performance depends on the properties of its environment. Vibrations can be transmitted to the laser head from the surface on which it rests, causing both beam pointing instability and power amplitude fluctuations. For the best results, the laser head should be placed on a stabilized optical table and isolated from mechanical contact with other equipment. Enough space should be left around the laser head so that the components inside can be reached easily for cleaning and maintenance.

The laser head and power supply are connected through a ten-foot umbilical cord. The laser head should be positioned so that the umbilical cord can hang freely. Allow at least 30 cm (12 inches) behind the power supply so that the umbilical cord will not be crimped and a minimum of 15 cm (6 inches) on the sides of the power supply, as well as the laser head, to ensure unobstructed air circulation. Figure 2-1 (Chapter Two, Description and Specifications) illustrates the recommended layout for the laser system.

Infinity Dimensions

Table 5-1. Dimensions of Infinity System

ITEM	LENGTH	WIDTH	HEIGHT
LASER HEAD	95 cm / 37.4"	48.6 cm / 19.2"	24.5 cm / 9.7"
POWER SUPPLY	68.6 cm / 27"	80 cm / 31.5"	53.4 cm / 21"
EMI FILTER	44 cm / 17.4"	36.5 cm / 14.4"	15.5 cm / 6.1"

Infinity Weights

The weight of the Infinity laser head is 150 lb (68 kg). The weight of the Infinity power supply without water is 235 lb (107 kg). With water the weight of the power supply increases to 272 lb (123 kg).

Ambient Temperature

The room in which your laser is located should be maintained at a temperature of 10-40°C (50-104°F). Stable operating temperature for the laser is around 30°C.

Cooling Water

The Infinity power supply incorporates a heat exchanger that regulates the temperature of a closed ("secondary") water loop through the laser head. The heat exchanger requires a flow of "primary" cooling water from a tap or other water source. The cooling water should meet the specifications given in Table 5-2 and Figure 5-1.

A Flow rate valve, located inside the power supply, regulates the amount of water consumed. Primary water flow rate may vary from 3.8 liters (1 gallon) per minute if the water temperature is very cold, 10°C (50°F), to 22.7 liters (6 gallons) per minute, if the water temperature is warm (maximum 30°C (90°F)), see Figure 5-1. For best laser performance the water should not fluctuate more than $\pm 1^{\circ}\text{C}$.

Two 7.6 meter (25 foot) lengths of reinforced garden hose with female connectors are supplied with the laser. These are connected at the rear of the power supply and to facility inlet and drain. Remember to attach the bracket, that will hold the umbilical, to the power supply first. After the hoses have been connected, remove the larger of the two power supply covers and open the main facility valves, drain first, and inspect the system for leaks.

Table 5-2. Cooling Water Specifications

PARAMETER	DESCRIPTION	
Flow rate (min.):	20 lpm @ 30°C	5 gpm @ 86 F
Incoming Temperature:		
Minimum	5 ° C	41 ° F
Maximum	30 ° C	86 ° F

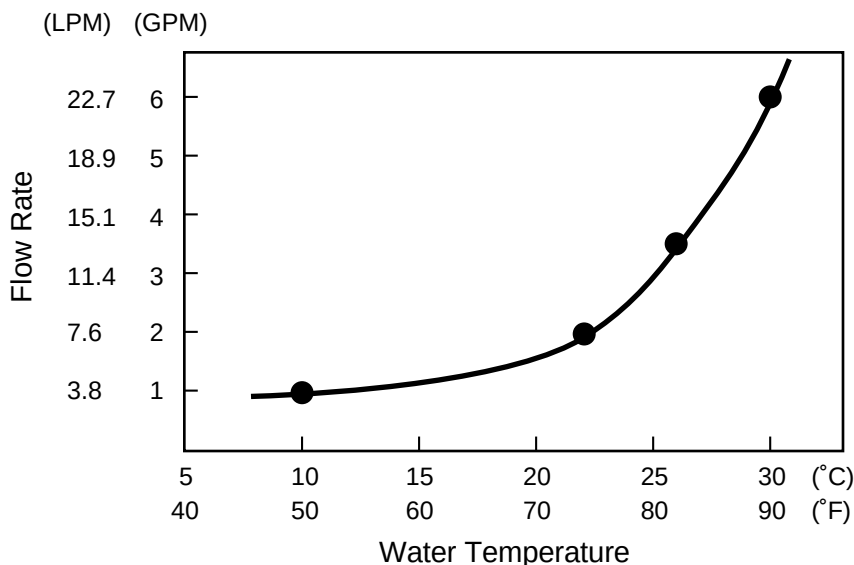


Figure 5-1. Primary Water Flow vs. Primary Water Temperature

External Heat Exchanger Requirements

An external closed loop heat exchanger used with the Infinity system needs to have a cooling capacity of at least 7.5 kW. This requirement is based on operation maximum repetition rate and at maximum lamp voltage, 550 V. Water temperature and flow specifications for the external cooling unit must be consistent with the requirements in Table 5-2.

Electrical Service

The AC electrical service for the Infinity power supply must meet the specifications in Table 5-3.

The electrical service may be either WYE or Delta connected, with the fourth wire to building ground. WYE is more tolerant to one line versus ground fluctuations and is therefore recommended. For safety reasons the circuit breaker should be located in the same room as the laser.

Attached to the power supply is a ten-foot power cable without a connector on the free end. The Infinity power cable is made up of 4x8G wires. The user must provide the hardware necessary to connect this cable to the local power service (e.g., Hubbell 4100P9W connector).

Table 5-3. Electrical Specifications

PARAMETER	DESCRIPTION	
Frequency (Hz)	60	50
Voltage (VAC),3-phase	208	380
Current-nominal (A)	20	15

System Computer

The Infinity laser is operated from an external (IBM compatible) computer using software from Coherent. Refer to Chapter Seven, System Software, for installation and hardware requirements.

The PC computer, on which the Infinity interface runs, connects to the RS 232 port on the power supply by a standard 9 pin cable. The COM port to which the cable is connected on the computer can be chosen in the Infinity software, see Chapter Seven.

Installation of External Interlock

The Infinity system provides the user with an external interlock circuit. It is located at the rear of the power supply, Figure 5-2. The system can only be turned on if this contact is short circuited. When the system is installed a short circuited RCA connector is provided for this purpose.

If it is desirable to connect the Infinity interlock to an external circuit an external switch which energizes a double pole single throw relay can be used, Figure 5-2.

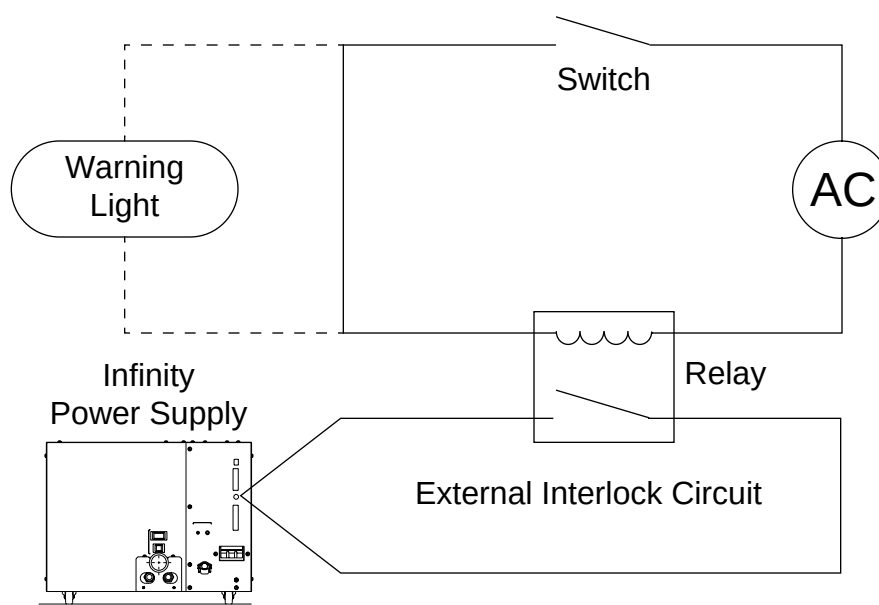
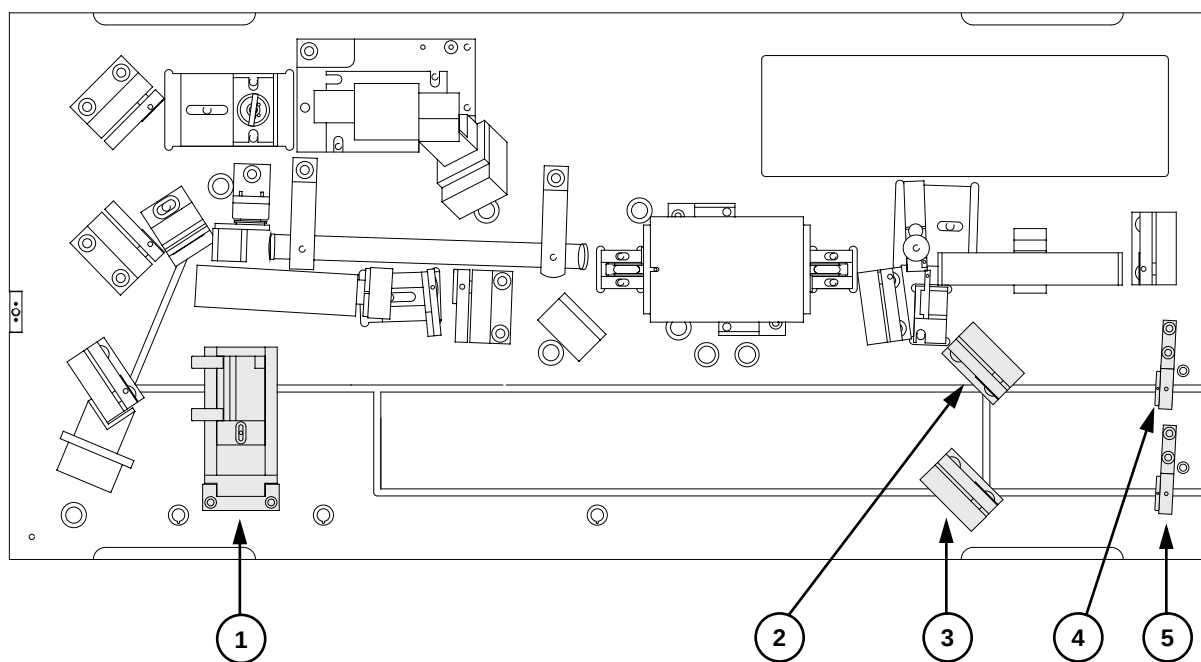


Figure 5-2. Connection of Infinity External Interlock to External On/Off Switch

Installation of SHG package

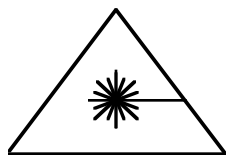
Initial installation of the Second Harmonic Generation package is done when the system is delivered or, if added later, by a Coherent field service engineer. If the package has been removed follow the procedure below to reinstall the SHG crystal. It is assumed that the necessary mounts and dichroics have been installed, see Figure 5-3.

If the residual IR is not to be utilized, the Infinity beam dump should be positioned in the location of the 1064 nm half waveplate, item 4 in Figure 5-3.



- 1 SHG assembly
- 2 Dichroic mirror.
Transmits 1064 nm and reflects 532 nm.
- 3 Dichroic mirror.
Transmits 1064 nm and reflects 532 nm.
- 4 Half wave plate for 1064 nm (optional)
- 5 Half wave plate for 532 nm (optional)

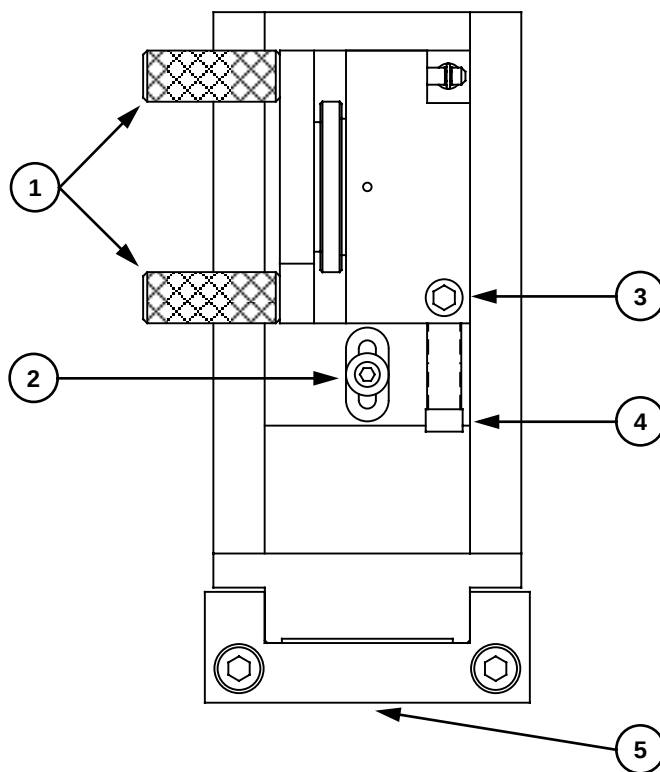
Figure 5-3. Layout for Second Harmonic Generation



The procedure described below requires the laser to be operated with the laser head cover open and the interlock defeated. Safety requires that everyone present in the room wears appropriate safety glasses and that access to the room be restricted.

SHG Installation Procedure

1. Open the head cover and install the interlock defeat. Place a power meter and expanding lens (e.g., $f = -10$ cm AR coated for 1064 and 532 nm) outside the Infinity head.
2. Run the Infinity control program. Set the repetition rate to 10 Hz, the IR pump energy to 1 mJ, and click on "On".
3. Using an IR viewing card, verify that the beam travels along the marked beam path on the base plate and at constant height above it (i.e., the pump beam is centered in the inboard exit aperture of the Infinity head cover). If not, use the vertical and horizontal tilt adjustments of the IR fold mirror assembly to position the pump beam.
4. Click on Standby. Install the alignment aperture in the cell holder. Restart the laser by clicking on On.
5. Adjust the horizontal and vertical tilt controls of the cell holder, 3 and 4 in Figure 5-4, so that aperture surface faces the incoming IR beam at normal incidence. Thereafter tilt the mount slightly around its horizontal axis so that a reflection from the cell window, when it is inserted, will strike the white diffuser, mounted above the IR fold mirror assembly. This is the position in which the crystal will be used and therefore the translational alignment should be carried out.
6. Increase the pump energy to 25 mJ. Adjust the vertical and horizontal translation stages on the SHG assembly, 2 and 5 in Figure 5-4, so that the beam is centered on the aperture.
7. Increase the power in the main control window in steps of 25 mJ. If scattered light appears anywhere along the edges of the aperture move the cell holder until the beam clears the aperture. Continue until 600 mJ can be transmitted. Use burn paper to verify that the beam is not being clipped by the aperture.



- 1 Lock nuts securing the cell to the gimbal mount
- 2 Lock screw for horizontal translation
- 3 Vertical adjustment screw controlling tilt around horizontal axis
- 4 Horizontal adjustment screw controlling tilt around vertical axis
- 5 Lock screw for vertical translation

Figure 5-4. Crystal Holding Assembly

8. Set the IR energy to 1 mJ and click on Standby.
9. Remove the alignment aperture and install the SHG crystal. Position a white paper in the 532 nm beam path.
10. Click on On and set the IR power to 25 mJ.
11. Tilt the crystal back and fourth, by hand, around its **vertical** axis. Look for a bright green spot to appear on the white paper. Use the horizontal control screw, controlling the tilt around the vertical axis (4 in Figure 5-4), to set the angle at which the 532 nm light appears with maximum brightness.

12. After the optimal phase-matching angle has been found, if required, tilt the crystal around its horizontal axis (3 in Figure 5-4) so that the return reflex from the entrance window strikes the white diffuser above the IR fold mirror assembly.



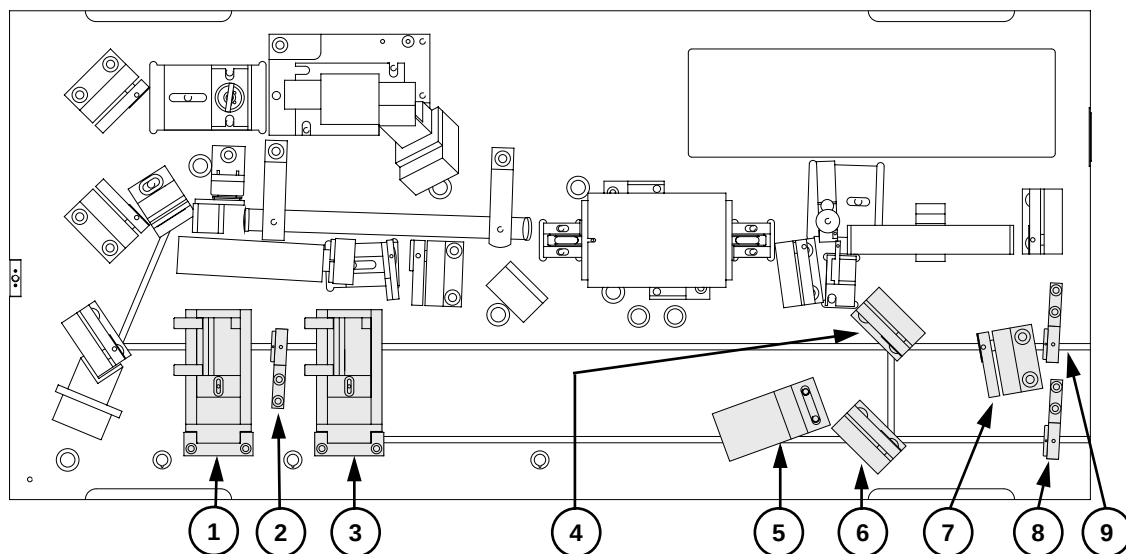
If the beam travels through the crystal off center irreversible damage may occur. Make sure at low powers that the 532 nm beam goes through the center of the crystal and that it is round and symmetrical.

13. Set the IR energy to 1 mJ and click on Standby. Remove the SHG crystal and reinstall the alignment aperture. Verify that the pump beam is still centered in the aperture. If it is not, the necessary corrections should be made (see above). This is done to ensure that the crystal is still centered on the beam after it has been tilted.
14. At low power, ~25 mJ, adjust the tilt angles of the first dichroic so that the green beam strikes the second dichroic in the center. Adjust the controls of the second dichroic so that the beam exits through the center of the outboard exit aperture in the Infinity head cover.
15. With the laser running at the desired repetition rate and pump energy, optimize the SHG phase matching angle.
16. Turn off the system, remove the interlock defeat and close the head cover. The system is now ready to be used.

Installation of THG package

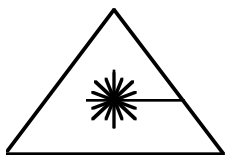
Initial installation of the Third Harmonic Generation package is done when the system is delivered or, if added later, by a Coherent field service engineer. If the package has been removed follow the procedure below to reinstall the THG crystal. It is assumed that the SHG option and the necessary mounts and dichroics have been installed, see Figure 5-5.

If the residual 532 nm radiation is not to be utilized, the Infinity beam dump (item 5 in Figure 5-5) should be positioned in the location of the dichroic fold mount and 532 nm half wave plate (items 7 and 9 in Figure 5-5).



- 1 SHG assembly
- 2 Wave plate
- 3 THG assembly
- 4 Dichroic mirror.
Transmits 1064 nm and 532 nm and, reflects 355 nm.
- 5 Beam dump for 1064 nm
- 6 Dichroic mirror.
Transmits 1064 nm and 532 nm and, reflects 355 nm.
- 7 Dichroic mirror (optional).
Transmits 532 nm and reflects 1064 nm.
- 8 Half wave plate for 355 nm (optional)
- 9 Half wave plate for 532 nm (optional)

Figure 5-5. Layout for Third Harmonic Generation



The procedure described below requires the laser to be operated with the laser head cover open and the interlock defeated. Safety requires that everyone present in the room wears appropriate safety glasses and that access to the room be restricted.

Installation Procedure for THG

1. Make sure that the two dichroics, 4 and 6 in Figure 5-5, are for 355 nm. Place a power meter and expanding lens (e.g., $f = -10$ cm AR coated for 532 and 355 nm) outside the Infinity head.
2. Run the Infinity control program. Set the repetition rate to 10 Hz, the IR pump energy to 1 mJ, and click on "On".
3. Position the waveplate holder, 2 in Figure 5-5, relative to the beam according to Figure 5-6 and Figure 5-7. Secure it to the base plate.

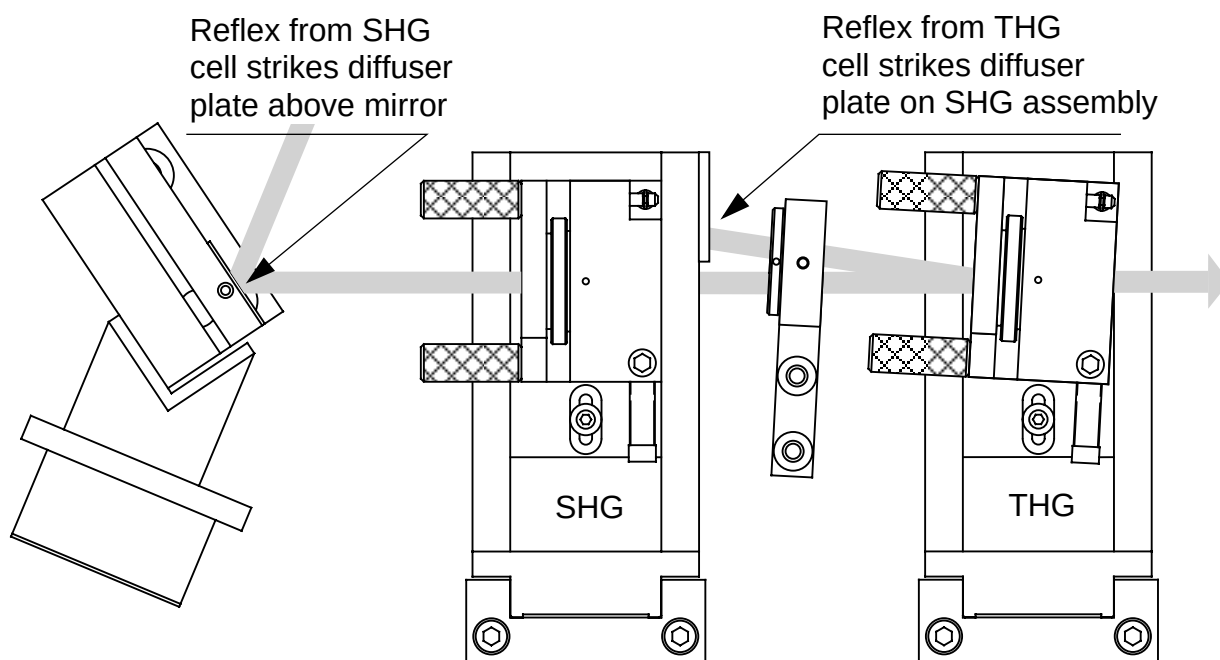


Figure 5-6. Beam Path through SHG and THG Optics

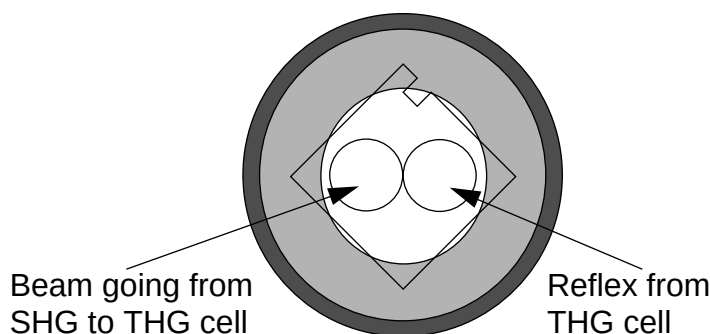


Figure 5-7. Beam Paths through THG Wave Plate, seen from the THG Crystal

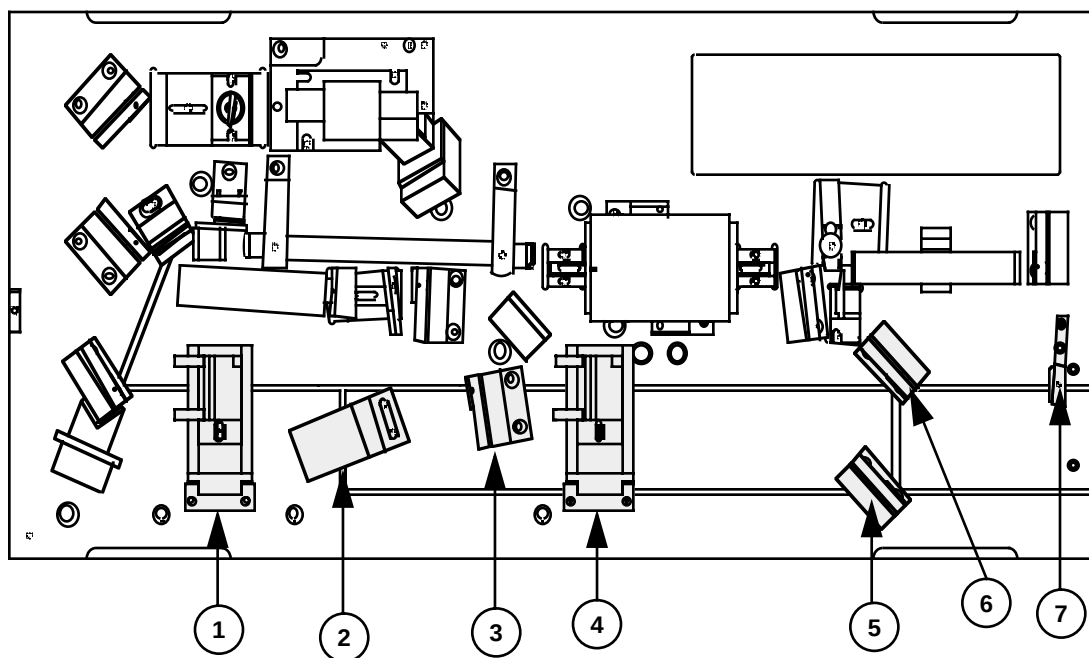
4. Click on Standby. Install the alignment aperture in the crystal cell holder. Restart the laser by clicking on On.
5. Adjust the horizontal and vertical tilt controls of the cell holder, 3 and 4 in Figure 5-4, so that aperture surface faces the incoming 1064/532 nm beam at normal incidence. Thereafter tilt the cell slightly around its vertical axis so that the reflection from the cell window, when the crystal is inserted, will strike the white diffuser, mounted on the side of the SHG assembly. This is the cell position in which the crystal will be used and therefore the alignment should be carried out.
6. Adjust the vertical and horizontal translation stages on the THG assembly, 2 and 5 in Figure 5-4, so that the beam is centered on the aperture.
7. Increase the power in the main control window in steps of 25 mJ. If scattered light appears anywhere along the edges of the aperture move the cell holder until the beam clears the aperture. Continue until 600 mJ can be transmitted. Use burn paper to verify that the beam is not being clipped by the aperture. Compare burn patterns taken before and after the aperture.
8. Turn down the power to 1 mJ and click on Standby.
9. Position a white paper in the 355 nm beam path.
10. Remove the alignment aperture and install the THG crystal.
11. Restart the laser by clicking on On. Set the IR power to 25 mJ. Optimize the intensity of the green light by adjusting the SHG crystal angle. Increase the IR power to 50 mJ.

12. Tilt the THG crystal back and fourth, by hand, across its full travel range around the **horizontal** axis. Look for a bright blue fluorescent spot to appear on the paper. Use the vertical control screw, controlling the tilt around the horizontal axis, to set the angle at which the blue light appears with maximum brightness.
13. After the phase-matching angle has been found, tilt the crystal around its vertical axis so that the return reflex from the entrance window strikes the white diffuser.
14. Set the IR energy to 1 mJ and click on Standby. Remove the THG crystal and install the alignment aperture. restart the laser by clicking on On. Repeat steps 6 – 9. This is done to ensure that the crystal is still centered on the beam after it has been tilted.
15. Adjust the tilt angles of the first dichroic so that the tripled beam strikes the second dichroic in the center. Adjust the tilt angles of the second dichroic so that the tripled beam exits through the center of the outboard exit aperture of the Infinity head cover.
16. Gradually increase the power while monitoring the (green) mode on the crystal surface. Adjust the phase matching angle so that the blue light appears with maximum brightness. If it becomes obvious, as the laser power is turned up, that the beam does not go through the center of the crystal go back and repeat the alignment procedure from step 2.
17. Make small rotational (a few degrees) adjustments to the wave plate until the energy at 355 nm is maximized.
19. In the tripling process one photon at 1064 nm is mixed with one at 532 nm. Optimum efficiency for the tripling process is therefore achieved when there is one photon at 532 nm available for every photon at 1064 nm, a 50/50 mix. To accomplish this it might be necessary to slightly detune the SHG crystal from the position at which the SHG efficiency is the highest. With the laser running at full power, optimize the tripled energy by slightly detuning the SHG crystal. Finally optimize the THG phase matching angle.
20. Turn off the system, remove the interlock defeat and close the head cover. The system is now ready to be used.

Installation of FHG package

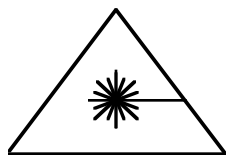
Initial installation of the Fourth Harmonic Generation package is done when the system is delivered or, if added later, by a Coherent field service engineer. If the package has been removed follow the procedure below to reinstall the FHG crystal. It is assumed that the SHG option and the necessary mounts and dichroics have been installed, see Figure 5-8.

If the residual 532 nm radiation is not to be utilized, the second Infinity beam dump should be positioned in the location of the 532 nm half wave plate (item 7 Figure 5-8).



- 1 SHG assembly
- 2 Beam dump for 1064 nm
- 3 Strip out dichroic
- 4 FHG assembly
- 5 HR 266 nm dichroic mirror
- 6 Dichroic mirror
Transmits 532 nm and reflects 266 nm
- 7 Half wave plate for 532 nm (optional)

Figure 5-8. Layout for Fourth Harmonic Generation



The procedure described below requires the laser to be operated with the laser head cover open and the interlock defeated. Safety requires that everyone present in the room wears appropriate safety glasses and that access to the room be restricted.

Installation Procedure for FHG

1. Open the Infinity head cover and install the interlock defeat. Install the stripout dichroic mount (item 3 in Figure 5-8) without the dichroic, the FHG crystal mount (item 4 in Figure 5-8) and the second beam dump (item 2 in Figure 5-8). Install the alignment aperture in the FHG crystal mount.
2. Place a power meter outside the Infinity head, in-line with the outboard beam path on the Infinity baseplate. Run the Infinity control program. Set the repetition rate to 10 Hz, the IR pump energy to 1 mJ, and click on "On".
3. Adjust the horizontal and vertical tilt controls of the FHG cell holder, items 3 and 4 in Figure 5-4, so that aperture surface faces the incoming IR beam at normal incidence. Thereafter tilt the cell slightly around its vertical axis so that the reflection from the cell window, when the crystal is inserted, will strike the white diffuser, mounted on the side of the SHG assembly, see Figure 5-9. This is the cell position in which the crystal will be used and therefore the alignment should be carried out.



The back reflection from the FHG cell needs to be aligned so that it will go through the IR stripout dichroic (when installed) without clipping the dichroic mount.

4. Adjust the vertical and horizontal translation stages on the FHG assembly, items 2 and 5 in Figure 5-4, so that the beam is centered on the aperture.

5. Increase the power in the main control window in steps of 25 mJ. If scattered light appears anywhere along the edges of the aperture move the cell holder until the beam clears the aperture. Continue until 600 mJ IR power has been reached. Use burn paper to verify that the beam is not being clipped by the aperture. Compare burn patterns taken before and after the aperture.
6. Turn down the Infinity IR energy to 1 mJ and click on Standby. Install the IR stripout dichroic and the FHG crystal, mounts 3 and 4 in Figure 5-8.
7. Set the repetition rate to 10 Hz and click on On. Increase the IR energy in 1 mJ steps until a distinct IR beam can be seen on the IR beam dump, item 2 in Figure 5-8. Reposition the beam dump and/or the strip out dichroic mount if necessary.

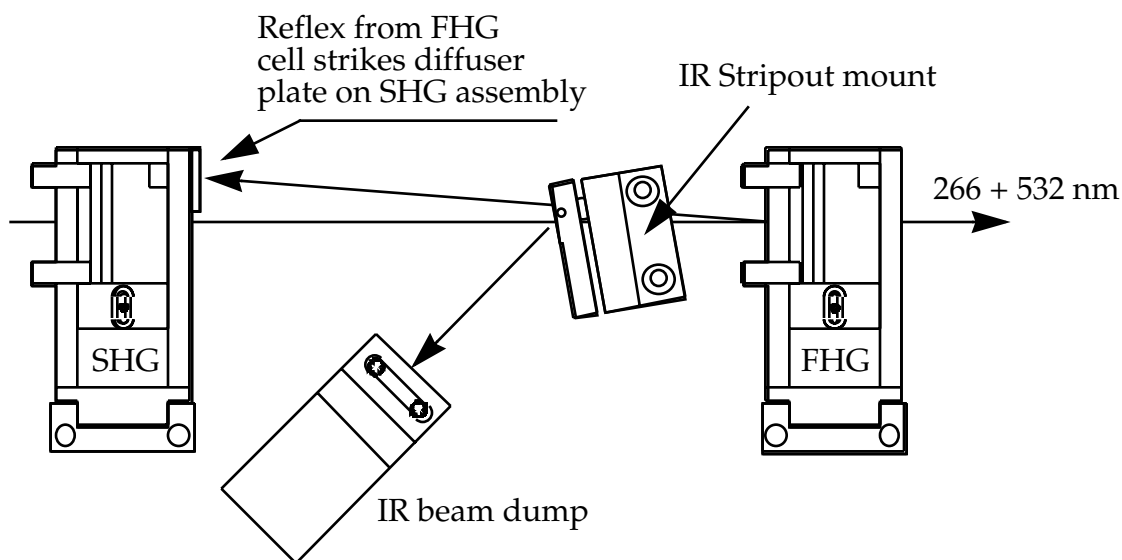


Figure 5-9. Beam Path through SHG and FHG Optics

8. Look for the 532 nm back reflection on the ceramic diffuser attached to the SHG mount. Adjust the vertical tilt of the FHG mount to position the back reflection on the diffuser if necessary.

9. Verify that both the SHG output beam, and that the back reflection from the FHG crystal, are not being clipped by the IR stripout dichroic mount.
 - a. If SHG beam is being clipped, the IR stripout mount will need to be repositioned.
 - b. If the back reflection from the FHG cell is being clipped the tilt angle of the harmonic mount will need to be corrected.

If large adjustments to the FHG mount are required, reinstall the alignment aperture and return to step 4.



The standard beam expanders can not be used with the FHG option. It will therefore be necessary to send the 266 nm output directly into a power meter head. This head should be equipped with a UV diffuser to prevent damage to the detector.

10. Place the system in Standby. Place a white card, or paper, in the beam path. Restart the laser by clicking on ON and increase the IR power to 50 mJ.
11. Tilt the FHG crystal back and fourth, by hand, across its full travel range around the **horizontal** axis. Look for a blue fluorescent spot to appear on the white paper. Use the vertical control screw, controlling the tilt around the horizontal axis (item 3 in Figure 5-4), to set the angle at which the 266 nm spot appears with maximum brightness.
12. After the phase-matching angle has been found, again verify that the return reflex from the FHG entrance window strikes the white diffuser mounted on the side of the SHG crystal assembly.
13. Verify that the back reflection is not being clipped by the IR stripout dichroic mount.
14. Set the IR energy to 1 mJ and click on Standby. Remove the FHG crystal and install the alignment aperture. Restart the laser by clicking on On. Repeat steps 4 and 5 with the stripout dichroic installed. This is done to ensure that the crystal is still centered on the beam after it has been tilted.

15. Adjust the tilt angles of the first turning dichroic (item 6 in Figure 5-8) so that the 266 nm beam strikes the second dichroic in the center. Adjust the tilt angles of the second dichroic so that the 266 nm beam exits through the center of the outboard exit aperture in the Infinity head cover.
16. Gradually increase the power while monitoring the (green) mode on the crystal surface. Adjust the phase matching angle so that the blue light appears with maximum brightness. If it becomes obvious, as the laser power is turned up, that the beam does not go through the center of the FHG crystal, go back and repeat the alignment procedure from step 2.
17. To generate the Fourth Harmonic, two photons of 532 nm are required. Therefore, it really should not be necessary to go back and make any adjustments to the phase matching of the SHG crystal. However, the fact that the SHG crystal is optimized should be verified.
18. Turn off the system, remove the interlock defeat and close the head cover. The system is now ready to be used.

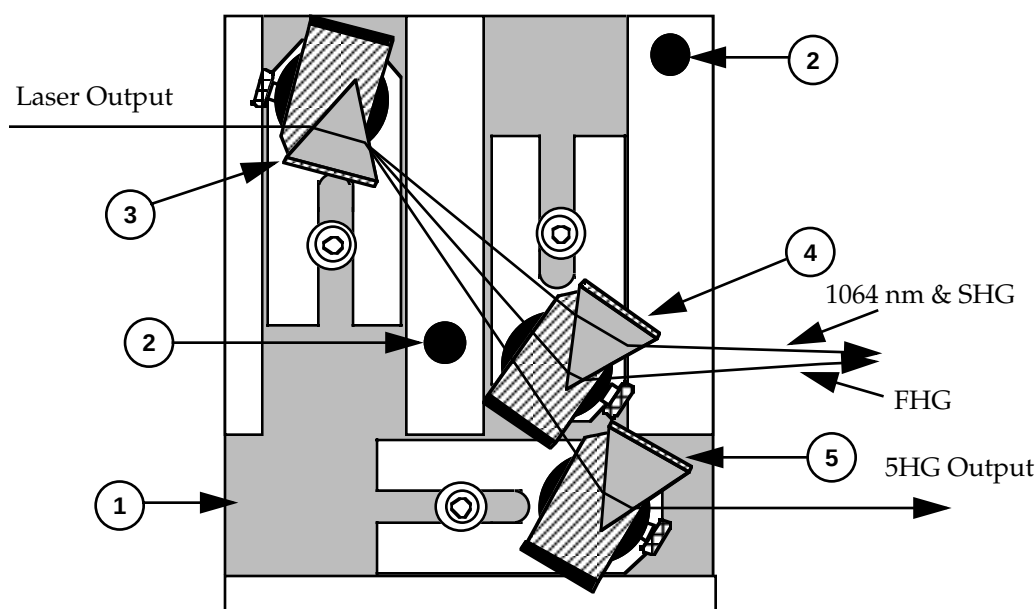


Due to the large nonlinear coefficient of BBO at 266 nm, 133 nm radiation can also be generated by the same crystal employed to generate the 266 nm radiation. BBO is not transparent to 133 nm radiation resulting in a significant temperature gradient as the pump energy and repetition rate of the laser system are changed. It is therefore necessary, and normal, to have to adjust the phase matching angle of the FHG crystal if the laser system operational parameters are changed by more than a few percent.

Installation of 5HG package

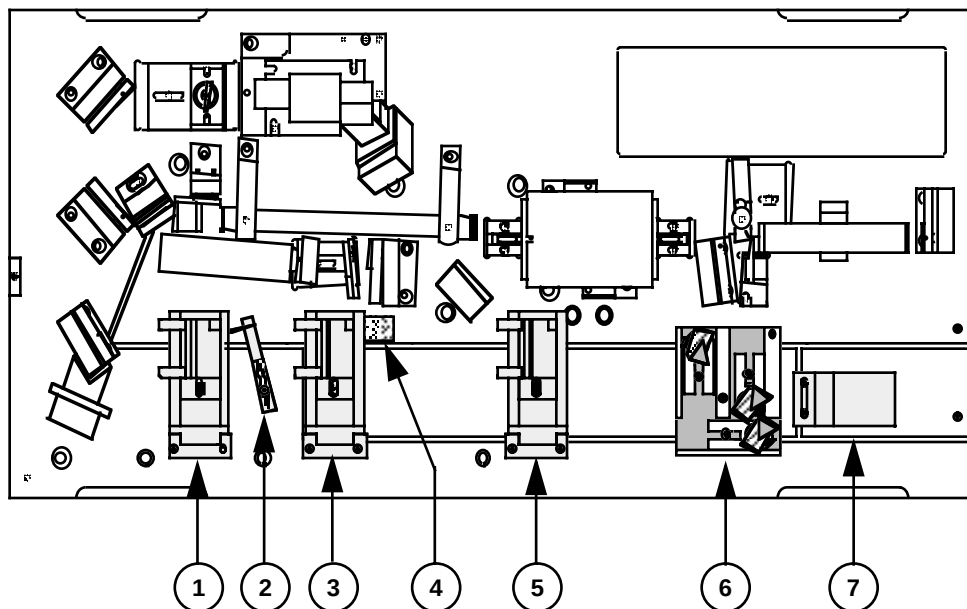
Initial installation of the 5TH Harmonic Generation package is done when the system is delivered or, if added later, by a Coherent field service engineer. If the package has been removed follow the procedure below to reinstall the 5HG crystal.

In the procedure which follows it is assumed that the system is operating in the FHG configuration. If not, install the FHG option first. Figure 5-10 gives an overview of the prism assembly employed to separate the 5TH harmonic from the 1064, 532 and 266 nm radiation. Figure 5-11 illustrates the proper mounting of the prism assembly as well as the 5TH harmonic crystal holder, the SHG and FHG retro-reflection ceramics, and the standard Infinity beam dump.



1	Base Plate
2	Securing Bolt Holes
3	Prism 1 (P1)
4	Prism 2 (P2)
5	Prism 3 (P3)

Figure 5-10. Prism Separation Assembly.



1	SHG Assembly
2	SHG Retro-reflection Ceramic
3	FHG Assembly
4	FHG Retro-reflection Ceramic
5	5HG Assembly
6	Prism Separation Assembly
7	Beam Dump

Figure 5-11. Layout for Fifth Harmonic Generation

Assembly Identification

Figure 5-12 gives a side and top view of a prism mount assembly. It is important to note the differences between the mounts, and to install each mount in its proper location. Depending on the version of your 5HG prism separation assembly there may be one or two differences between the prism mounts.

Common to all versions of this assembly is the fact that the mount for P1 is the mirror image of the mounts for P2 and P3 (see Figure 5-10). The second difference, found only in early versions of the assembly, is that the mount for P2 is longer (1-3/16") than the mount used for P3 (1"). This dimension is labeled "A" in the right hand illustration of Figure 5-12. In later versions of the assembly the mounts for P2 and P3 are the same length. The left hand illustration of Figure 5-12 indicates that a ceramic beam block is attached to each the

prism mounts. In Figure 5-10 these beam blocks were intentionally omitted for clarity in the drawing.

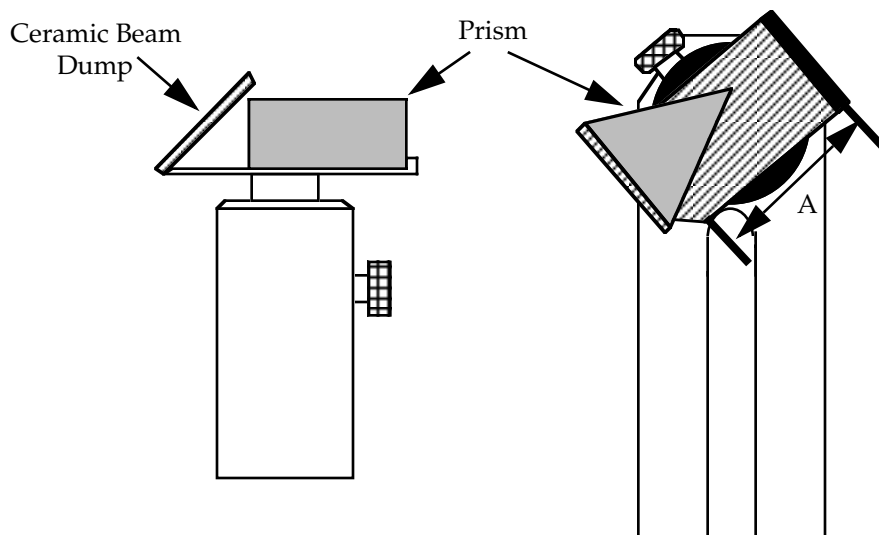


Figure 5-12. Prism Mount Assembly.

Installation Procedure for 5HG

To determine the angle between a prism face and the input beam, a card should be used to determine the position of the prism which results in the minimum deviation of the beam of interest (i.e., the position of the prism where the beam is least distorted along the horizontal axis). In this case, the beam of interest is the 213 nm beam. When the condition of minimum deviation is obtained the entrance and exit angles of the laser beam with respect to the prism faces will be identical and the largest possible aperture through the prism is realized. In this orientation the lowest possible reflection losses due to the prism faces is obtained.

1. Open the Infinity head cover and install the interlock defeat. Install the 5HG crystal cell holder and the alignment aperture.
2. If the 5HG option has not been previously installed, install the ceramic diffuser into its slide mount and attach the slide mount to the front of the FHG mount (see Figure 5-11). Note, if the FHG mount is an old style mount there will "not" be a screw hole for attachment of the ceramic slide mount. If this is the case the FHG harmonic mount will need to be switched with the 5HG mount. The ceramic should be positioned slightly to the

right of center (towards the amplifier stage) with respect to the center of the FHG crystal.

3. If the standard ceramic diffuser is attached to the SHG crystal stage (i.e., the diffuser employed for THG), remove this diffuser and install the FHG retro-reflection diffuser, item 2 in Figure 5-11.
4. Place a power meter outside the Infinity head, in line with the outboard beam path on the Infinity baseplate. Run the Infinity control program. Set the repetition rate to 10 Hz, the IR pump energy to 1 mJ, and click on "On".
5. Position a beam dump, on the Infinity head baseplate, to collect the output beam (1064, 532 and 266 nm radiation). Increase the voltage until the threshold for FHG is reached. Position the FHG retro-reflection ceramic so that the beam is centered on the ceramic and is elongated to cover as large an area on the ceramic as possible, see Figure 5-13.

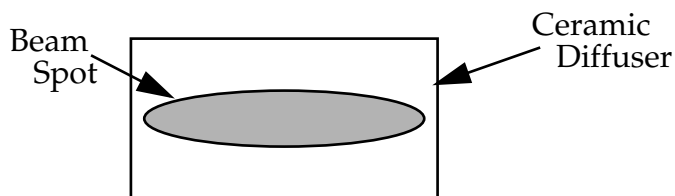


Figure 5-13. Beam Image on Properly Aligned Retro-Reflection Ceramic.

6. Adjust the horizontal and vertical tilt controls of the 5HG cell holder, 3 and 4 in Figure 5-4, so that aperture surface faces the incoming IR beam at normal incidence. Thereafter tilt the cell slightly around its horizontal axis so that the reflection from the cell window, when the crystal is inserted, will strike the ceramic diffuser, mounted on the FHG assembly. This is the cell position in which the crystal will be used and therefore the alignment should be carried out.
7. Adjust the vertical and horizontal translation stages on the 5HG assembly, 2 and 5 in Figure 5-4, so that the beam is centered on the aperture.
8. Increase the power in the main control window in steps of 25 mJ. If scattered light appears anywhere along the edges of the aperture move the cell holder until the beam clears the aperture. Continue until 600 mJ can be transmitted. Use burn paper to verify that the beam is not being clipped by the aperture. Compare burn patterns taken before and after the aperture.

9. Turn down the power and click on Standby.
10. Position P1 as illustrated in Figure 5-10. Place a white card, mounted on a beam dump, in the expected beam path due to P1 (i.e., near the location of P3 in Figure 5-10 or 5-11).
11. Remove the alignment aperture and install the 5HG crystal.
12. Restart the laser by clicking on "On" and increase the IR energy to 25 mJ. Position P1 so that the laser beam is striking the center of prism with respect to both the vertical and horizontal axis. Slowly increase the IR power until the 266 nm (FHG) radiation can be seen on the beam dump card as well as the 532 nm radiation. Increase the IR power until the 266 nm beam is stable (approximately 100 mJ IR energy). Optimize the phase matching angle of the FHG cell. Adjust the angle of P1 to minimize the deviation of the 532 and 266 nm beams.
13. Tilt the 5HG crystal back and fourth, by hand, across its full travel range around the **vertical** axis. Look for a second blue fluorescent spot to appear on the white card. Use the horizontal control screw, controlling the tilt around the vertical axis (4 in Figure 5-4), to set the angle at which the second blue spot appears with maximum brightness.
14. After the phase-matching angle has been found, tilt the crystal around its horizontal axis so that the return reflex from the entrance window strikes the white diffuser mounted below the FHG crystal assembly. If this is the first time the 5HG option is being installed, it may be necessary to reposition the ceramic. Re-establish the 5HG phase matching angle.
15. Set the IR energy to 1 mJ and click on Standby. Remove the 5HG crystal and install the alignment aperture. restart the laser by clicking on On. Repeat steps 7-14 to ensure that the crystal is still centered on the beam after the final tilting of the harmonic stage.

16. If necessary increase the IR energy to stabilize the 5HG beam. Adjust the angle of P1, with respect to the laser beam, to minimize the deviation of the “5HG” beam. Position P2 as illustrated in Figure 5-10. When properly installed the SHG beam should be at the top of the prism and the FHG near the bottom. The 5HG beam should clear the prism, and the prism mount, as illustrated in Figure 5-10. It may be necessary to raise the mount of P2 to get the beams centered vertically.
17. Rotate P2 to minimize the deviation of the 532 and 266 nm beams. Position the Infinity beam dump as illustrated in Figure 5-11. Verify that the 213 nm beam is not being clipped by P2 or its mount.
18. Position P3 as illustrated in Figure 5-10. Position the prism so that the deviation of the 213 nm beam is minimized and the beam exits cleanly out of the Infinity head cover aperture.



It is important to note that the normal 355 expansion lens can not be used for the 213 nm radiation generated by this option. It is therefore very important to use a (low energy) pulsed power meter equipped with a UV diffuser.

19. Verify that the power meter, equipped with a UV diffuser, in the path of the 213 nm beam. Gradually increase the power while adjusting the phase matching angle of the 5HG stage.



Typical energies for the 5TH harmonic are: 5 mJ at 100 Hz and 400 mJ IR and 15 mJ at 10 Hz and 600 mJ IR. The documentation which came with the 5HG option, or the laser system if the 5HG option was part of the original purchase, should be referenced to determine the energies specified by Coherent at the time of manufacturing.



Due to thermal loading, changes in the repetition rate or IR power by more than 5 to 10% may require that the phase matching angles of the FHG and 5HG crystals be optimized as well as the three prism angles (i.e., the minimum deviation of the SHG, FHG and 5HG beams). The latter can be most easily accomplished using a white card to follow the beam paths from one prism to the next, starting with P1.

20. To generate the 5TH harmonic, one photon of 266 nm is mixed with one at 1064 nm. Optimum efficiency for the process is therefore achieved when there is one photon at 266 nm available for every photon at 1064 nm, a 50/50 mix. To accomplish this it will be necessary to slightly retune the SHG and the FHG crystals from the position at which the FHG efficiency was the highest.

Set the repetition rate to the desired value (i.e., typical rate at which the system will be used). Set the IR energy to its maximum value. Optimize the 213 nm energy at these system parameters by slightly retuning the SHG crystal and then the FHG crystal. Finally, optimize the 5th harmonic phase matching angle.

Close the head cover and allow the system to run for approximately 10 minutes. Repeat the phase matching procedure until the maximum 213 nm energy is obtained. If less 213 nm radiation is desired decrease the IR energy before repeating the optimization procedure.

Installation of Telescope Assembly

The following is the procedure for installing and aligning the Infinity telescope. There are three different optic combinations which can be installed in the Infinity telescope. The optic numbers, focal lengths, AR coatings, and resulting magnification are given in Table 5-4. Note that optic "A" (installed in mount #1) is the optic closest to the harmonic stage and optic "B" (installed in mount #2) is the optic closest to the exit window in the Infinity head cover.

Table 5-4. Infinity Telescope Optics

	Optic "A"	Optic "B"	Coatings	Magnification
1	0167-912-00 (f = 148 mm)	0169-830-00 (f = 289 mm)	532/355	2x (11 mm)
2	0167-912-00 (f = 148 mm)	0167-913-00 (f = 270 mm)	532/355	1.8x (9.9 mm)
3	0167-910-00 (f = 148 mm)	0167-911-00 (f = 270 mm)	1064/532	1.8x (9.9 mm)

1. An overview of the installed telescope baseplate, location in head, is given in Figure 5-14. Align the mount baseplate so that the two attaching holes, indicated in Figure 5-15, line up with threaded holes in the Infinity baseplate. Secure the telescope mount baseplate to the Infinity head baseplate.
2. There will be a focused back reflection from the first positive lens of the Infinity telescope. This reflection will strike the second turning dichroic mount just below the optic. To prevent damage to both the mount and the dichroic two ceramic beam blocks should be stacked and mounted (double sided tape works well) just below the turning dichroic.
3. Construct two laser beam targets. Turn on the Infinity and adjust the IR energy until there is stable harmonic production. If an IR telescope is being installed then the IR power should be adjusted to 5 - 10 mJ.

Position one of the targets as far from the Infinity head as possible (4 to 5 meters). If necessary use external turning optics. Position the second target just outside the Infinity head. Both targets should be positioned so that the laser beam strikes the center of the targets.

4. The optics of the Infinity telescope are plano-convex. There is a penciled arrow on the side of each optic which points towards the convex side. The optics are mounted in the telescope so that the arrows (convex sides) point outward (away from the center of the telescope assembly).



The lens retaining ridge of lens mount #1 is vertically tilted. The telescope optic should be installed so that the convex side of the optic is orientated downwards and is facing towards the amplifier stage of the Infinity.

- Install optic "A" into lens mount #1. Tighten the set screw to secure the lens. Adjust the vertical and horizontal position of the lens so that the beam is centered on the optic, and the beam position on the target closest to the Infinity laser head is not perturbed (i.e., the expanding beam is still striking the center of the target).
5. Install the vacuum cell on the telescope mount baseplate so that the window on the vacuum cell is as close to lens "A" as possible. This will prevent damage to the cell window from internal reflections in lens "A".
 6. Install lens "B" into the mount #2 and secure the lens mount to the telescope baseplate. Adjust the position of lens mount #2 so that the laser beam is centered on the target closest to the Infinity head. Adjust the translational position of the mount (with respect to the beam axis) so that the resulting beam diameter is as close as possible to the value given in Table 5-4.
 7. Verify collimation of the laser beam by comparing burns taken at the laser head and at the target farthest from the laser system.

Verify that the collimation adjustment has not moved the laser beam off center on the target. If laser beam is found to be off center, return to Step 6.
 8. Verify that the back reflection from lens #1 is striking the second dichroic mount on the stacked ceramics.

If the laser beam is centered, the installation and alignment of the Infinity telescope assembly is complete.

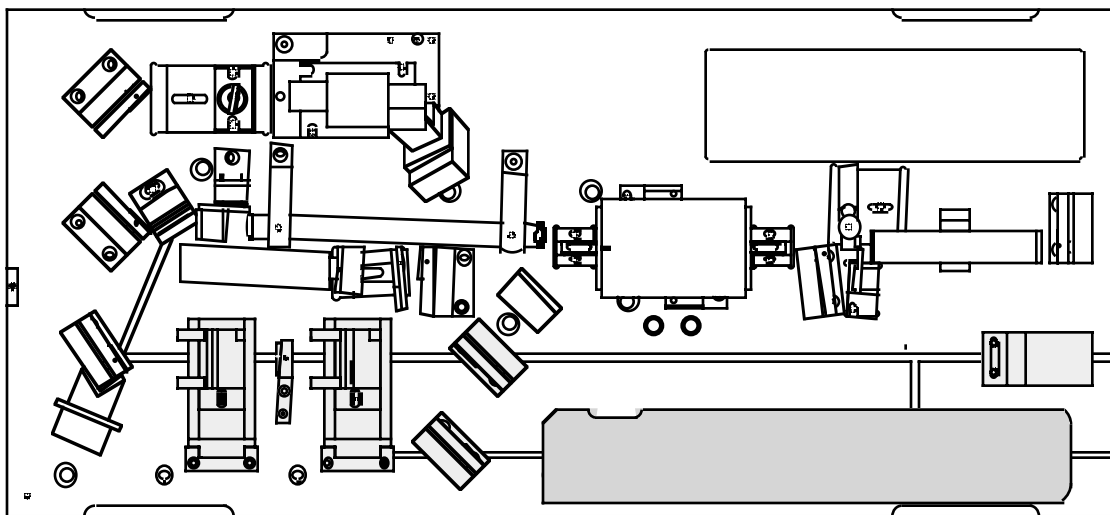


Figure 5-14. Orientation of steering dichroic mounts, IR beam dump, and Infinity Telescope base plate.

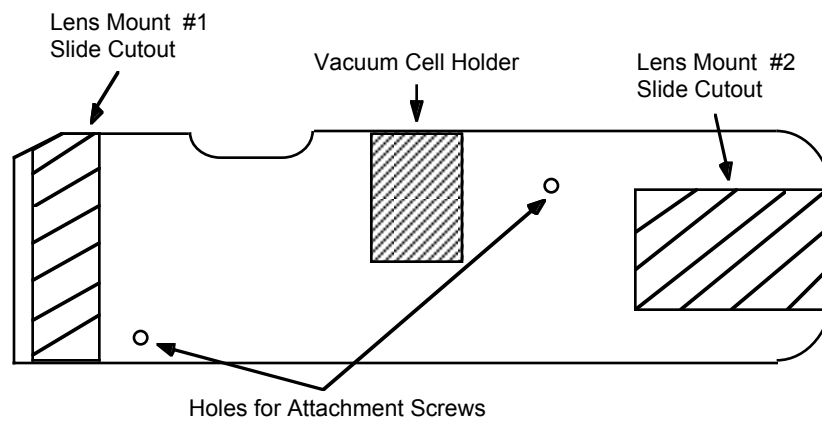


Figure 5-15. Overview of Telescope Mount Baseplate

OPERATOR'S MANUAL

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CHAPTER SIX **MAINTENANCE**

Introduction

This chapter contains preventive maintenance procedures and instructions for how to replace the flash lamp, set the system timing and change the filters in the water cooling system. The chapter concludes with a listing of instruments which will aid in the maintenance the Infinity laser system.

It is advisable that the optics in the laser are checked for burn marks, dust and scratches once a month. By the time a problem with an optical component is severe enough to affect the laser's performance it is usually too late to restore it.

Optics Cleaning and Inspection

The most common reason for performance degradation in laser systems is contaminated optics. Laser optical components are routinely exposed to very high energy levels. When optical coatings are clean, this energy will either be reflected or transmitted. If the surface is contaminated, part of the energy will be absorbed, creating hot-spots which can burn the precision coating and dramatically reduce reflection efficiency. Absorption caused by contaminated optical surfaces will degrade performance and shorten component life. Once absorption starts to take place, the rate of surface destruction rapidly increases. Thus even a very small contamination can rapidly give rise to considerable damage. Contaminants which can cause absorption include a variety of particles which may fall on the optical surface or condense from surrounding vapors. Even lens tissue fibers, and plastic gloves can be a source of contaminants. In order to maintain optimum performance, proper cleaning is an absolute necessity. Even though it is advisable to regularly clean the optics, there is also a risk of scratching the components in the process. Operating the laser with the cover on will reduce the need to clean the optics. It is also a good habit to keep the laboratory as clean as possible.

Optics Inspection

A small flashlight is useful for optics inspection. Illuminate the surfaces at various angles and look for streaks and dust particles. If the surface does not appear uniformly colored or if there are signs of dust on the surface, use the cleaning procedure for installed optics outlined below.

General Cleaning Procedure

For Installed Optics:

1. Remove the laser head cover.
2. Neatly fold a sheet of lens tissue into a rectangular shape (approximately 1/4 in x 3/4 in.). Take care not to contaminate the tissue with soiled or oily fingers.

3. Clamp and hold the tissue with hemostats. To avoid damage to the optics, ensure that the hemostats are not close to the edge of the tissue.
4. Moisten the tissue with a few drops of reagent grade methanol or acetone.
5. Gently wipe across the optic in one direction, see Figure 6-1. One pass should be sufficient. Repeat if necessary with a clean tissue.

This procedure should work well for most components in the laser. Different procedures are required for optical components that are difficult to access.

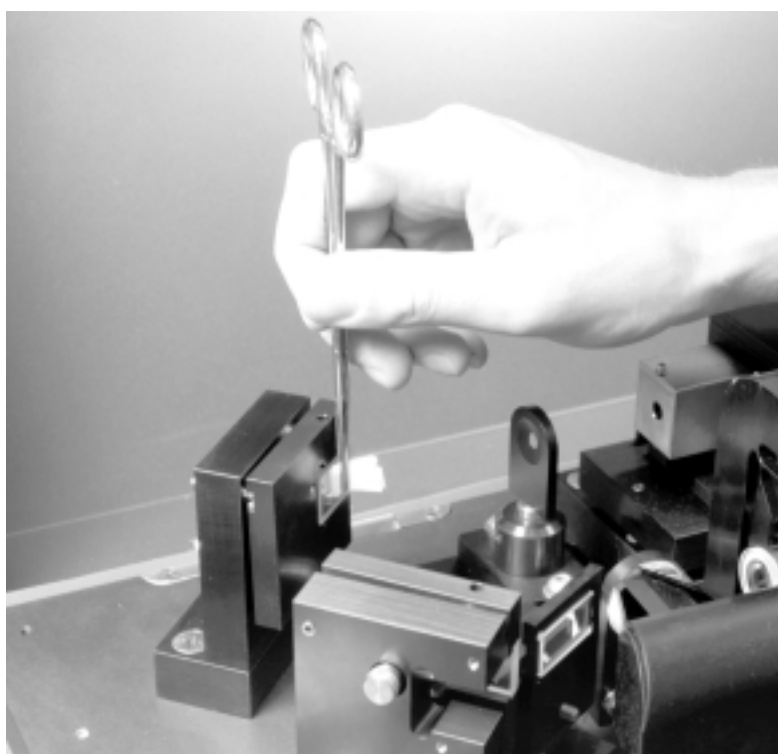


Figure 6-1. Cleaning of Installed optics

90° Polarization Rotator

The plate holding the 90° rotator can be taken out of the laser by removing the spring loaded tensioning screw (see Chapter Three) that holds the moveable part of the mount. The optic can thereafter be cleaned using the procedure described above, see Figure 6-2. Monitor the amount of depolarized energy in the Monitors window after the plate has been reinserted. Adjust the vertical and horizontal tilt angle controls of the mount to bring down the depolarized energy if necessary.



Figure 6-2. Cleaning of 90° Rotator

Relay Imaging Lenses

Removing the vacuum cell in the relay imaging system provides access to the 1 inch concave lens and the relay lens surfaces facing it, Figure 6-3. The windows of the vacuum cell can also be cleaned while the cell is removed. Monitor the amount of depolarized energy in the “Monitors” window after the cell has been reinserted. Follow procedure 4 in Chapter Four to minimize the depolarized energy if it is above 1.5 % of the IR output power.

When reinstalling the vacuum cell, the teflon set screw should not be tightened with a ball driver or Allen key (i.e., the set screw should be “finger” tight). Over tightening the set screw could easily cause the cell to crack.

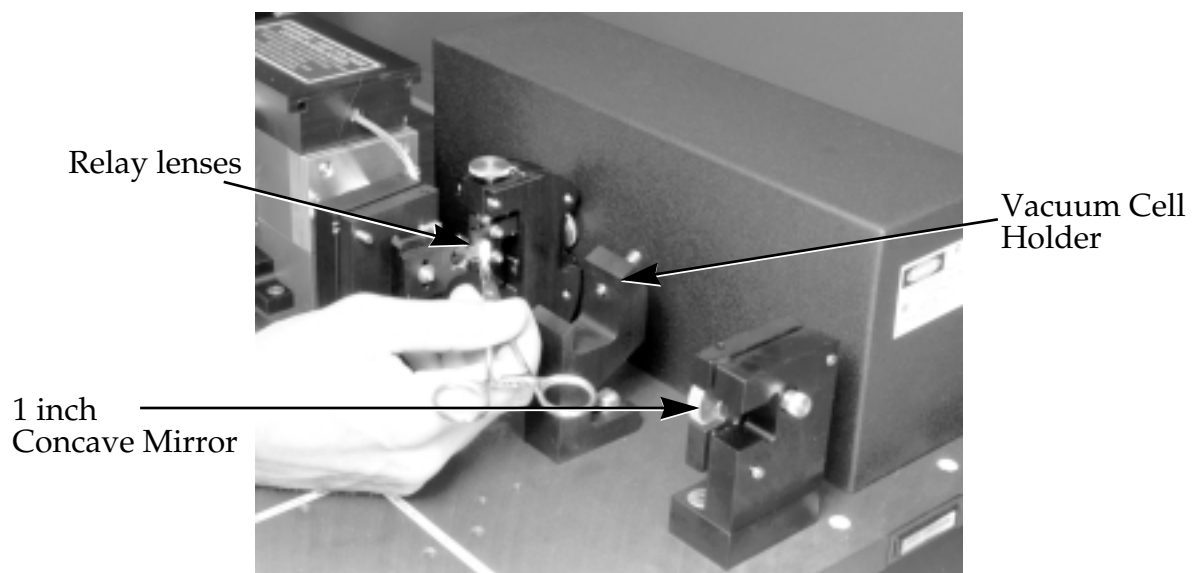


Figure 6-3. Cleaning of Relay Lens #1

Water Cooling System

The water cooling system removes excess heat from the flash lamp, the laser rods and from some of the electrical components in the power supply and the laser head. If the cooling system were to malfunction, the temperature of the laser system would increase, potentially causing irreversible damage. In order to protect the laser from such an event, several safety features have been built into the system. Sensors located at different places in the water loop continuously monitor water level, temperature, flow and resistivity (see Chapter Three, Controls and Indicators). If any of these readings fall outside their preset safety ranges, the microprocessor in the power supply will turn the system off.

To ensure optimal performance and prevent unnecessary downtime it is advisable to periodically check water hoses and water connections for wear and leaks. Make sure that there is enough water in the internal cooling loop. The main reservoir should be at least 90% full. If necessary refill it with "steam" distilled water.

The capacity of the deionization cartridge decreases with time. It should be replaced when it is no longer possible to keep the water resistivity within the desired range, 2 – 6 MΩ cm, by adjusting the flow through the cartridge. See Chapter Three for information about how to adjust the water resistivity. It is advisable to replace the particle filter and the water when the deionization cartridge is replaced.



Use only steam distilled water in the internal water cooling system. Do not use deionized tap water.

For optimum system performance, it recommended that the water and deionization filters be replaced yearly.

Changing the Water and Deionization Filter

1. Turn the key switch on the power supply to off and turn off the main facility circuit breaker. Unscrew and remove the larger of the two covers on the power supply.

Particle Filter Replacement:

- a. Unscrew the filter holder, item 1 in Figure 3-7, and remove the filter inside.
- b. Rinse the new filter thoroughly in distilled water
- c. Insert the new filter and screw the holder back on. It helps if the O-ring is moistened with water first.

Deionization Cartridge Replacement:

- a. Remove the water hoses from the top and bottom of the old cartridge, item 7 in Figure 3-7.
- b. Insert a new cartridge, flow arrow pointing up, and connect the water hoses.
2. Turn on the water pump in the Service window, accessed by clicking on "Service..." in the Infinity control window. Check for leaks. Let the water run until all sounds of bubbles in the system have disappeared and there are no bubbles in the water flowing through the particle filter.
3. Allow the system to run for approximately 1 hour. During this time adjust the deionization cartridge bypass valve, item 3 in Figure 3-7, as necessary to regulate the water resistivity between 2 – 6 MΩ cm.
4. Reinstall the heat exchanger cover. The laser is now ready for normal use.



If the internal water cooling system is drained, remove the deionization cartridge and seal it so that it does not dry out. If the cartridge is allowed to dry it will lose its capacity to maintain a high enough water resistivity. Do not insert a new cartridge until you are ready to fill the system with distilled water.

Replacing the Flash Lamp

Change the flash lamp when specified power can no longer be achieved or if the laser energy exhibits large shot to shot fluctuations. Before changing the flash lamp, verify that the shot counter reading is in agreement with the expected life time for the lamp, approximately 50 million shots. If this is not the case there might be a different reason for the performance degradation. See Chapter Four, Procedures 2 and 3, for more information.

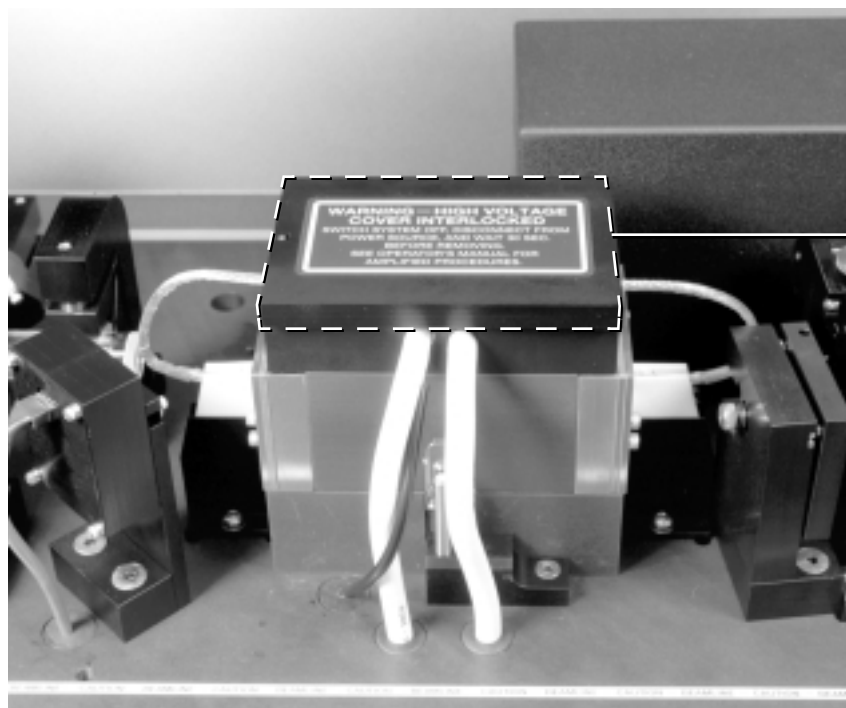


When changing the flash lamp avoid skin to glass contact as this may leave grease marks! Contaminating the Nd:YAG rod end surfaces with finger grease will lead to a drastic increase in absorption that might damage the coatings. Dirt on the glass envelope will reduce the flash lamp efficiency.

Removing the Old Flash Lamp.

Please complete, and check off, each step before proceeding. Numbers in the text are given with reference to Figure 6-4.

1. Turn the power supply key switch to Off position and switch off the main facility circuit breaker.
2. Expose the lamp terminals by sliding the top plastic cover to the side, Figure 6-4.



Slide top cover off to expose flash lamp leads.

Make sure that the power supply is turned off and disconnected from the main AC line!

Figure 6-4. The Infinity POT Assembly with the Flash Lamp Lead Housing on Top

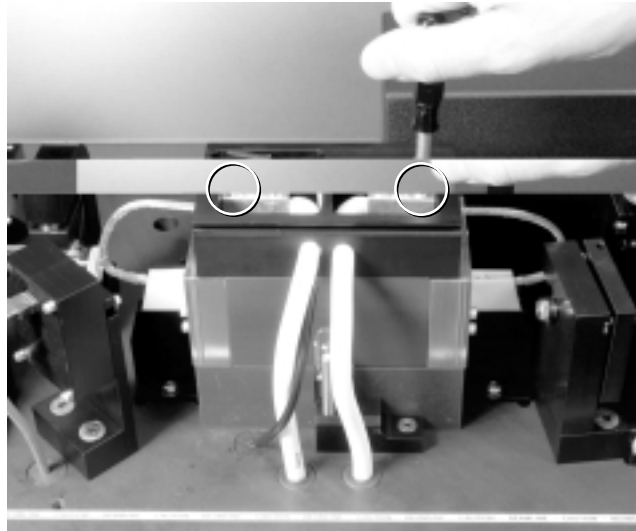


Figure 6-5. Loosening the Flash Lamp Terminal

3. Loosen the lamp terminal, encircled in Figure 6-5, on both sides using the 3/16" nut driver from the maintenance kit.
4. Remove leads from side holes and bend the lamp terminal box on top of the POT out of the way, Figure 6-6.

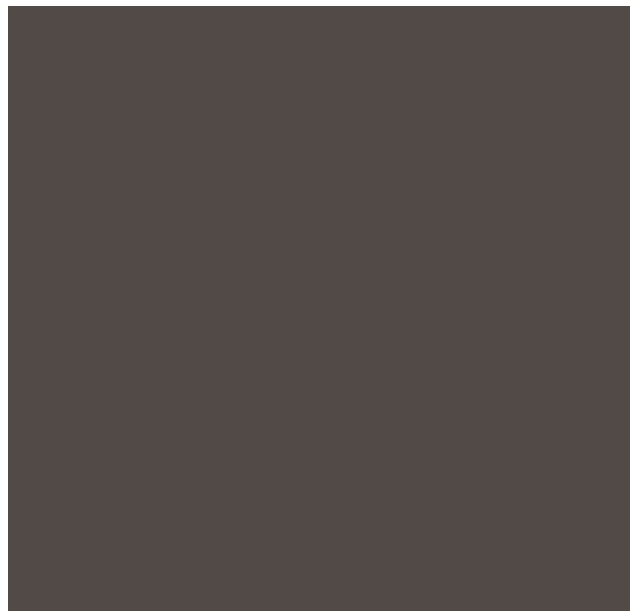


Figure 6-6. Flash Lamp Leads Released and the Lamp Terminal Housing Bent to the Side

5. Prepare a work area with a soft surface on which the POT can be positioned.

6. Remove the two vertical dowel pins, 1 in Figure 6-7, release the latches holding the POT, 2 in Figure 6-7, tilt it sideways a few degrees to drain the water out of it and then lift the POT out of the laser, 3 in Figure 6-7.
7. Put the POT on a clean soft surface so that the green lamp lug (cathode) is on the right-hand side.



Be careful not to scratch the bottom surface of the POT! Doing so might impair the proper function of the two O-ring seals.

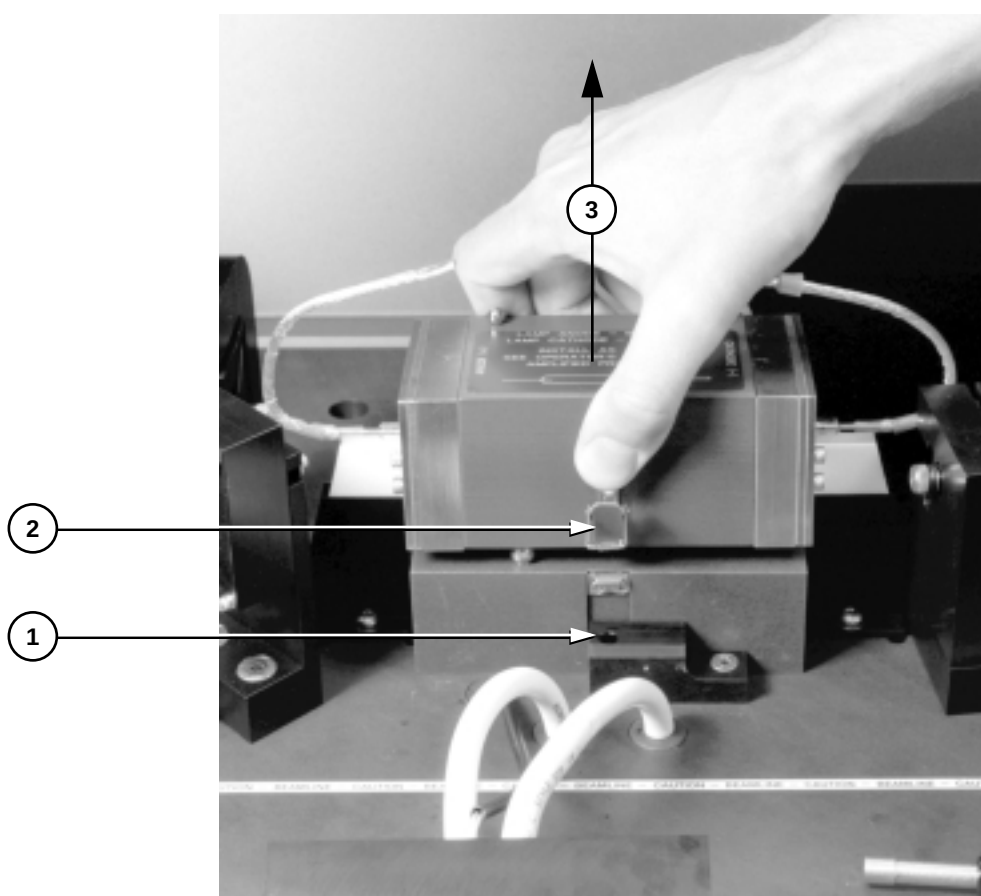


Figure 6-7. Dowel Pins Removed and Latches Released.

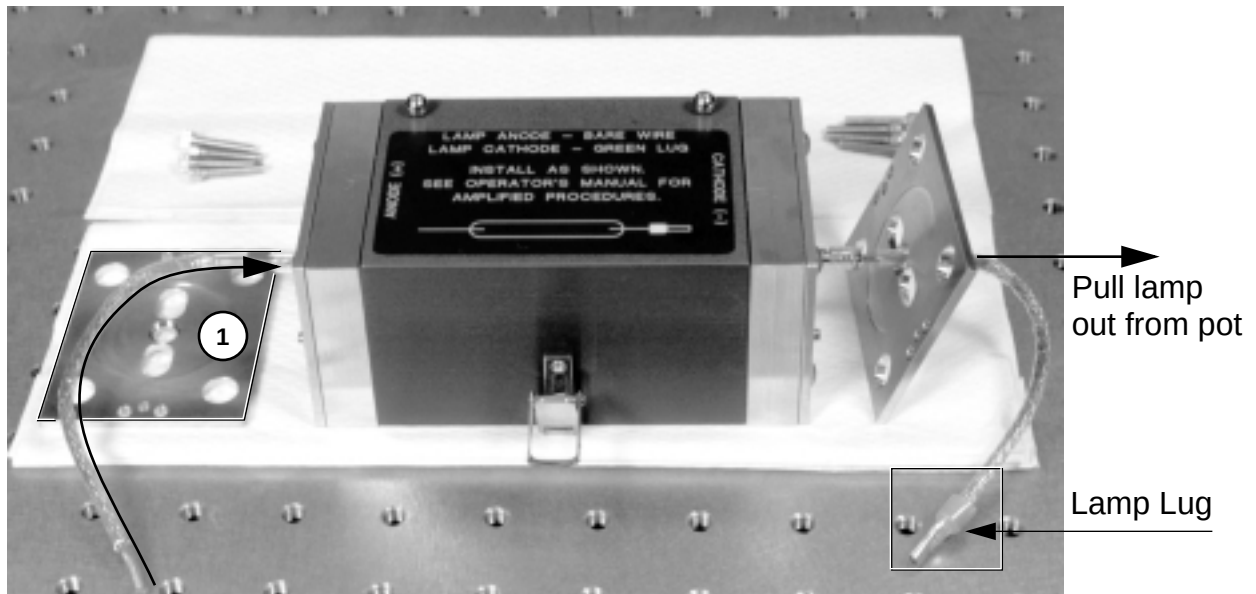


Figure 6-8. POT with the Left Endplate Removed

8. Remove the left endplate, 1 in Figure 6-8. Note that because of the lug attached to the flash lamp lead on the right side the lamp must be taken out of the pot before the right endplate can be removed, 2 in Figure 6-9.
9. Using the finger cots supplied, carefully remove the O-ring on the left-hand side.
10. Slide the flash lamp out of the POT to the right, Figure 6-8. Slide off the remaining O-ring and thereafter the second endplate, 1 and 2 in Figure 6-9, respectively.

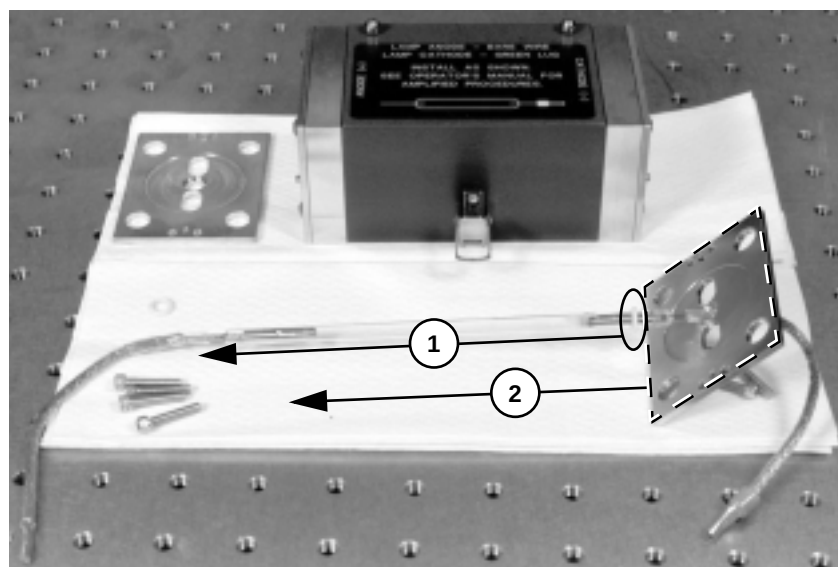


Figure 6-9. Remove the O-ring and the Endplate as Shown

Inserting the New Flash Lamp

Open the box with the replacement lamp and check that the following items are present: 1 Flash lamp, 2 O-rings, 1 O-ring seater and 6 Finger cots.

Please complete, and check off, each step before proceeding.

1. Put on the finger cots supplied with the flash lamp.
2. Take the flash lamp out of the plastic bag.
3. Thread the lug-free end through the right end plate. The side on which the lamp hole is beveled goes against the POT, Figure 6-10.
4. Thread one O-ring over the left lead and slide it across the lamp to a position about 5 mm from the right end of the glass envelope, Figure 6-10. Use the piece of plastic tubing supplied with the flash lamp to push the O-ring across the lamp envelope.
5. Slide the lamp through the POT.
6. Align the plate and POT side slots. Screw the right endplate back on, finger tight.
7. Use the plastic tubing to push the other O-ring into place on the left side, Figure 6-11.
8. Put the left endplate back in place, with the slot on the side of it aligned with that on side of the POT.
9. Center the lamp in the POT such that an equal amount protrudes on each side.

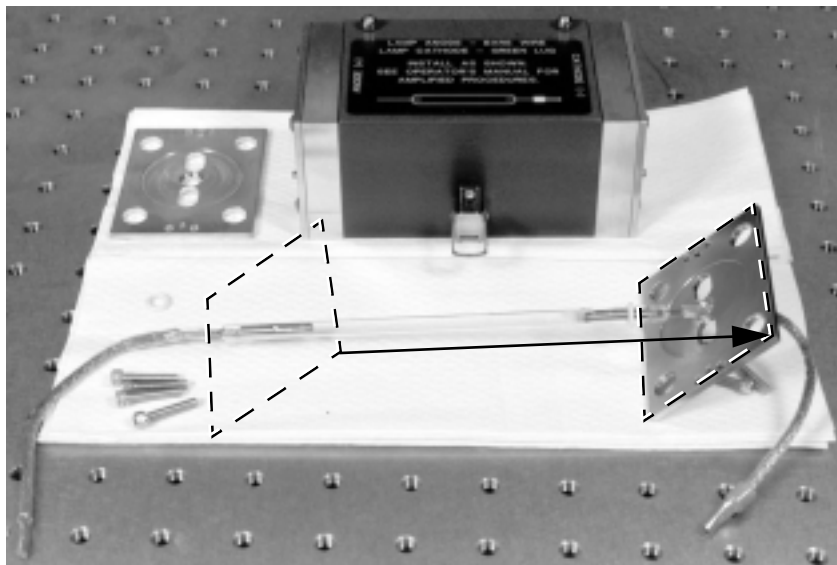


Figure 6-10. Position the Endplate and the O-ring as Shown

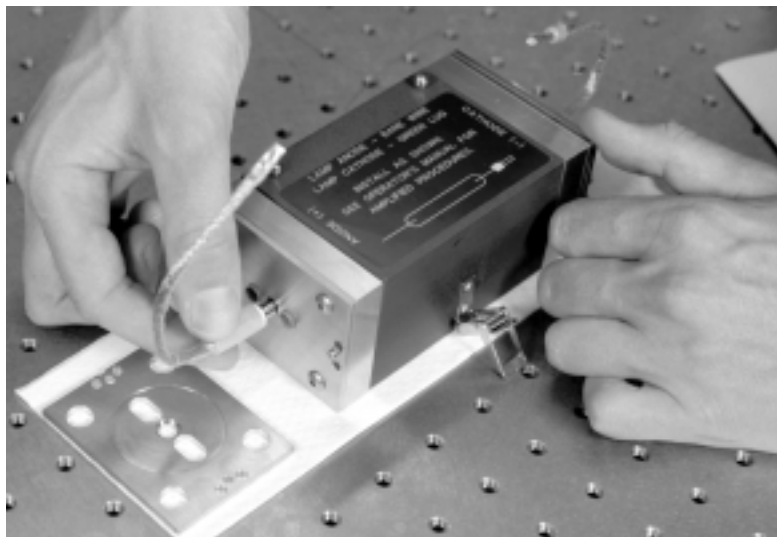


Figure 6-11. Position the Second O-ring as Shown Using the Plastic Tubing Supplied with the Flash Lamp

10. Tighten the screws crosswise, on both sides.
11. Put the POT back into the laser head. Close the latches and make sure that they are properly secured before inserting the dowel pins.
12. Push the wires into the terminal box as far as possible and tighten the 3/16" nuts firmly.
13. Put the lamp terminal box cover back on.
14. Turn on the water pump and check for leaks before starting the laser.
15. Increase the power slowly and, if necessary, minimize the depolarized energy. See Chapter Four, procedure 4, for additional information.

Replacing the EPROM

The Infinity software comprises two separate programs:

1. The Control Program, which is stored permanently on an EPROM inside the power supply and
2. The Graphical User Interface program, INFINITY.EXE, that runs on a standard IBM-style personal computer (PC).

The Control Program is responsible for the management of all low-level hardware signals, fault detection, pulse energy monitoring and servo systems. It is important to realize that all direct access to the hardware is done by the Control Program stored in the EPROM. If the communications fail between the power supply and the PC, the laser remains in a safe operating state since the Control Program knows how to "take care of itself."

The User Interface program accepts input from the user in order to change the laser's state. It also displays diagnostic and status information about the laser. Its primary purpose is to interact with the user; it has no ability to exercise direct control over the laser hardware.

When the control program is upgraded it may be necessary to replace the EPROM in the power supply. Use the following procedure to do this.

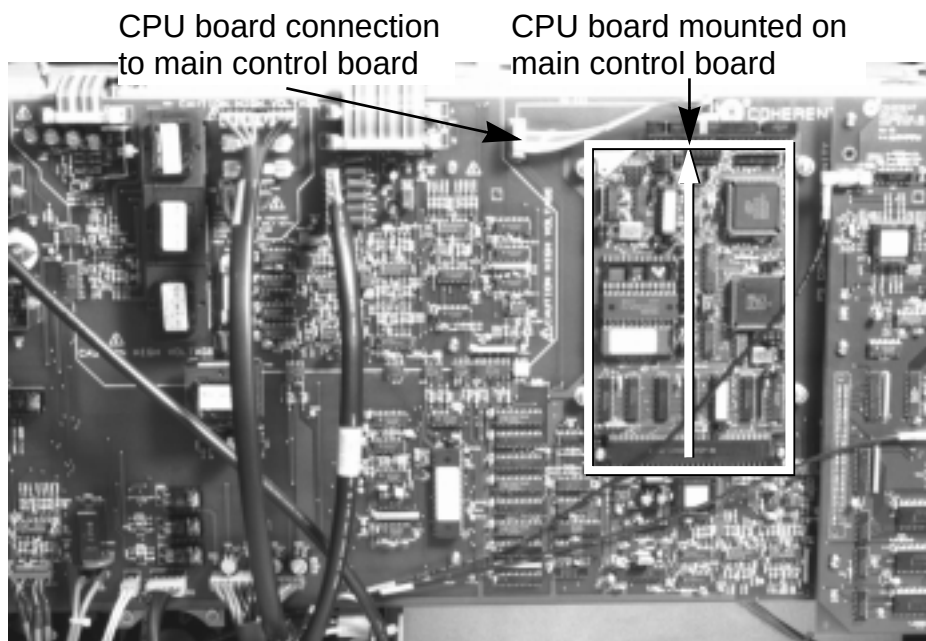
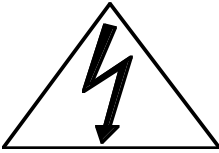


Figure 6-12. Side View of Control Board and CPU Board in Power Supply

1. Turn “off” the power supply, open the breaker on the back of the power supply, and switch off the main facility circuit breaker.



It is important to note that there is 3-phase power on the power supply control board even when the key switch is in the “off” position. Therefore, to minimize the risk of electrical shock, the fused breaker on the back of the Infinity power supply should be opened before the protective cover is removed.

2. Exit the Infinity program and turn off the computer.
3. Remove the smaller of the two power supply covers.
4. Locate the CPU board on the large control board, Figure 6-12. Disconnect the cable from the control board and slide the CPU board up and out of the power supply.
5. Carefully note how the old EPROM is mounted in its socket, Figure 6-13. The notch on the EPROM should match up with the drawing on the board and the EPROM must be positioned as far towards the left as possible in the socket.

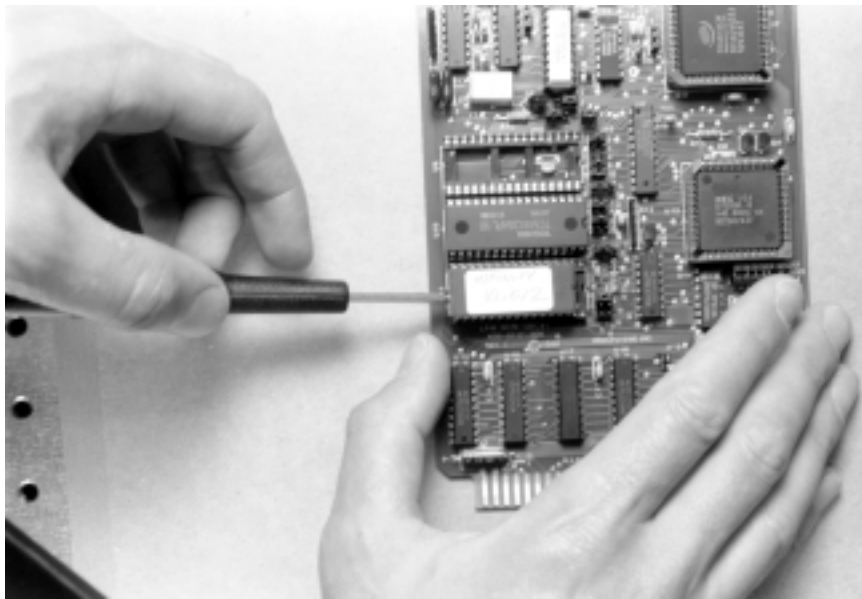


Figure 6-13. Removing the EPROM from the CPU Board

6. Remove the old EPROM using e.g. a small screwdriver as shown in Figure 6-13.

7. Insert the new EPROM in the socket the same way as the old chip was mounted.
8. Put the CPU board back on the control board. Connect the cable to the control board.
9. Put the power supply cover back on.
10. The EPROM replacement procedure is now completed and the laser is ready to be turned on again.

Lamp Timing

The Infinity lamp flash pulse width, and relative DPSS Q-switch pulse position, are set at the factory so that the laser performs optimally. Under normal usage of the laser there should be no need to change them.

The following procedure describes how to restore these parameters if they are lost. Note that these parameters can also be found on the software floppy disks and the final test “performance” sheet delivered with the laser system.



Modifying the parameters contained in the lamp timer window will affect laser performance. Deviation from factory settings may result in lower output power and unstable DPSS laser performance.

Due to internal delay in the large transistor that discharges the capacitor bank through the flash lamp, the set and the actual lamp pulse width differ, see Figure 6-14. For precise timing control an oscilloscope with one channel connected through inductive coupling to the “cathode” flash lamp lead and one to a photodiode monitoring the Q-switch pulse, can be used. This setup is illustrated in Figure 6-15.

It is desirable to set the Q-switch delay so that it opens just before the lamp is turned off. It has been found that positioning the Q-switch $\sim 4 \mu\text{s}$ before the falling edge of the lamp current is optimal.

Procedure

1. Open the Infinity head cover and install the interlock defeat. Tape a piece of electrical wire, ~ 8 cm long, in parallel to the white cathode flash lamp cable, Figure 6-15.
2. Click on the “Service...” button in the main window and then select “Lamp” from the Service window menu. In the resulting drop down menu select “Timing...”. Dismiss the warning window which opens when the Timing window is selected.
3. Connect the “Lamp Trig Synch” BNC output, on the front panel of the Infinity power supply, to the trigger channel of the oscilloscope. Set the scope trigger channel for 50Ω termination and to trigger off the falling edge of the power supply trigger signal.

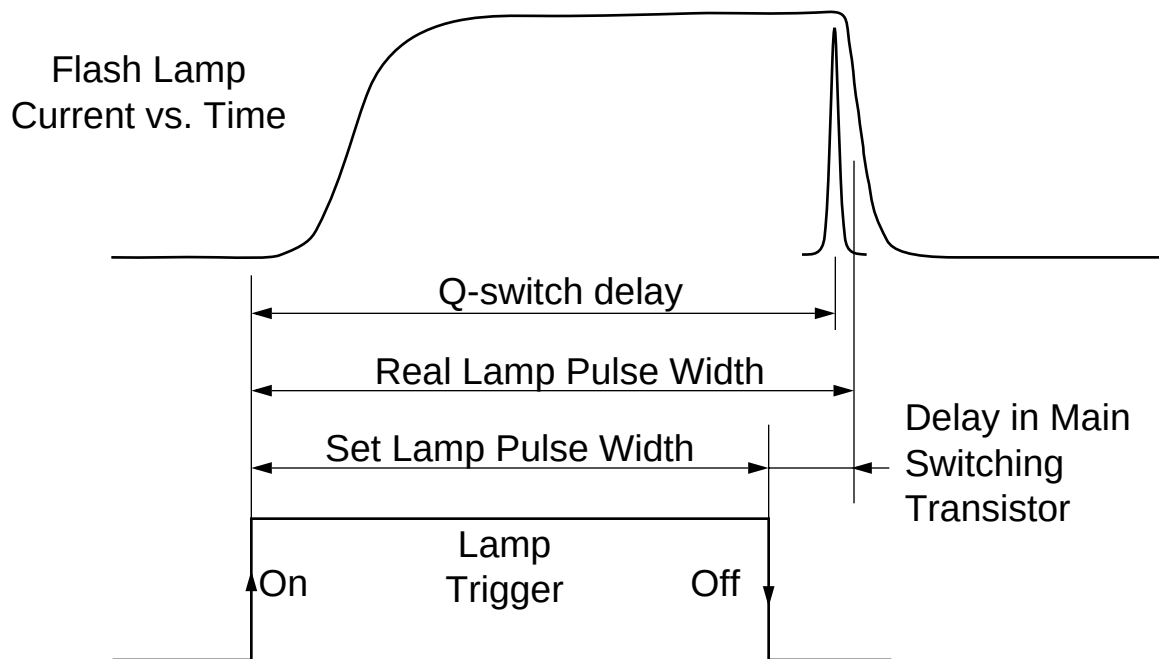


Figure 6-14. Timing of Lamp Trigger, Lamp Current and Q-switch.

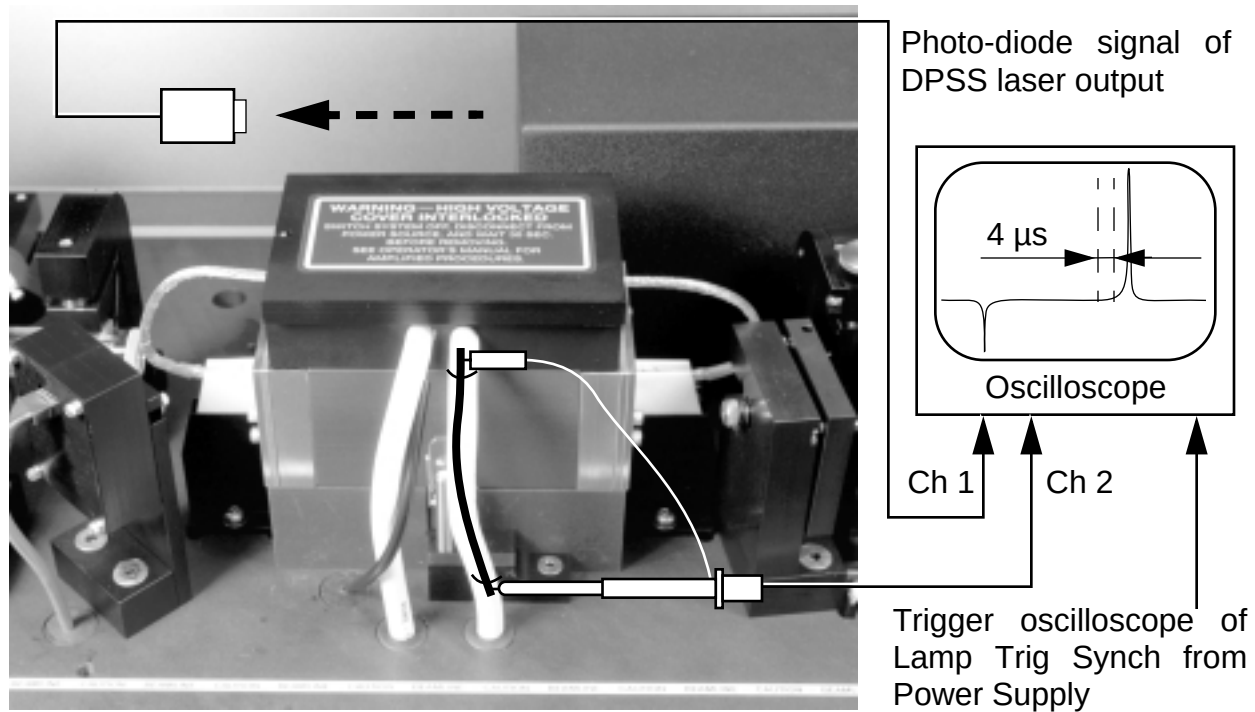


Figure 6-15. Setup to Determine Timing Settings for Lamp Pulse and Q-switch.

4. Connect an 1x oscilloscope probe across the wire attached to the lamp cathode lead. Terminate the probe at 1 M Ω . Place the IR viewing card (or a white business card) between the Faraday Isolator and the Collimating lens to block the DPSS output.
5. Set the system repetition rate to its maximum value and set the IR energy to 1 mJ. Start the Infinity laser system. The solid trace in Figure 6-15 shows the derivative of the lamp current (inductive coupling). This trace should be observed on the oscilloscope.
6. Using the "Lamp pulse width" slide bar of the timing window, set the lamp pulse to 250 μ s (the time between the lamp current turn on and turn off "spikes").



To see the Q-switch pulse on the photodiode, it may be necessary to block the light from the flash lamp.

7. Hook the photodiode to the oscilloscope and slowly zoom in on the lamp turn off spike while looking for the Q-switch signal from the DPSS.
8. Using the Q-switch delay slide bar, position the laser pulse so that it occurs 4 μ s before the lamp current turns off.

After setting the system timing, closing the timing window will automatically save the new parameters.

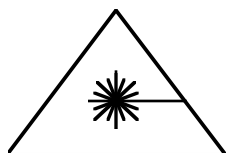
9. Turn off the Infinity laser system and remove the beam block from between the Faraday Isolator and the Collimating lens. Remove the 1x oscilloscope probe and the photodiode if it was placed inside the laser head. Remove the interlock defeat and close the Infinity head cover. The system should now be ready for use.

Safety and Diagnostic Equipment

The following is a list, including descriptions, of equipment which would prove to be helpful in the maintaining of the Infinity laser system.

Power Meters

An external power meter for the Infinity system should be able to handle an average power of 50 W and the high peak powers associated with pulsed Nd:YAG lasers. Be aware that a power meter can break even when exposed to intensities less than the specified damage threshold if not handled correctly. If suddenly exposed to a high power laser pulse it might break from the thermal shock. The recommended procedure is to position the meter in the beam at low power and then slowly turn up the power. If a power meter is available but does not have the necessary damage threshold, either a fraction of the beam can be split off using a glass slide or the beam can be expanded before entering the power meter. If the latter option is chosen it is advisable to use a negative focal length lens since this will avoid bringing the beam to a focus.



For some power meters the sensitive surface is highly reflective. Be aware of where the reflected beam is going, when positioning a power meter or an optical component in the beam path!

It is a good habit to keep a log book in which the laser power, lamp voltage and burn patterns of the fundamental and the harmonics, measured at regular intervals and after any changes, such as replacing the flash lamp, have been made to the system, are recorded. Power meters well suited to work with the Infinity system are sold by Coherent. Call 408-764-4983 or fax 408-764-4800 (within the U.S.) or your local Coherent representative for more information.

IR Viewing Cards

These cards provide an inexpensive means of determining the location of the invisible IR beam from the oscillator and a way to track it through the system. They are not suitable for the IR pulse since the power of this is enough to permanently burn the coating of the card. When using a IR card in the beam, keep it in constant motion to avoid burning it. As with all reflective surfaces put into the beam path, be aware of where the back reflex is going.

IR Viewing Scopes

These instruments usually look and are handled like a small camera. They produce an image in a view-finder that shows both the visible and the near infrared light present. Even though this device is not needed for the maintenance of the system, it is useful for making sure that no harmful IR beams are present in the room.

Beam Diagnostic Instruments

In order to characterize the beam and align the laser with high precision it is necessary to have an instrument showing the mode of the beam in detail. Such instruments usually consist of a small CCD camera, a frame grabber card and a computer. If the system is bought as a package it will also come with the software needed to read out the camera and to manipulate the image on the computer. A CCD camera system well suited to work with the Infinity system is sold by Coherent. Call 916-888-5107 or fax 916-885-6039 for more information.

Photo-Diodes

To monitor the temporal behavior of the laser a fast photo diode connected to an oscilloscope is sufficient. The photo diode should have a rise time of less than 1 ns and the oscilloscope a band width of 350 MHz or faster. Be aware that photo-diodes can be very sensitive and may easily be damaged by intense laser pulses. It is often sufficient to just position the photo diode in the vicinity of the laser where there is enough stray light to obtain a signal.

For advice on finding and selecting diagnostic and safety instruments for the Infinity laser system, please contact your local Coherent representative.

OPERATOR'S MANUAL

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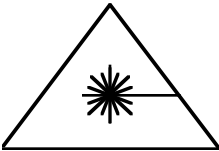
CHAPTER SEVEN SYSTEM SOFTWARE

Introduction

This chapter provides an overview of the Infinity laser system Graphical User Interface (GUI). It is divided into three major sections. The first section contains general interface information, including computer hardware requirements. Section two presents a detailed description of the interface component windows. The chapter concludes with a discussion of the Infinity Command Language. It is advisable to read the first two sections of this chapter before attempting to start the laser for the first time.



The reader is directed to the “readme.txt” file and the “Doc” folder, located on the Infinity CD, for software release notes and installation instructions.



By manually increasing the flash lamp voltage the Infinity system can deliver output energies significantly higher than it is specified to provide. Doing this will, however, permanently damage optical components. Do not run the system at higher energies than what the system is rated for.

General Interface Information

The Graphic User Interface (GUI) controlling the Infinity laser system is divided into five main windows. When the program is started only the Infinity Control window is initially displayed. The other windows, Infinity Application Interface, Infinity System Monitor, Infinity Service and Infinity About are accessed by clicking on their respective buttons, “App...”, “Monitor...”, “Service...”, and “About...” in the main control window. The organization of these windows is shown in Figure 7-1.

Even though all of these windows can be open at the same time only one can be active at a time. The information displayed is continuously updated in both the active and the inactive windows. A window is made active by positioning the mouse cursor anywhere on the window and depressing the left mouse button. In this manual the term “clicking” is used to refer to this maneuver. Clicking on a button such as “Off” while the window in which it is located is inactive will not only activate the window, it will also perform the action attached to the button. Just as with any Windows program the Infinity windows can be moved on the desktop and resized. This is particularly useful when multiple windows are open.

Figure 7-2 shows general controls available in every Windows window. These are used to resize, move, open and close windows. They are accessed by clicking on the symbol in the upper left corner. A window can also be minimized, maximized, and closed by clicking on the icons in the upper left corner of each window.

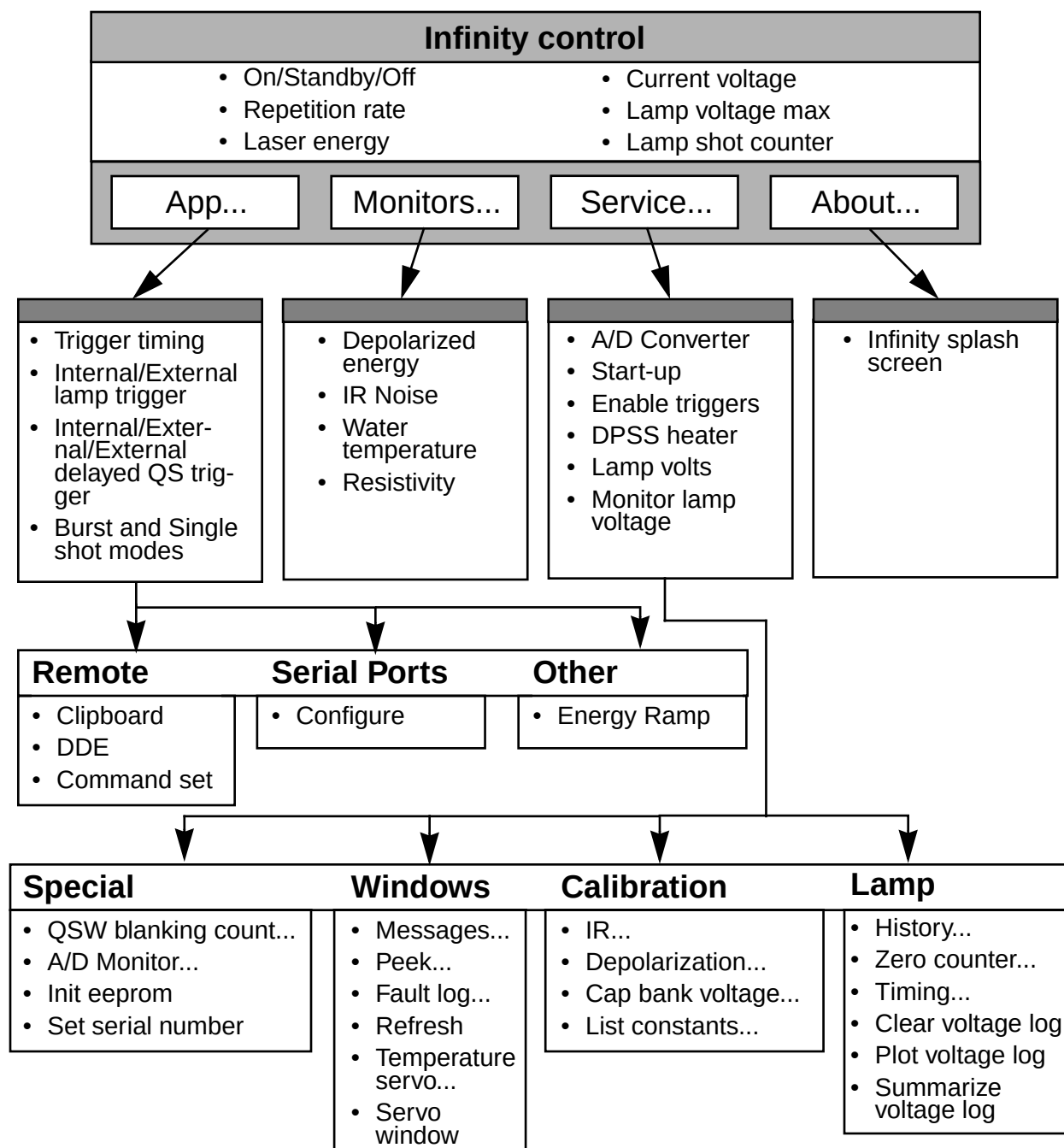


Figure 7-1. Infinity GUI Window Hierarchy.



Figure 7-2. General Window Controls.

Menu Bars

Some of the Infinity windows have a menu bar, see for example Figure 7-3. The “drop down” menus are activated clicking on them or by depressing the “Alt” key on the keyboard and simultaneously depressing the underlined letter. Once a drop down menu is activated the desired function can then be chosen in the same manner. Depressing the “Esc” key will deactivate the window menu.

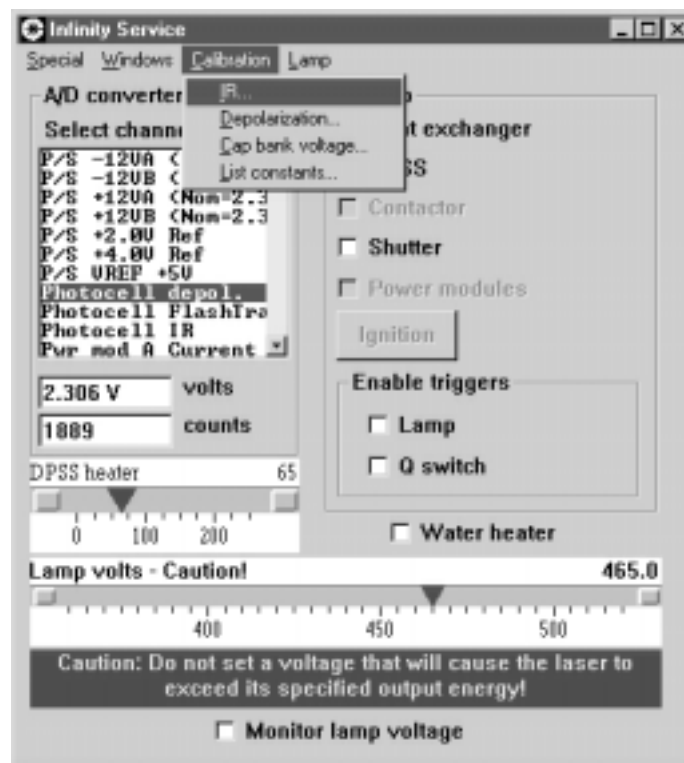


Figure 7-3. Menu Available under Calibration in the Service Window.

Slide Bars

Most parameters in the Infinity windows are controlled through the use of slide bars such as the one shown in Figure 7-4. This particular panel is taken from the main control window and it is used to set the repetition rate. It can be done in three different ways. The numbers given below are only valid for this particular panel. In other panels the parameter value will change by different amounts when the same operation is carried out. The relationship between the magnitude of the changes is roughly the same.

1. Positioning the mouse cursor on the triangular indicator, item 1 in Figure 7-4, and depressing the mouse button allows the indicator to be dragged along the slide bar with a resolution of 1 Hz.
2. Positioning the mouse cursor on the area to the left or right of the indicator, item 2 in Figure 7-4, and clicking on the mouse button will increase or decrease the repetition rate by 10 Hz. If the repetition rate is set to a non-integer number it will change to the nearest even multiple of 10. Holding the mouse button down will move the repetition rate rapidly in steps of 10 Hz.
3. Positioning the cursor on the left or right box, item 3 in Figure 7-4, at the end of the scale and clicking on the mouse button will increase or decrease the repetition rate by 0.1 Hz. Holding the mouse button down will move the repetition rate rapidly in steps of 0.1 Hz.

The number set is displayed in the upper right corner.

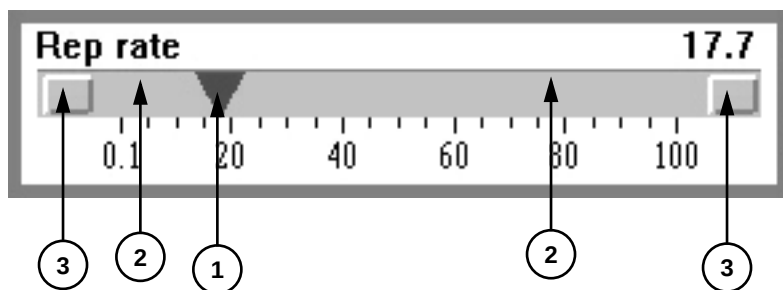


Figure 7-4. Active Areas in Infinity Slide Bar.

Computer Hardware Requirements

The Infinity software is designed to run under the Microsoft® Windows 95 and NT operating systems on an IBM compatible computer. In order to install and run the Infinity software the computer system should meet the following hardware requirements:

- Minimum of a 486/DX4 processor. Although Windows will run on such a computer, a Pentium processor is strongly recommended.
- A minimum of 16 megabytes (MB) of RAM.
- A minimum of 30 MB free hard disk space.
- A CD ROM drive.
- A Microsoft® compatible mouse or trackball. The program uses a Graphical User Interface (GUI) which can only be controlled by the use of a mouse.

Program Windows

The following section presents each of the windows, as well as the functionality of each of the window controls, in the Infinity GUI.

Main Control Window

This window is automatically displayed when the program is run. It is used for switching on and off the laser, for setting the repetition rate and output energy, and for accessing all other windows. It also contains a status display and two shot counters for the flash lamp, one which can be reset and one which is accumulative. The content of this window is described in Table 7-1. Termination of the program is achieved by either pressing Alt and F4 at the same time or clicking on the symbol in the outer right corner. Note that the laser system serial number is displayed in the title bar.

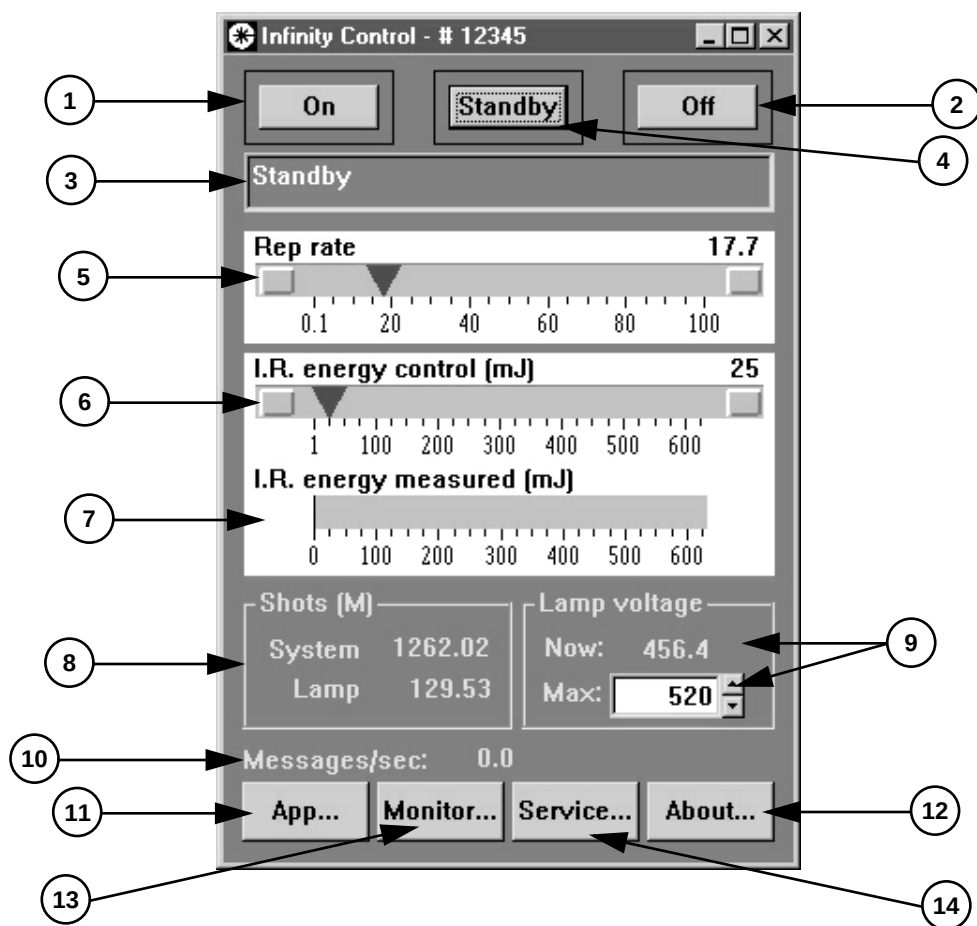


Figure 7-5. Infinity Main Control Window.

Table 7-1. Functions in the Infinity Main Control Window.

#	CONTROL	FUNCTION
1	On	Pressing the On button initiates a chain of events that after a ~5 sec. delay starts the laser. The laser will lase at the repetition rate and power level set on the slide bars in panels 5 and 6, respectively. The On button can be pressed directly if the laser is turned off; it is not necessary to go to standby mode first.
2	Off	Pressing the Off button will shut down all sub-systems. Lasing is terminated immediately.
3	Display	System status messages are displayed in this window.
4	Standby	Pressing the Standby command will turn on the water pump, master oscillator (DPSS), and simmer the flash lamp (assuming that the laser is Off). In Standby mode all systems are active except that the flash lamp and Q-switch triggers are disabled.
5	Repetition rate panel	The repetition rate of the laser is set in this panel. It can be done in three different ways. 1. Positioning the mouse cursor on the triangular indicator and depressing the mouse button allows the indicator to be moved along the slide bar with a resolution of 1 Hz. 2. Positioning the mouse cursor on the area to the left or right of the indicator and clicking on the mouse button will increase or decrease the repetition rate by 10 Hz. An exception to this occurs if the repetition rate is a non-integer multiple of 10, e.g. 83 Hz or 20.1 Hz. Clicking in either area will then change the repetition rate to the nearest higher multiple of 10, i.e. 83 Hz changes to 80 Hz or 90 Hz and 20.1 Hz changes to 20 Hz or 40 Hz. 3. Positioning the cursor on the left or right box at the end of the scale and clicking on the mouse button will increase or decrease the repetition rate by 0.1 Hz. When external lamp triggering is chosen the color of the slide bar changes from white to grey.
6	Pulse energy panel	In this panel the pulse energy of the laser is set. It can be done in three different ways. 1. Positioning the mouse cursor on the triangular indicator and depressing the mouse button allows the indicator to be moved along the bar with a resolution of 5 mJ. 2. Positioning the mouse cursor on the area to the left or right of the indicator and clicking on the mouse button will increase or decrease the pulse energy by 25 mJ. 3. Positioning the cursor on the left or right arrow at the end of the scale and clicking on the mouse button will increase or decrease the pulse energy by 1 mJ. The number set is displayed in the upper right corner. Setting the output energy in this panel automatically enables the LightLoop servo. If the lamp voltage is set manually in the service window Light Loop is deactivated, the pulse energy is set to 0 mJ in panel 6 and the color of the slide bar changes from white to grey.

#	CONTROL	FUNCTION
7	Energy meter	Displays the infrared energy as measured by the built-in energy meter. It can be different than the set energy, displayed in panel 6. A calibration window is available in the menu bar in the Service window, see Chapter Four for IR calibration details.
8	Lamp shot counter	System. Number of shots the system has fired measured in 10E6. This counter can not be changed by the user and is unaffected by any changes in the system. Lamp. Number of shots the flash lamp has fired measured in 10E6. Reset the counter each time the flash lamp is changed.
9	Lamp voltage	Now: The actual voltage applied to the flash lamp. It is set either manually in the Service window or by the on-board microprocessor. In the latter case the processor sets the voltage to what is necessary in order to produce the output chosen in panel 6. In neither case can the voltage be set higher than the lamp max voltage, shown in the same panel. When the difference between actual and max voltage is less than 5 V the background color changes to red. Max: This voltage represents the maximum that can be applied to the flash lamp. Clicking on the up or down arrow will increase or decrease it by 5 V. Minimum setting is 500 V and maximum 570 V.
10	Messages/sec:	The average number of messages transferred over the serial cable between the power supply and the computer.
11	App...	Opens a window in which 1. the timing of two independently variable synchronization signals can be set (the pre-trigger and the high precision trigger). 2. external or internal triggering of the flash lamp and the Q-switch is chosen. See Table 7-3. 3. the delay time for internal delayed trigger mode is set. See Table 7-3. 4. the burst and single shot modes of operation can be selected. 5. the COM ports and communication with another computer are configured.
12	About...	Displays the Infinity "Splash". Gives the software version number and copy right information.
13	Monitor...	Opens a window containing bar graphs displaying depolarized energy, pulse energy noise, water temperature, and water resistivity.
14	Service...	Opens the Infinity Service window. Sub-systems can be switched on and off separately, the lamp voltage and the DPSS temperature can be set manually and various test point values read out.

Timing, Triggering, Burst and Single Shot modes

This window is accessed by pressing the “App...” button in the main control window. The controls in it allow you to:

- Choose between internal and external triggering of the flash lamp and the Q-switch
- Set the timing of the two outgoing synchronization pulses, available on the power supply
- Set the internal delay when using the laser in “External delay” mode
- Set the system to provide a single shot or “n” number of shots on demand
- Configure the external communication settings of the system

Resizing this window also resizes the timer panels. To set the timers with optimum precision enlarge the window by clicking on the center window symbol in the upper right hand corner. See Chapter Four, Daily Operation, for details about timing and triggering.

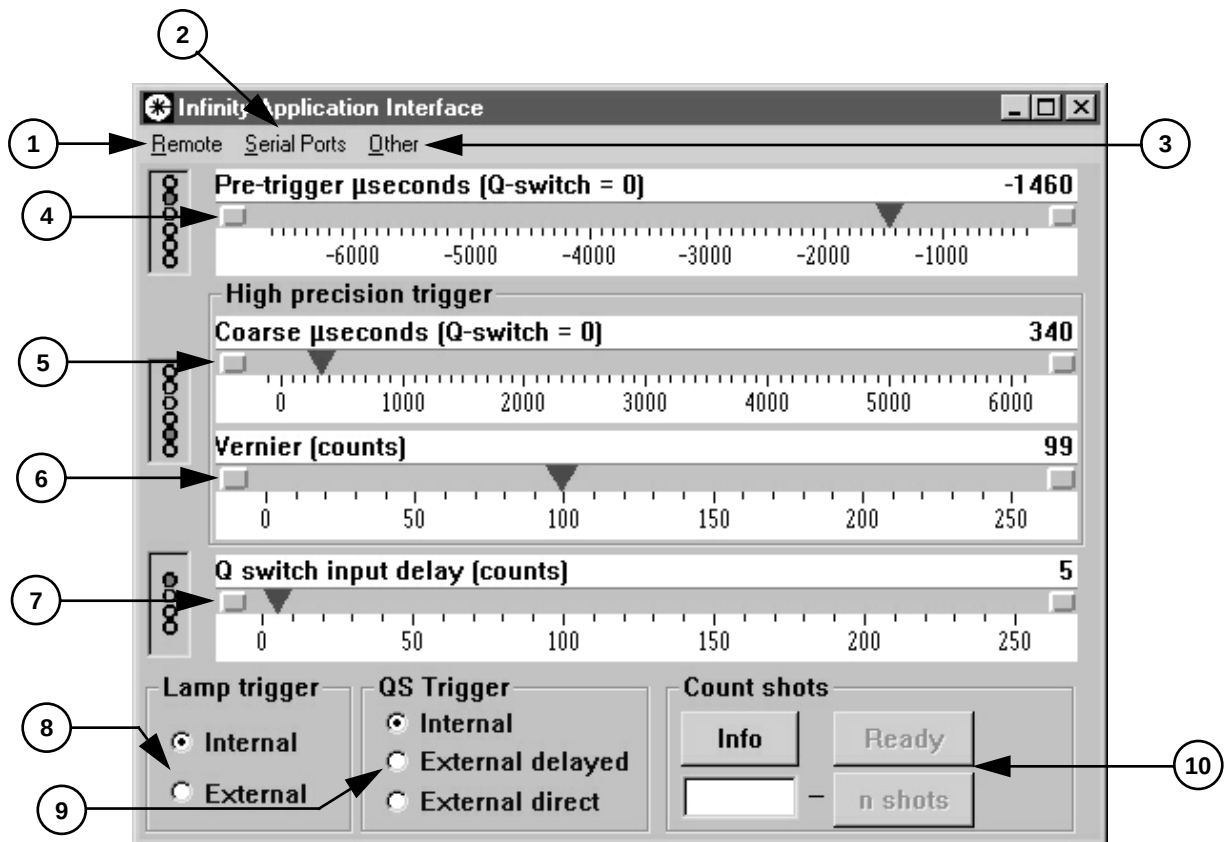


Figure 7-6. Timing and Triggering Window.

Table 7-2. Functions in the Timing and Triggering Window.

#	CONTROL	FUNCTION
1	Remote	Clipboard Sends the content of a pasted clipboard as commands to the laser. The clipboard can be created in any word processor or text editor. All commands are processed sequentially. DDE (Dynamic Data Exchange) Toggles between DDE on and off. In the On mode the Infinity system responds to DDE commands. Command set Opens a new window in which all commands available for remote control of the laser are listed.
2	Serial ports	Configure Opens a new window in which • The COM port that the laser is controlled from can be chosen • The communication protocol and which COM port the remote is connected to can be set. See Figure 7-7 and Table 7-3.
3	Other	Energy Ramp Opens a new window in which the Infinity laser system can be programmed to ramp its IR energy. The user can set the start and stop energies, as well as the time of the total ramp and the dwell time at the final energy. See Figure 7-8 and Table 7-4.
4	Pre-trigger	Programmable outgoing trigger referenced to the Q-switch trigger. It can precede the Q-switch trigger by between 6.77 ms and ~240 μs. Available at the LAMP TRIG SYNCH BNC on the power supply.
5	High precision trigger	Programmable outgoing trigger referenced to the Q-switch trigger. It can precede the laser pulse by a maximum of 150 μs and supercede it by a maximum of 6.15 ms. Available at the VAR QS SYNCH BNC on the power supply.
6	Vernier	High-resolution control of the outgoing Q-switch sync trigger. One count corresponds to ~0.5 ns. The total range is ~100 ns.
7	Q-switch input delay	This control allows the user to delay the externally supplied Q-switch sync pulse before it is allowed to trigger the Q-switch. One count corresponds to ~0.1 ns. The total range is ~25 ns.
8	Lamp Trigger	These buttons toggle between Internal and External triggering of the flash lamp. Internal. The laser provides the flash lamp trigger pulse. External. The laser waits for a user-supplied TTL trigger to the Lamp Trig BNC on the power supply. Triggering occurs on the falling edge of the pulse.

#	CONTROL	FUNCTION
9	QS Trigger	This button toggles between Internal, External and External delayed triggering of the Q-switch. Internal. The laser provides the Q-switch trigger. The timing of it is set in the Lamp timing window available from the Service window. External. The laser waits for a user-supplied TTL trigger to the QS Trig BNC on the power supply. Triggering occurs on the falling edge of the pulse. External delayed. Offers the possibility of delaying the external trigger before it triggers the Q-switch. The internal delay time is set using the slide bar (7) described above.
10	Count shots	These buttons allow you to fire a singleshoot or to program the laser system to provide “n” number of shots. The labels presented on these buttons will depend on the EPROM loaded into the power supply. If a v1.022 EPROM is not used, the buttons will appear as indicated in Figure 7-6 and the software option will not be implemented. If a v1.022 EPROM is used, the buttons will be labeled as indicated below (arrow indicates label change for proper EPROM): Ready or Standby: Pressing of this button places the laser system in “armed” mode. Info → 1 shot: Pressing this button causes the laser to fire 1 shot at the pulse energy set in the main control window. n shots. Pressing this button causes the laser to fire “n” shots, set in the input box below the 1 shot button, at the pulse energy and repetition rate set in the main control window.

Serial Port Configuration

The controls in this window allow the user to configure the external communication parameters of the laser system. See the end of this chapter for additional information on remote controlling the Infinity laser system.

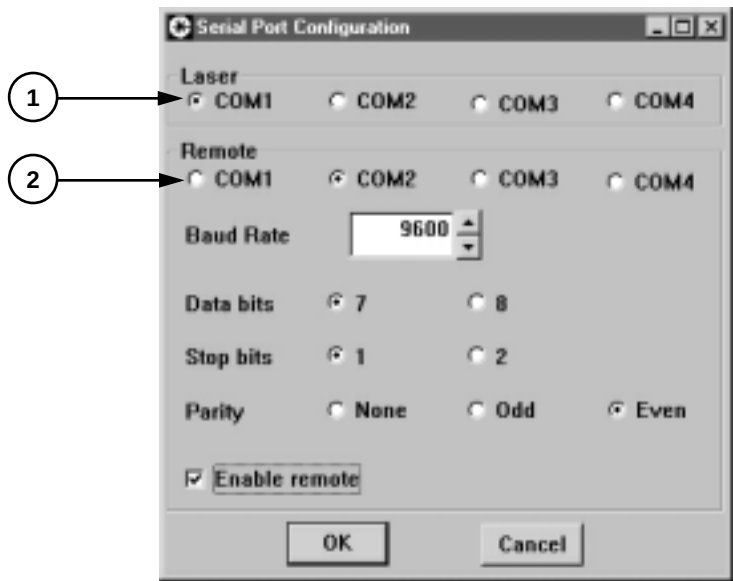


Figure 7-7. Serial Port Configuration Window.

Table 7-3. Functions in the Serial Ports Window.

#	CONTROL	FUNCTION
1	Laser	COM Selects the COM port to which the power supply is connected.
2	Remote	COM Selects the COM port to which the remote computer is connected. Baud Rate Selects the Baud rate used for communication between the remote and the PC. Data bits Selects the number of data bits used for communication between the remote and the PC. Stop bits Selects the number of stop bits used for communication between the remote and the PC. Parity Selects the parity used for communication between the remote and the PC. Enable remote When this box is checked the Infinity PC is controlled from the remote PC.

Energy Ramp

The controls in this window allow the user to configure the Infinity so that it ramps its IR energy.

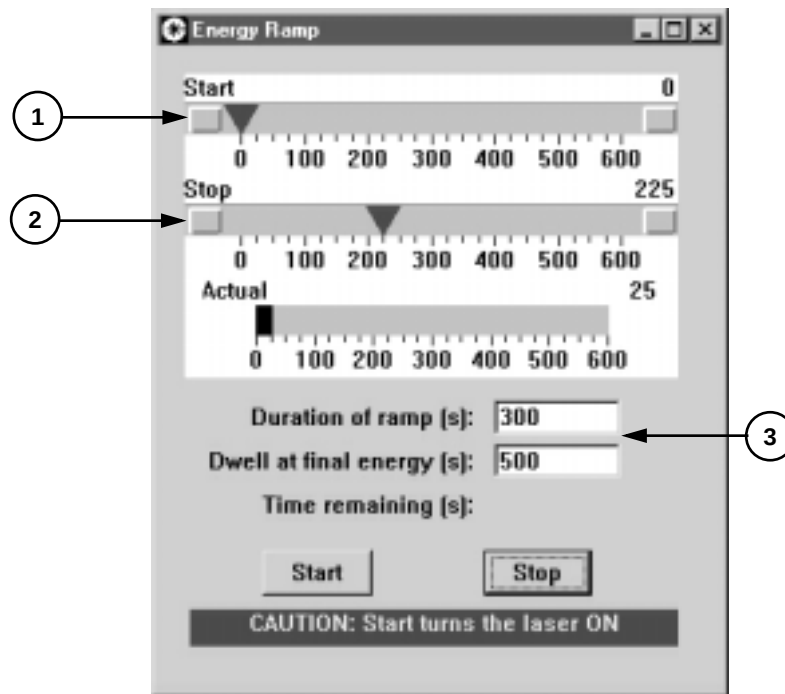


Figure 7-8. Energy Ramp Window.

Table 7-4. Functions in the Energy Ramp Window.

#	CONTROL	FUNCTION
1	Start	Allows the user to select the start energy for the ramp.
2	Stop	Allows the user to set the end energy for the ramp.
3	Duration and Dwell Times	Allows the user to set the total time, in seconds, for the energy ramp as well as the time spent at the "stop" energy before the laser is switched to Standby.

Monitor

This window is accessed by pressing the “Monitor...” button in the main control window. It contains four bar graphs, shown in Figure 7-9, displaying depolarized energy, system energy IR noise, water temperature and water resistivity.

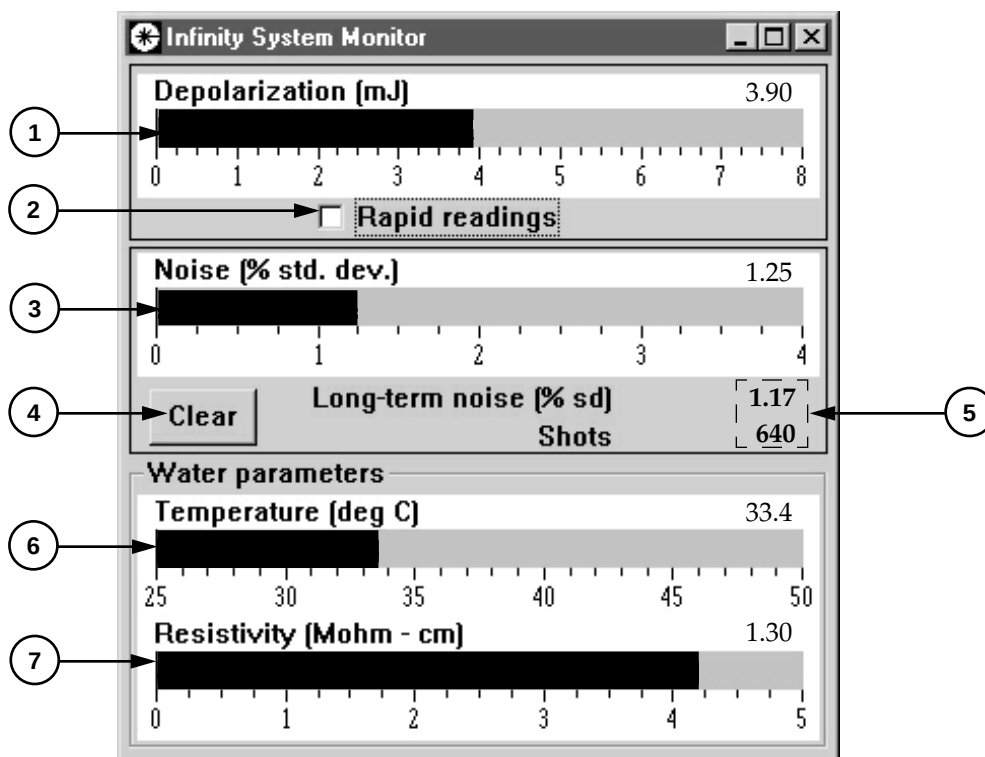


Figure 7-9. Infinity Monitor Window.

Table 7-5. Functions in the Infinity Monitor Window.

#	CONTROL	FUNCTION
1	Depolarization	The laser radiation that after amplification, instead of being reflected of the thin-film polarizer, is transmitted through it and directed back into the Faraday isolator is called depolarized energy. A calibrated photodiode detects this energy and the result is displayed in panel 1. If the depolarized energy is above 6 mJ/pulse see Procedure 4 in Chapter Four, Daily Operation. See Procedure 11 for the calibration procedure.
2	Rapid readings	Activating rapid readings stops the transfer of noise data and uses the freed capacity to speed up the transfer rate of depolarized energy data. This mode of operation is intended to be used when the laser controls are adjusted to minimize the depolarized energy.
3	Noise	The measured standard deviation of the shot-to-shot fluctuations of the IR energy is displayed in panel 3. The noise is only specified for the system when it is operated at maximum energy. If the system is found to be noisier than specifications see Procedure 3 in Chapter Four, Daily Operation.
4	Clear	Clears the long term noise data and restarts the measurement.
5	Long term noise	Shot-to-shot intensity noise calculated from the number of shots shown. After the shot counter has reached 640 the noise data represents a moving average of the most recent 640 shots.
6	Temperature	Temperature of the water in the internal cooling loop. It is measured by a thermocouple located in the reservoir. If the temperature rises above 45°C the laser is turned off.
7	Resistivity	Resistivity of the water in the internal cooling loop. The resistivity should be higher than 3 MΩ/cm. Replace the filter when the resistivity falls to or below 2 MΩ/cm with the adjustment valve fully open. If the resistivity falls below 1.5 MΩ/cm the laser will not operate.

Service Window

This interface is accessed by pressing the “Service...” button in the main interface. It allows the user to switch on and off different components of the system individually, monitor system test points, set the master oscillator temperature and flash lamp voltage. Setting the lamp voltage manually in this window automatically disables flash track and sets the energy read out in the main control window to zero.

In order to access this window the test file “Service.txt” must be present in the Infinity directory. If this file is not present the message “Special authorization is needed to access the Service Window!” will be displayed. The content of the Service.txt file is unimportant.



By manually increasing the flash lamp voltage, output energies significantly higher than specified for the system can be produced. Doing this will permanently damage optical components. Do not run the system at higher energies than what the system is rated for.

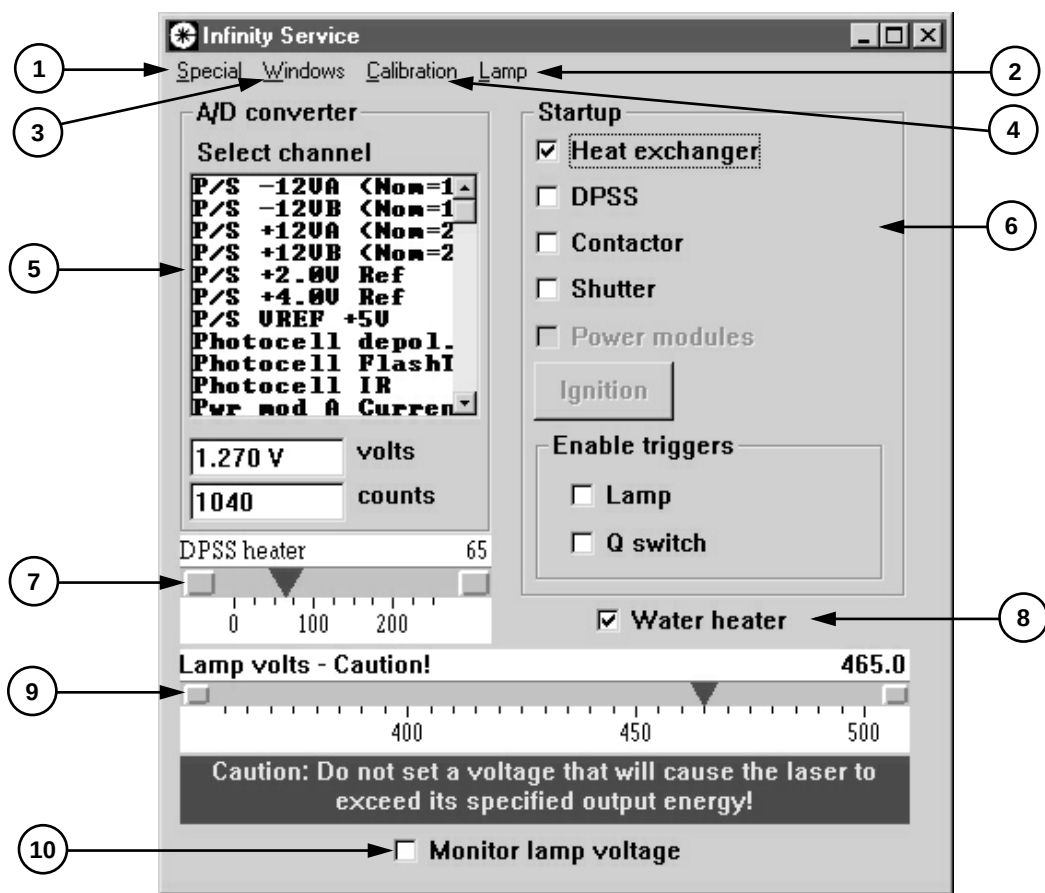


Figure 7-10. Infinity Service Window.

Table 7-6. Functions in the Infinity Service Window.

#	CONTROL	FUNCTION
1	Special	QSW blanking count... Sets the number by which the flash lamp repetition rate will be divided to yield the repetition rate of the laser output. The default value is 1. A/D Monitor... Allows a A/D channel value to be averaged. Parameter is selected from the list in the window. The default cycle value is 50. Init eeprom After a new eeprom has been installed in the power supply this program is run to initiate it. The laser will automatically tell you if it is necessary. Set serial number... Writes the laser systems serial number to the EPROM.
2	Lamp	History... Returns a log of the number of shots fired by each flash lamp used in the laser. Zero counter... Resets the flash lamp shot counter to zero. Reset it after every flash lamp change. Timing... See Figure 7-12 and Table 7-8. Clear voltage log Restarts the collection of lamp voltage data. Plot voltage log Returns a plot of lamp voltage as a function of time. Summarize voltage log Opens a window displaying the lamp voltage change as a function of lamp shots.
3	Windows	Messages... Returns a log of all messages relating to changes in operating parameters sent between the computer and the laser system. Peek... Returns a listing of the status of all digital control lines. Fault log... Opens a window containing a list of all fault messages displayed since the system was installed. Refresh Updates all information in all windows. Temperature servo... Opens a window in which the parameters for the SHG and the THG temperature servos are set. See Figure 7-11 and Table 7-7. Servo Window... Opens the servo window for FlashTrack. See Figure 7-13 and Table 7-9.

#	CONTROL	FUNCTION
4	Calibration	IR... Opens a window in which the built in IR energy meter is calibrated. See Procedure 10 in Chapter Four for the procedure. Depolarization... Opens a window in which the built in depolarized energy meter is calibrated. See Procedure 11 in Chapter Four for the procedure. Cap bank voltage... Opens a window from which the automatic capacitor bank voltage calibration procedure is initiated. It is important that the laser has been on for at least 10 minutes before this procedure is used. List constants... Lists the constants that are used to compute calibrated values for IR energy, depolarized energy and capacitor bank voltage. This window is for reference only. To change the constants, use the calibration procedures 10 and 11 in Chapter Four.
5	A/D Converter	Clicking on a line in the list provides the test point value of the chosen parameter in the display panel below it. The test point is only read once.
6	Startup	The buttons controlling the following systems are automatically enabled or disabled to prevent unsafe operation. It is thus not possible to start the sub-systems in the wrong order. Heat exchanger. Switches on (checked) or off (blank) the pump that circulates the internal cooling water. DPSS. Turns on the laser diode that pumps the master oscillator. Contactor. Engages the main contactor in the power supply. Shutter. Opens the mechanical shutter, located between the thin film polarizer and the spatial filter, in the laser head. No laser light can be emitted when the shutter is closed. Using the shutter allows the user to eliminate lasing without turning off the flash lamp or the oscillator. Going to Standby mode or disable the Q-switch trigger will also temporarily eliminate lasing. Power modules. Turns on the two booster boards that supply the charging voltage to the capacitor bank. Ignition. Simmers the lamp. If the cover is off (requires the cover interlock to be defeated) this can be seen as a steady intense glow from the lamp. Simmering of the lamp means that an ionized channel has been established between the anode and cathode in the lamp. Lamp. Enables or disables the lamp trigger. Q-switch must be off. Q switch. Enables or disables the Q-switch trigger. The lamp must be turned on before the Q-switch in order for the amplifier to reach partial thermal equilibrium before the Q-switched pulse is injected into it.
7	DPSS Heater	Slide bar used to change the temperature of the master oscillator. Optimum operating temperature is set at the factory. It only needs to be changed if problems with mode jumps occur. See Chapter Four, Procedures 3 and 9. The temperature range is 35°C–55°C.

#	CONTROL	FUNCTION
8	Water heater	Turns on the heater in the water reservoir. This is always done automatically by the system if needed.
9	Lamp volts	Allows the flash lamp voltage to be set manually. Lightloop is automatically disabled. It is enabled by setting the output energy in the main control window (mJ). By manually increasing the flash lamp voltage the Infinity system can deliver output energies significantly higher than what it is specified to give. Doing this will permanently damage optical components. Do not run the system at higher energies than what the system is rated for.
10	Monitor lamp voltage	Toggles between lamp monitor off and on. When On, lamp voltage as a function of time is monitored and stored in a file. The content of the data file can later be plotted using the plot command under the lamp menu title in the Service... window. This feature is useful for tracking lamp aging. Note that this feature is automatically disabled if the Service window is closed.

Crystal Temperature Window

The crystal mounts temperature servo set point is set in the Crystal temperature window. This window is accessed from under the Windows menu in the Service window. The temperature (DAC counts) is factory set.

If the servo after 15 minutes has not been able to regulate the crystal mount temperature it will assume that there are no mounts present and set the drive signal to zero.

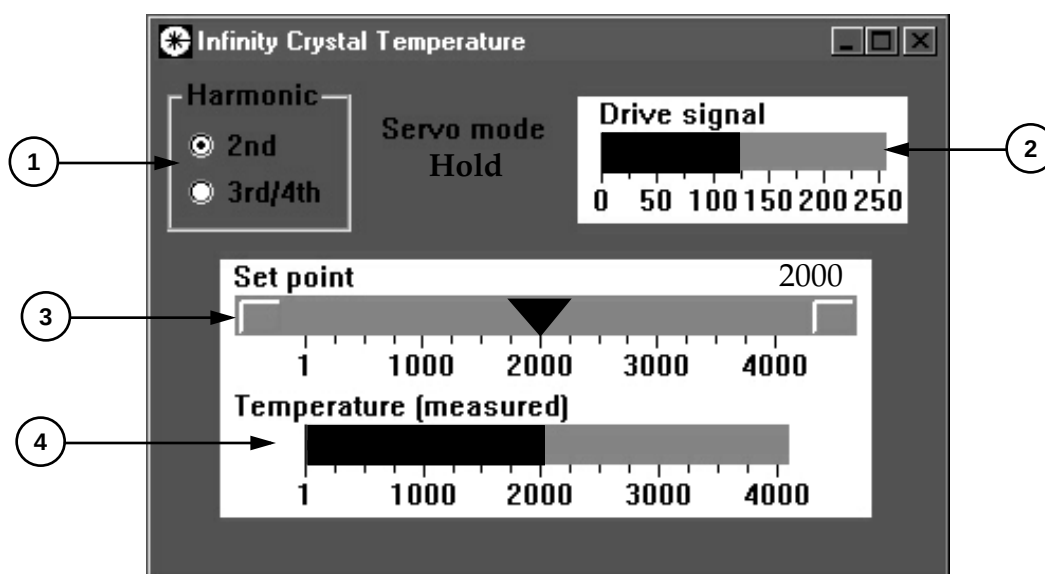


Figure 7-11. Crystal Temperature Window.

Table 7-7. Functions in the Crystal Temperature Window.

#	CONTROL	FUNCTION
1	Harmonic	Selects which mount the window controls addresses.
2	Drive signal	Proportional to the difference between set and actual temperature.
3	Set point	The temperature set point for the crystal mount.
4	Temperature	The actual temperature of the crystal mount. When the servo is working properly the temperature should be equal to the set temperature.

Lamp Timing Window

The controls in the lamp timing window allow the user to customize the flash lamp pulse width and the Q-switch trigger position. They are set at the factory so that the laser performs optimally. Under normal usage of the laser there should be no need to change them. See Chapter Six, Maintenance, for additional information on setting the “System Timing”.



Modifying the parameters contained in the lamp timer window will affect laser performance. Deviation from factory settings may result in lower output power and unstable DPSS laser performance.

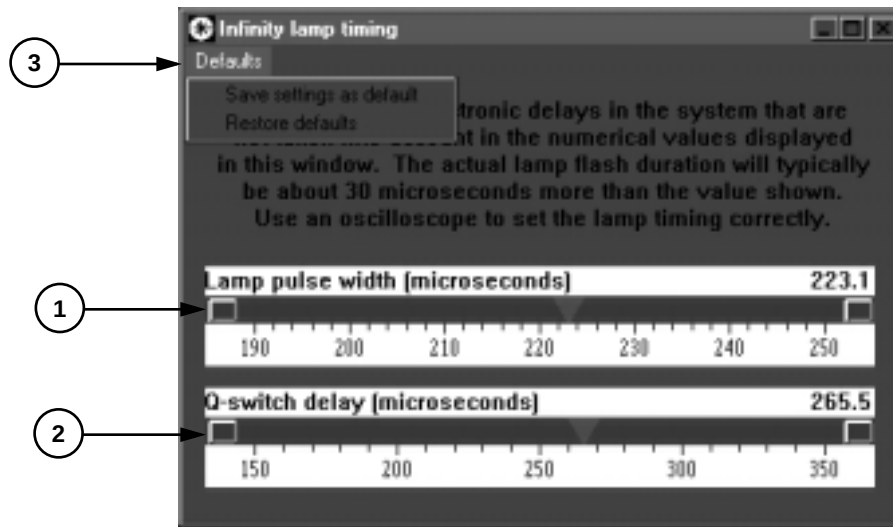


Figure 7-12. Lamp Timing Window.

Table 7-8. Functions in the Lamp Timing Window.

#	CONTROL	FUNCTION
1	Lamp pulse width	Sets the lamp emission time by controlling the duration of the current pulse from the capacitor bank. The width, as measured on an oscilloscope, should be 250 μ s.
2	Q-switch delay	The setting of the Q-switch delay determines when, during the flash lamp emission, the Q-switch is triggered. The optimum setting which is set at the factory, is 4 μ s before the flash lamp is turned off.
3	Defaults menu	Allows the user to reset the timing values to the default values and to save the current values as default values.

FlashTrack Servo Controls

FlashTrack is a control system that reduces shot-to-shot intensity fluctuations. Figure 7-13 gives the control window for this feature. See Chapter Two for additional information on this feature.

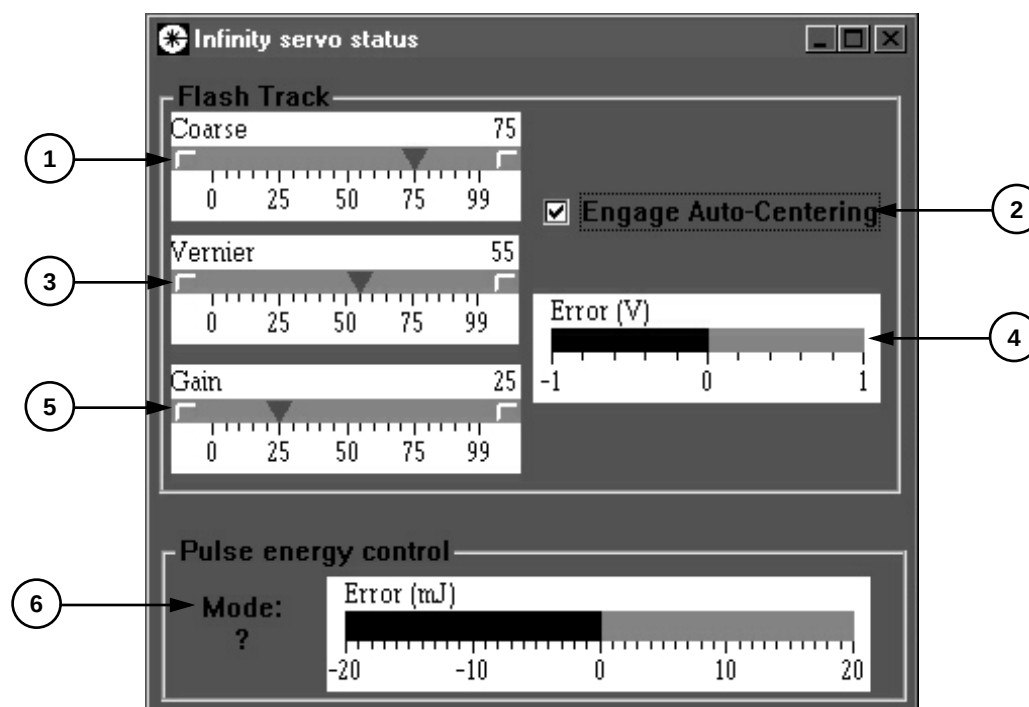


Figure 7-13. Flash Track Window.

Table 7-9. Functions in the Flash Track Servo Window.

#	CONTROL	FUNCTION
1	Coarse	Coarse adjustment for level shift of the photo diode signal. Flash Track can be turned off by setting coarse to zero. It is automatically optimized when auto-centering is engaged.
2	Engage Auto Centering	Automatically optimizes the coarse and vernier settings. Optimum settings yield a photocell Flash Track voltage of close to 2.5 V (center of the A/D converter range). This voltage can also be read out through the A/D converter in the Service window.
3	Vernier	Fine adjustment for level shift of the diode signal. It is automatically optimized when auto-centering is engaged.
4	Error	Voltage proportional to the energy correction applied.
5	Gain	The gain setting determines the amount of energy correction applied for a given error signal. Optimize it by monitoring the intensity noise, panel 3 in Figure 7-9.
6	Pulse energy control	The difference between measured energy and set energy. Not active when the lamp voltage is set manually.

Laser Control Methods

The Infinity control software provides a message language for controlling the laser system without directly using the Graphical User Interface (GUI). There are three ways to use the message language: First, from another Windows application by means of Dynamic Data Exchange (DDE, an operating system feature); second, from an external computer by RS-232; or third, from command scripts pasted to the Windows clipboard. In each case the set of possible messages is the same.

Messages consist of a string of characters from the ASCII character set. When a message is received by the Infinity GUI, it is parsed by a command processor. If the message is found to be syntactically correct it is dispatched to the rest of the application for processing. Most messages will cause a change to the laser state, which will be reflected in the on-screen user interface.



The user interface remains active at all times. It is therefore possible to freely mix mouse input (normal mode of communicating with the GUI) with command processing.

Messages are of three types: command, query, and control. Commands are used to change the laser state by setting parameters such as repetition rate, pulse energy, and timer values. Queries are used to interrogate the laser for status information. A control message is used to terminate a DDE session or list of clipboard commands.

In the following sections the clipboard, RS-232 and DDE methods of communication are discussed.

Infinity Language Commands

Table 7-10 gives a listing of all of the Infinity Language commands. The commands are presented in the first column of Table 7-10. In column two a brief description of the command is given. Note that command arguments are of two types: *float* and *integer*.

Table 7-10. Infinity Command Language.

Command	Description
End	Terminates message exchange. In the case of RS-232, this closes the receiving port. In the case of DDE, this terminates the DDE conversation.
On	Turns the Infinity laser system on.
Off	Turns the Infinity laser system off.
Standby	Places the Infinity laser system in standby mode.
HighPrecCoarse <i>float</i>	Units are microseconds.
HighPrecVernier <i>integer</i>	Units in DAC counts.
Input Delay <i>integer</i>	Sets Q-switch input delay, in nanoseconds.
LampTrigger <i>integer</i>	Sets lamp trigger mode: 0=internal, 1=external.
PreTrigger <i>float</i>	Sets the Applications window Pre-trigger value.
PulseMj <i>float</i>	Sets Infinity IR output energy.
QswTrigger <i>integer</i>	Sets Q-switch trigger mode: 0=internal, 1=external direct, 2=external delayed.
RepRate <i>float</i>	Sets Infinity repetition rate.
qDepol	Returns the depolarized energy in millijoules. If this query is sent when the Monitor window is closed, the window will open automatically. Because of the time required to open the window a reply of zero may occur. On subsequent invocations of this query the reply will be correct.
qFaultN <i>integer</i>	Returns the test of one active fault from the indexed list. The argument indexes into the list; the number of active faults can be obtained by the query "qFaults". The list is 1-based (i.e., the first fault is # 1). If an invalid argument is sent with this query an empty string will be returned.
qHighPrecCoarse	Queries High precision coarse trigger value, see the Applications window. Value returned is in microseconds.
qHighPrecVernier	Queries High precision trigger vernier value, see the Applications window. Value returned is in DAC counts.

Command	Description
qInputDelay	Queries High precision trigger Q-switch input delay value, see the Applications window. Value returned is in DAC counts.
qLampShots	Units are in millions of shots.
qLampTrigger	Returns current lamp triggering state of the Infinity laser system: 0=Internal, 1=external.
qLampVMargin	Returns the difference between the current lamp voltage and the maximum lamp voltage as set in the main control window of the GUI. Units are in volts.
qLampVMax	Returns the maximum lamp voltage as set in the main control window of the GUI. Units are in volts.
qLTNoise	Returns the long term noise as a percent. If this query is sent when the Monitor window is closed, the window will open automatically. Because of the time required to open the window a reply of zero may occur. On subsequent invocations of this query the reply will be correct.
qMjActual	Returns the last measured laser power reading. Units are in millijoules.
qNFaults	Returns the number of active faults, defining the valid range of arguments for the query "qFaultN". If there are no faults active "0" will be returned.
qNoise	Returns the short term noise, based on the last 80 shots, in percent form.
qPreTrigger	Returns the current value for the Q-switch pretrigger. Units are in microseconds.
qPulseMj	Returns the pulse energy set point (NOT the actual laser power which is returned by qMjActual).
qQswTrigger	Returns the current Q-switch trigger mode: 0=internal, 1=external direct, 2=external delayed.
qRepRate	Returns the current repetition rate.
qShots	Returns the total number of shots on the system in millions.
qState	Returns the current state of the laser system in the form of a string which matches the one shown in the display area of the GUI main control window.
qVersion	Returns the software version number shown in the header of the GUI main control window.
qVolts	Returns the actual lamp voltage.

Command	Description
qWaterRes	Returns the water resistivity in megaohm-cm. If this query is sent when the Monitor window is closed, the window will open automatically. Because of the time required to open the window a reply of zero may occur. On subsequent invocations of this query the reply will be correct.
qWaterT	Returns the water temperature in degrees centigrade. If this query is sent when the Monitor window is closed, the window will open automatically. Because of the time required to open the window a reply of zero may occur. On subsequent invocations of this query the reply will be correct.

Sending Clipboard Messages

An easy method for checking command syntax is to send messages to the GUI by means of the Windows clipboard. A command can be placed into the clipboard from any available text editor (the "Notepad" utility found in the Windows Accessories folder works well for this purpose). Once the test editor has been launched type in the command of interest. Then:

1. using the mouse high-light the command
2. select the "Edit" menu item from the text editor main menu. From the resulting popup menu select "Copy".

To direct the Infinity program to read the contents of the clipboard, in the GUI, open the "Applications" window. From the menu bar select "Remote" and then, from the resulting drop down menu, select "Clipboard". If the command syntax is correct the GUI will produce an audible beep and then make the requested change in state. If the syntax of the command is wrong a message box will appear on the screen. The heading of this box will read "Remote syntax error:". The body text of the message box will contain the message "Command X not understood". Where "X" is the complete command as read from the clipboard.

When using this option it is important to keep in mind that the GUI will not provide any feed back other than the syntax error message box. In other words, no response is generated to queries or to command arguments that are out of range.

Sending DDE Messages

The Infinity program is a fully functional DDE server capable of communicating with other Windows applications. This section describes the protocol used and is intended to support application programming. It assumes a familiarity with Windows DDE concepts and terminology. For further information, see the Microsoft Windows Software

Development Kit (SDK) help or an equivalent reference source.

The Infinity program acts as a DDE Server supporting a single application/topic pair:

Application:'Infinity'

Topic:'Control'

To start the Infinity DDE server, open the Application window and select the DDE item from the Remote pull-down menu. Before commands can be sent to Infinity, the client program must obtain the window handle of the Infinity DDE server. This is accomplished by means of a **WM_DDE_INITIATE** message containing the application/topic pair ('Infinity/'Control') just given.

Once communication is established, Infinity processes DDE transactions of type **WM_DDE_REQUEST**. This message takes two parameters:

Format: The only supported format type is **CF_TEXT**.

Item: A string containing the command to be sent to Infinity. The strings are identical to the RS-232 commands as documented in the Infinity User's Manual. A terminating CR/LF is not required.

In response, the Infinity program sends a DDE transaction of type **WM_DDE_DATA** back to the client window. The returned item has a format type of **CF_TEXT**. The content of the responses is the same as for RS-232 commands, see below.

Sending Messages by RS-232

To enable the Infinity program to respond to RS-232 messages you first must configure the serial port on which the Infinity software is to poll for messages, see Figure 7-7 and Table 7-3.

After configuring the serial port, the Infinity program will try to open the port and then begin to poll the port for incoming messages. If there is a problem with the port initialization an error box will appear on the screen. If this occurs the configuration of the serial port will need to be verified (i.e., base address and IRQ conflicts). The information entered in the Serial Port Configuration window is saved when the program is terminated, so it is not necessary to repeat this procedure each time the GUI is started.

RS-232 Message Syntax

Each RS-232 message to the laser consists of a string of printable ASCII characters terminated by a carriage return (ASCII 13). Leading and trailing "white space" characters are

discarded before the message is processed. Replies from the laser will consist of printable ASCII characters followed by a carriage return and a line feed (ASCII 13 + ASCII 10).

Each message begins with the keyword followed by 0 or 1 arguments, see Table 7-10. If an argument is required, use a space character (ASCII 32) between the keyword and the argument. Keywords are case sensitive. Floating point numbers, when required, consist of an optional minus sign, at least one digit (possibly 0) followed by an optional decimal point and an optional string of trailing digits (i.e., [-]n[.][m]). Scientific notation is not supported. Integers are represented by a sequence of digits with no decimal point.

When the program receives a message that begins with an invalid keyword, the words "Syntax error:" (without the quotes) are transmitted back in the reply followed by the entire contents of the invalid message. When the program receives a message that contains a valid keyword, but an invalid argument, the words "Command error:" (without the quotes) are transmitted back in reply followed by the entire contents of the invalid message. This occurs, for instance, if an attempt is made to set a parameter to a value outside of its valid range. For example, "RepRate 135.2" would cause the reply "Command error: RepRate 135.2" because the upper limit of the repetition rate is 100.0.

If the message is received correctly and executed, the word "OK" is transmitted back in reply.



Every message will cause a reply of some kind. It is very important not to transmit a new message until the previous message has been acknowledged by the GUI.

Some messages query the laser for information (these are identified by a keyword that begins with a lower case "q"). In reply to a query, the requested information will be transmitted back in a message-specific format as described in Table 7-10.

Example RS-232 Program

The following is an example program of how to use a second computer to remotely control the Infinity laser system. To obtain the required number of serial ports, a SIIG serial expansion card (model 101812) with a single DB9 port was installed in the Infinity computer. The new Com port was given the base address 03E8 and was assigned interrupt (IRQ) 5, which is normally reserved for a second parallel port.

The 3-serial ports, and the Windows operating system, were configured to allow:

- the remote computer to be on Com port 1
- the Infinity power supply to be on Com port 2
- the mouse to be on Com port 3

The program was written and executed with the QBasic utility program provided with the Windows 3.1 operating system, installed on a Toshiba laptop computer.

When the example program first starts it polls the Infinity software to determine the current system state, repetition rate, IR pulse energy setting, lamp voltage setting, and number of active faults. Following this the function key menu is displayed, F1 through F7. The action initiated by each key is explained in the menu. If input is required from the user a task specific prompt appears below the function key menu.

When the user terminates the program, the laser system is automatically sent to the "Off" state, the repetition rate is set to 10 Hz, and the IR pulse energy is set to 1 mJ. For reasons of safety, it is always advisable to minimize the repetition rate and IR pulse energy before shutting down the laser system.

```
' Example remote Infinity program
```

```
' Coherent Laser Group (1997)
```

```
DECLARE SUB Setcomport ()
```

```
DECLARE SUB laserstate ()
```

```
DECLARE SUB menu ()
```

```
COMMON SHARED string1 AS STRING * 80
```

```
COMMON SHARED string2 AS STRING * 80
```

```
COMMON SHARED flag1%
```

```
KEY(1) ON ' Turn on monitoring of function keys
```

```
KEY(2) ON
```

```
KEY(3) ON
```

```
KEY(4) ON
```

```
KEY(5) ON
```

```
KEY(6) ON
```

```
KEY(7) ON
```

```
CALL Setcomport ' Initialize com1
```



```
start:
    CALL laserstate      ' Get current laser state information
    CALL menu           ' Print function key menu to screen
    flag1% = 0
    DO WHILE flag1% = 0
        '   Defining function key F1 as laser to Standby event.
        ON KEY(1) GOSUB lasertostandby
        '   Defining function key F2 as laser to On event.
        ON KEY(2) GOSUB turnonlaser
        '   Defining function key F3 as laser Off event.
        ON KEY(3) GOSUB turnofflaser
        '   Defining function key F4 as set repetition rate event.
        ON KEY(4) GOSUB setretrate
        '   Defining function key F5 as set pulse energy event.
        ON KEY(5) GOSUB setpulseenergy
        '   Defining function key F6 for screen refresh.
        ON KEY(6) GOSUB redoscreen
        '   Defining function key F7 for program termination.
        ON KEY(7) GOSUB exitroutine
    LOOP
END

                                ' Main menu functions

lasertostandby:
    string1 = "Standby"
    PRINT #1, string1
    string1 = ""
    INPUT #1, string1
RETURN start

turnonlaser:
    string1 = "On"
    PRINT #1, string1
    string1 = ""
    INPUT #1, string1
RETURN start
```

turnofflaser:

```
string1 = "Off"
PRINT #1, string1
string1 = ""
INPUT #1, string1
```

RETURN start

setrepreate:

```
string2 = ""
INPUT "Enter new repetition rate: ", string2
string1 = ""
string1 = "RepRate " + string2
PRINT #1, string1
string1 = ""
INPUT #1, string1
```

RETURN start

setpulseenergy:

```
string2 = ""
INPUT "Enter new IR pulse energy: ", string2
string1 = ""
string1 = "PulseMj " + string2
PRINT #1, string1
string1 = ""
INPUT #1, string1
```

RETURN start

redoscreen: RETURN start

exitroutine:

' Shut off laser

```
string1 = ""
string1 = "Off"
PRINT #1, string1
INPUT #1, string1
```

' Set IR energy to minimum value

```
string1 = ""
string1 = "PulseMj 1"
PRINT #1, string1
INPUT #1, string1
```

```
' Set repetition rate to 10 Hz

string1 = ""
string1 = "RepRate 10"
PRINT #1, string1
INPUT #1, string1

' Disable Com port and set termination flag

COM(1) OFF
flag1% = 1
RETURN

' Program subroutines

SUB laserstate
CLS 0
PRINT ""
PRINT " Sample RS-232 remote control program for the Infinity laser
system."
PRINT ""

' Print last Infinity system message

PRINT "Last Infinity system message was: " + string1

' Get current laser state

string1 = ""
string1 = "qState"
PRINT #1, string1
INPUT #1, string2
string1 = ""
string1 = "Current state of the laser system: " + string2
PRINT string1

' Get current repetition rate

string1 = ""
string1 = "qRepRate"
PRINT #1, string1
INPUT #1, string2
string1 = ""
string1 = "Current Rep Rate is (Hz): " + string2
PRINT string1
```

```

                                ' Get IR pulse energy
string1 = ""
string1 = "qPulseMj"
PRINT #1, string1
INPUT #1, string2
string1 = ""
string1 = "Current IR pulse energy is (mJ): " + string2
PRINT string1

                                ' Get current lamp voltage
string1 = ""
string1 = "qVolts"
PRINT #1, string1
INPUT #1, string2
string1 = ""
string1 = "Current lamp voltage setting is (V): " + string2
PRINT string1

                                ' Get current number of active faults
string1 = ""
string1 = "qNFaults"
PRINT #1, string1
INPUT #1, string2
string1 = ""
string1 = "Current number of active faults is: " + string2
PRINT string1
PRINT ""
END SUB

```

SUB menu

```
PRINT "          Active Function Keys"
PRINT ""
PRINT "          F1 : Turn Laser to Standby"
PRINT "          F2 : Turn Laser On"
PRINT "          F3 : Turn Laser Off"
PRINT "          F4 : Set Repetition Rate"
PRINT "          F5 : Set Pulse Energy"
PRINT "          F6 : Refresh Screen"
PRINT "          F7 : Exit Routine"
PRINT ""
PRINT "Make a Function Key Selection to Begin"
```

END SUB

SUB Setcomport

```
OPEN "COM1:9600,E,7,1,CD0,CS0,DS0,OP0,RS,TB2048,RB2048" FOR
RANDOM AS #1
```

```
COM(1) ON
```

END SUB

OPERATOR'S MANUAL

.....

CHAPTER EIGHT

THEORY OF OPERATION

Introduction

This chapter begins by presenting some of the underlying physics incorporated into the Infinity™ Nd:YAG laser system. Special attention will be given to those processes and features that are unique to the Infinity laser. It is, however, not necessary to read this chapter in order to operate the laser on a day to day basis. The chapter concludes with a review of the process of harmonic generation; specifically for the second, third and fourth harmonic configurations as they are employed in this laser system.

Laser Principles

By rearranging the configuration of their surrounding electrons, atoms and molecules can acquire energy from and give energy to their environment. Since the configurations the electrons can assume are limited to certain patterns according to the laws of quantum mechanics, the flow of energy is also limited to occur at certain discrete values in these processes. Following the absorption of an energy quantum the system (the word system will be used interchangeable for atoms and molecules) is said to be in an excited state. Since the system strives to return to its ground state (i.e., the state with the lowest possible energy) it will dispose of its excess energy. This can be done in several different ways, but if the system is left undisturbed it will do so by emitting one or more photons whose total energy will correspond to the energy difference between the excited state and the ground state. This process is called spontaneous emission and it occurs with a probability equal to the inverse of the lifetime of the excited state. In the literature this quantity is written as $A_{12} [s^{-1}]$, where the index refers to the energy levels involved. The process is illustrated in Figure 8-1a.

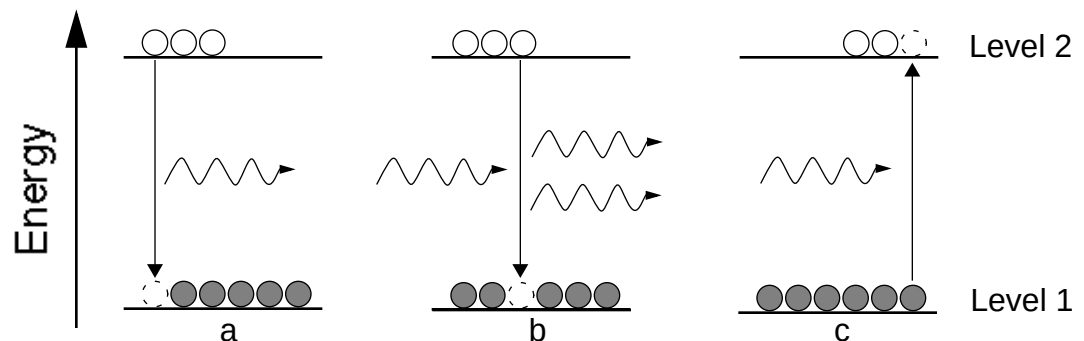


Figure 8-1. Illustration of Emission and Absorption Processes in Two Level System.
a. Spontaneous Emission, b. Stimulated Emission and c. Absorption.

The total rate of systems undergoing spontaneous de-excitation can now be written as $A_{21} N_2$, where N_2 is the number of systems in the excited state. If the excited system instead is disturbed by a radiation field in its vicinity, and the frequency of this external field is close to the resonance frequency of the excited system ($\omega = (E_1 - E_2)/2\pi\hbar$), the excited system can be "stimulated" to return to the ground state as shown in Figure 8-1b. When this occurs a photon with identical properties (the same energy, direction and polarization) to the incoming one is emitted in the process. This process is called stimulated emission and it is the underlying mechanism of all laser action. The reason for this is that it provides a way of generating large numbers of identical photons in a controlled way (i.e., a coherent amplification mechanism). Stimulated emission occurs with a probability of B_{21} which is related to the A_{21} factor introduced in the previous section through the relationship: $B_{21} = A_{21} 2\pi c^3 / (\hbar \omega^3)$. Now, if there are N_2 numbers of excited systems, the de-excitation rate due to stimulated emission can be written as $B_{21} N_2 I$, where I [$\text{W}/\text{cm}^2 \text{ Hz}$] is the energy density in the external field. This process is illustrated in Figure 8-1b.

If the system instead is in the ground state when it is exposed to an external field oscillating at its resonance frequency, it can absorb an energy quantum and end up in an excited state through a process called stimulated absorption. This process is illustrated in Figure 8-1c. The two processes, stimulated absorption and emission are symmetrical with respect to each other. This means that an incoming photon does not distinguish between the upper and lower energy state and they hence occur with equal probability, $B_{21} = B_{12}$.

The three processes described above are called radiative since they involve the creation or annihilation of a photon. In addition to these processes there are also various non-radiative de-excitation and excitation mechanisms such as collisions with other systems and electron impact. Under normal conditions all of these processes are present simultaneously.

Since stimulated emission is the process in which laser light is generated we would like to arrange things so that this process dominates over the others. The only way to do this is to have a greater number of systems in the higher lying state so that $B_{21} N_2 > B_{12} N_1$, which is the same as the criteria $N_2 > N_1$. The situation when there are more systems in the higher lying state is called population inversion and it is a necessary condition for lasing action to take place. It is not possible to

achieve population inversion in a two level system of the type shown in Figure 8-1. The reason for this is that the faster systems are pumped into the excited state, the faster they will also undergo stimulated emission back to the ground state. Hence at least three energy levels are needed in a system where population inversion is to be created. A disadvantage of a three level system is that the lower level in the lasing transition will coincide with the ground state. This means that the population difference $N_2 - N_1$ will be smaller than if the lower level would have been an originally empty state. It is therefore advantageous to instead use a gain medium in which four levels are used, the situation illustrated in Figure 8-2.

The material used in the Infinity system, Nd:YAG, can be described by this four level scheme and the corresponding energy levels in this material and the spectroscopic nomenclature, are shown in Figure 8-2. Besides having the advantage of being a four level system, Nd:YAG also possesses a number of other attractive properties that make it suitable as a laser gain medium. After diamond and sapphire it is the hardest material known to man, its thermal conductivity is excellent and rods of high optical quality can be produced. Because of its narrow fluorescence line width it displays a low lasing threshold and high gain.

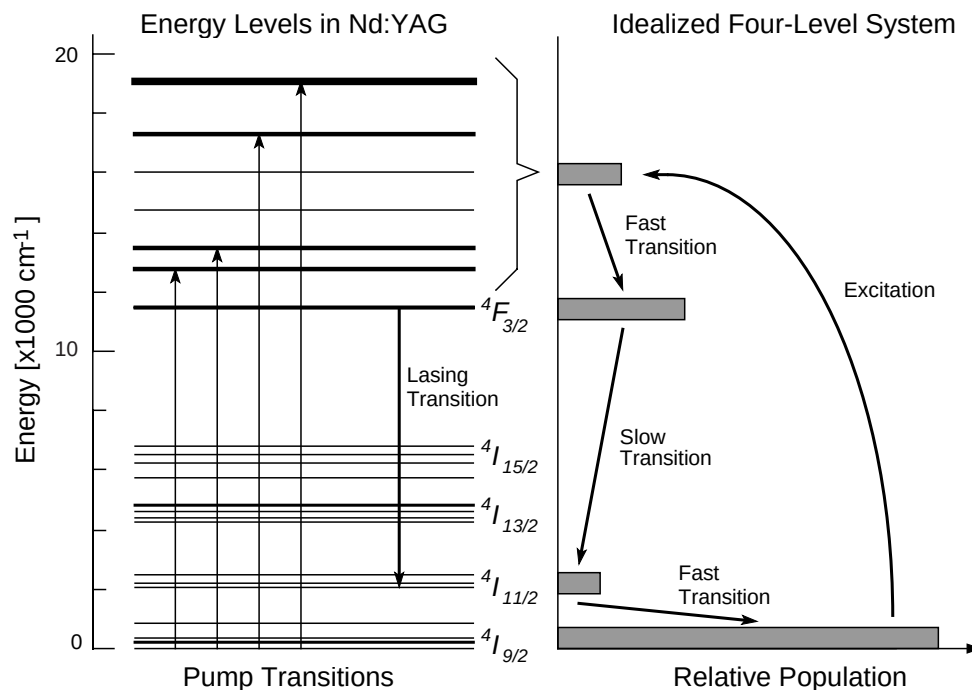


Figure 8-2. Energy Levels for a Nd:YAG System (Left) and a Four Level System (Right).

Laser Physics

Since the invention of the laser in 1960, it has been put to use in a great variety of applications. Despite the enormous multitude of laser types, they all share some common features. In order for a laser to produce coherent light a minimum of three conditions needs to be fulfilled. These three items are:

A Gain Medium

This is simply a medium in which energy from an external energy source, such as an electrical current or another laser, can be converted into a population inversion. If a population inversion is present in the medium then an incoming photon, having a frequency close to the corresponding energy difference, will with a greater probability be amplified through stimulated emission than absorbed through stimulated absorption. A gain medium is thus a coherent amplifier for photons.

A Pump Source

In order to prepare the system so that a state of population inversion is achieved, energy needs to be supplied. There are many different ways by which this can be done. The two most common are optical excitation, used in dye lasers, and electrical excitation (electron impact for example) used in gas discharge lasers like HeNe and Ar⁺ lasers. In the Infinity system the excitation energy is supplied by a flash lamp positioned close to the laser rods inside a highly diffusive cladding which ensures excellent optical coupling of the light from the flash lamp into the laser rods.

A Resonator

The function of the resonator is to provide the feedback necessary for the stimulated emission process and to confine the laser field within the boundaries of the gain medium. The simplest type of resonator consists of two plano parallel mirrors positioned with the gain medium between them as shown in Figure 8-3.

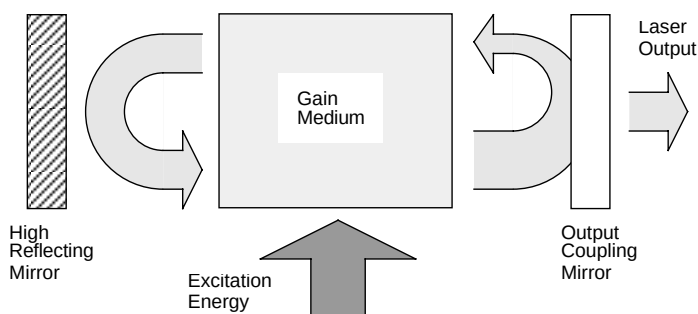


Figure 8-3. Illustration of Laser Resonator.

One of the mirrors has a reflectance of very close to 100% for the laser wavelength while the other allows some of the generated radiation to leak out of the cavity and hence has a lower reflectance, around 80% for example. The mirrors are called the high reflector and the output coupler, respectively. Some lasers have more complicated resonator arrangements incorporating more than two mirrors and/or other optical elements. The idea is still the same though, to confine the laser field inside the cavity, in a manner so that it reproduces itself after one cavity round-trip.

Even though an arrangement incorporating the three items described above is sufficient for producing laser emission other components are often added to it in order to control the polarization of the output, the spatial and temporal mode as well as the frequency width of the output. Once the pulse has left the oscillator its fundamental temporal and spatial properties are established. By passing it through one or more amplifying stages its brightness can, however, be increased (An amplifier is here simply a gain medium without a resonator). When doing so it is important to design the amplifier so that it does not introduce new or amplify present distortions. In Nd:YAG material this means that thermal effects need to be carefully controlled. Due to the nature of the phase conjugate mirror arrangement used in the Infinity laser, the distortions induced in the beam during the amplification process are kept at a minimum.

Laser Light Characteristic

Due to the nature of the process in which the laser light is generated, it has a number of unique properties which set it apart from conventional light sources. It is instructive to take a closer look at how the laser radiation compares to the light from an ordinary light source like an incandescent bulb. In comparison lasers are:

- **Brighter.** This means that the number of photons that a laser can provide per area and solid angle [$\text{W}/\text{cm}^2\text{sr}$] surpasses that of any other known source of light. This is an important parameter in optical systems since, using only passive components such as lenses, the brightness can not be increased outside the radiation source.
- **Coherent.** A coherent light source is one which maintains a fixed phase relationship between different points. Usually a distinction is made between spatial and temporal coherence. Spatial coherence is a measure of how well the phase can be predicted at different points in space at a given time while temporal coherence is a measure of how well the phase can be predicted at a

given point in space as a function of time. Coherence is hence a measure of the degree of phase correlation. A perfectly monochromatic light source would have infinite coherence length (time).

- nearly **Monochromatic**. The combination of narrow bandwidth and high peak power exhibited by lasers is one of their greatest assets. It allows the user to manipulate matter with an extreme degree of selectivity, and to perform advanced experiments in fields such as interferometry, holography and non-linear optics.
- **Tunable**. Using modern dye lasers, optical parametric oscillators, Raman shifting and various non-linear optical frequency conversion schemes, continuously tunable wavelengths from below 200 nm up into the infrared can be generated.
- **Directional**. This might be one of the laser's most striking characteristic. Laser beam propagation is limited only by diffraction. As a consequence laser beams can transport energy over enormous distances (e.g., to the moon and back) and be focused to very small spot sizes.

Even though the output from an ordinary light source, such as an incandescent lamp could be filtered to yield the same spatial and temporal coherence as any given laser source, the laser would still be more coherent. This is because the laser exhibits higher order coherency due to its different photon statistics. Photons from an ideal laser follow Poisson statistics while photons from a classical light source follow Bose-Einstein statistics. What this means is that the photons exiting from the laser can be viewed as statistically independent (i.e., detecting one photon does not aid in predicting when the next one will come). The filtering required to achieve the same temporal and spatial coherence as a laser would also result in an enormous loss of brightness.

Diode Pumped Lasers

Diode pumped solid state lasers are attractive for a number of reasons. They can be made very small, they convert electricity to laser light efficiently, they can be designed to lase in a single longitudinal mode and they are less sensitive to changes in their environment than flash lamp pumped systems. Furthermore the output from GaAs laser diodes, lasing in the region around 800 nm, have an excellent match with the main Nd:YAG absorption band (the $4F^{5/2}$ band). Flash lamps in comparison emit their light over a wide spectral range, covering the near ultraviolet, the visible and near infrared which means that most of the flash lamp output is converted into heat in the laser head. The fact that the output from laser diodes conveniently can be focused into the gain medium also contributes to reducing thermal effects in the rods. Because of the lower thermal load on the rods, problems with thermal lensing and thermally induced birefringence (thermally induced depolarization) are much less severe than in flash lamp pumped oscillators. As a result small diode pumped solid state lasers exhibiting excellent spatial and temporal behavior can be built. The life span of laser diodes, as measured in number of pulses fired before failure, is also significantly longer than that of flash lamps. State of the art flash lamps can fire up to several hundred million shots compared to more than one billion for laser diodes.

DPSS/DPMO

The Infinity master oscillator consists of two concave mirrors, a Nd:YAG prism and an acousto optical modulator arranged so that they form a ring resonator, see Figure 8-4.

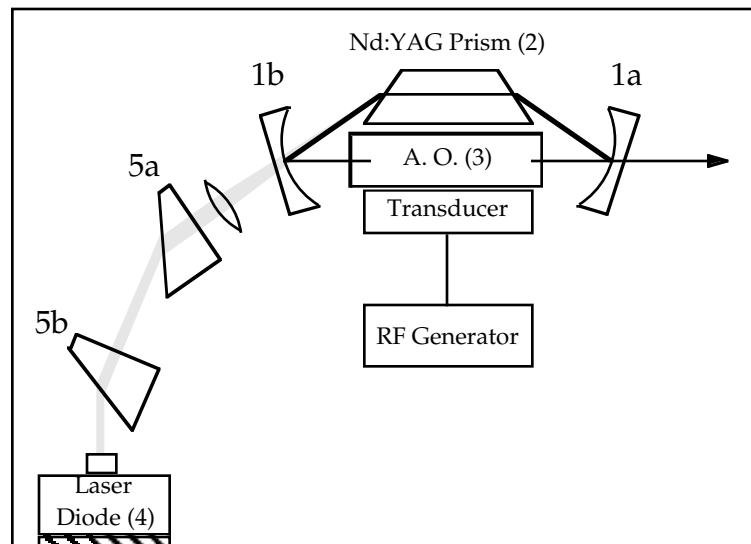


Figure 8-4. Infinity Master Oscillator Optical Layout.

In a ring formed cavity, instead of having a standing wave as in a linear cavity, two counter propagating waves can be present. Since these beams will leave the cavity in different directions, as shown in Figure 8-5, only one is usually useful. It is hence desirable to eliminate one of the beams so that all the stored inversion energy is made available for the direction used. This can be accomplished by introducing an additional loss mechanism for one of the directions. In the Infinity system the same device that is used to Q-switch the cavity, the acousto optical modulator, AOM, is used for this purpose.

Acousto-Optic Modulator

The acousto-optic modulator serves two purposes. It allows the round trip cavity losses to be regulated, providing the oscillator with a Q-switch. It also introduces a somewhat higher loss in the beam path for one of the two lasing directions. The last property, in combination with careful design of the oscillator geometry, ensures its unidirectional operation.

The underlying mechanism by which the modulator regulates the round trip losses is by diffraction of the optical wave against the acoustical wave induced in the material. As the modulator is switched on an acoustic wave is set up in the medium. The acoustic wave gives rise to a periodic variation in the medium's refractive index through the photoelastic effect. When the optical wave strikes this refractive index grating two new beams are created, one shifted up, the other down by the frequency of the acoustic wave. The directions of the new beams are given by conservation of momentum and energy in the photon - phonon interaction. Since the diffracted beams only can travel in a direction in which they undergo positive interference the optical path length difference between two adjacent rays must satisfy the following relationship

$$\lambda = 2\Lambda \sin(\Phi_B)$$

where λ is the wavelength of the optical radiation, Λ the wavelength of the acoustic wave and Φ_B the angle between them. Depending on if the incoming optical wave is scattered against or away from the acoustic wave the frequency of the diffracted beam will either be higher, $\omega_0 + \Delta\omega$, or lower, $\omega_0 - \Delta\omega$, than the incoming. These two processes are illustrated in Figure 8-5.

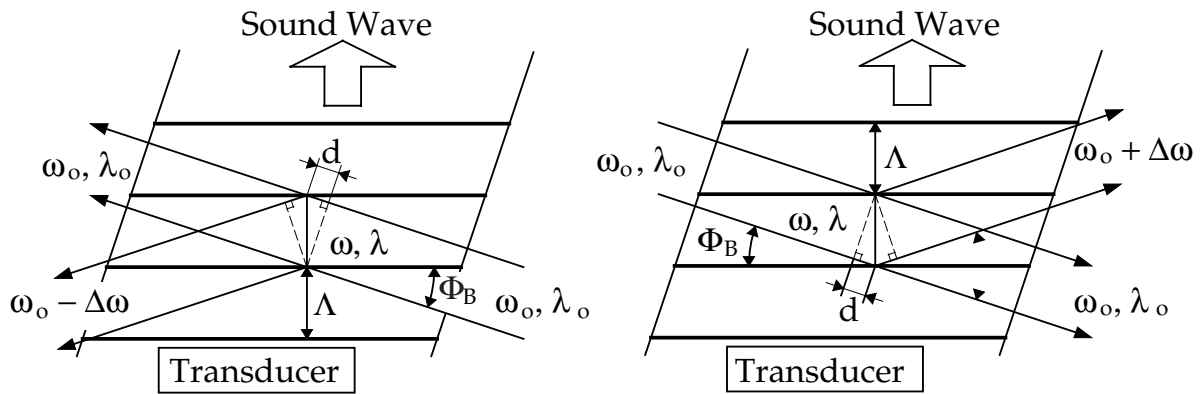


Figure 8-5. Geometry for Bragg Diffraction.

By switching on the acoustic field, energy is diffracted out of the main beam, introducing an additional loss mechanism in the cavity. When the transducer is momentarily switched off the acoustic wave decays, full transmission through the AOM crystal is restored and a laser pulse is created. A special feature of the design in the Infinity is that even when the transducer is on, the Q-switch is not fully closed but allows the oscillator to lase at a few milliwatts of CW power. This prelasing, which, as mentioned above, only occurs in one direction, provides the initial seed photons from which the next laser pulse builds up. This design enables the laser to be operated with exceptional stability with respect to energy, power, directionality and frequency.

Power Amplifier

By adding one or more amplifying stages after the master oscillator, a combination of high power, excellent beam quality and narrow band width can be achieved. The main function of the amplifier is to increase the brightness of the beam and to do so without introducing new or amplifying already present distortions.

Factors limiting the size of a Nd:YAG amplifier are a combination of crystal growth techniques and issues such as rod cooling and thermally induced stress and birefringence.

When designing an amplifier there is a trade-off between achieving high energy extraction or high gain. In order to extract most of the energy stored in the gain medium, the incoming pulse needs to already have an appreciable energy. If on the other hand high gain is desired then the incoming pulse should have an energy much smaller than the stored. In the case when the input energy is so small that the energy in the amplified pulse is still small compared to the stored, the amplification factor is called "small signal gain". In the

opposite case, when the input energy is comparable to the stored energy the amplification factor will approach unity and the gain is said to be saturated.

In the Infinity system a multi-pass arrangement is used. This allows the amplifier to be used both as a small signal amplifier, exhibiting high gain but poor energy extraction (first pass) and as a low gain and high energy extraction efficiency amplifier (second pass).

An important consideration in high gain systems such as Nd:YAG amplifiers is to suppress the amplification of spontaneous emission (ASE) which will decrease the energy available for coherent emission and increase the incoherent background emission. ASE can also propagate backwards into the oscillator, affect its performance and potentially damage it. Methods that have been developed to deal with ASE are spatial filters, since ASE diverges more rapidly than laser emission, and Faraday rotators acting as optical diodes. In the Infinity system suppression of ASE is solved by a combination of careful design of the optical elements in the amplifier and by the use of a SBS mirror. Until threshold is reached its reflectance is essentially zero. As it is turned on by the incoming laser field, the energy stored in the laser rods is rapidly consumed by the coherent laser emission.

Polarization Control

A thermal gradient is created across the rod diameter as the radiation from the flash lamp is absorbed in it. The temperature gradient in turn gives rise to a stress gradient because the hotter regions in the center of the rod will try to expand more than the cooler surface regions. The resulting stress gradient gives rise to a refractive index gradient across the rod surface. Since the radial and the tangential stress components are different, the refractive indices along these two directions will be different as well. As a consequence the electro-magnetic vector components of the incoming radiation along the radial and the tangential direction, will experience different refractive indices and hence travel at different speeds through the rods.

The pulse exhibits horizontal polarization as it enters the first amplifying rod. Due to the difference in refractive index along the radial and tangential direction, the pulse exiting has acquired an elliptical polarization in all points across its spatial mode except for those that coincide with the two diameters that are parallel and perpendicular to the polarization state of the incoming pulse.

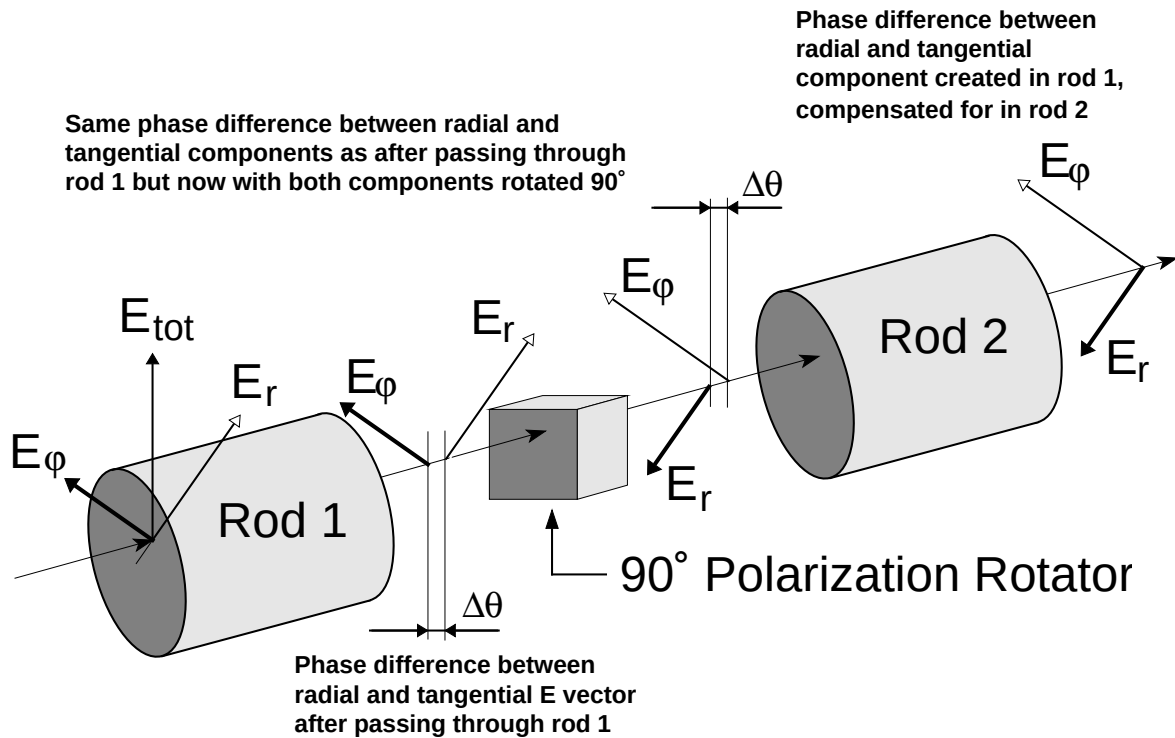


Figure 8-6. Polarization States of the Laser Pulse Before, Between and After Passing Through the Two Nd:YAG Rods.

If the pulse exiting the first rod, exhibiting elliptical polarization, is allowed to directly propagate through the second rod, the degree of ellipticity will increase, i.e. double. It is important to avoid this for the following reason.

After the pulse has been reflected in the SBS cell and passed through the two rods a second time the vertically polarized component of it will be reflected out of the laser amplifier by the thin-film polarizer. If the polarization state is poorly defined as the beam strikes the thin-film polarizer a large fraction of it will end up in the Faraday isolator. This energy is hence not available to the user and it can also, potentially, disturb the stability of the oscillator. It is therefore important that the beam is in a well defined polarization state as it goes through the quarter-wave plate, positioned between the second rod and the SBS cell.

To compensate for the phase difference induced by the birefringence in the first rod, a 90° polarization rotator is positioned between the two rods, see Figure 8-6. Light entering the first rod (from the left), having vertical polarization (horizontal in the laser, vertical used in the figure for illustration purpose only), will experience a phase

retardation between the radial, E_r , and tangential, E_f , component. Before the pulse enters the second rod the polarization is rotated 90° which means that the component that was radial in the first rod, now is the tangential and the tangential component in the first rod is now the radial. This means that the phase retardation that was induced in the first rod will be compensated by the second rod and the net result will be light that is again linearly polarized but now rotated 90° . The same correction will take place as the pulse goes through its second pass through the rods and the net result will be vertically linearly polarized light.

Relay Imaging

In order for the polarization control scheme described in the previous paragraph to work, the pulse traveling through the amplifier must experience the same relative environment in the two rods. Assuming that the two rods are close to identical and symmetrically loaded, this requires that the beam in the first rod is imaged into the second rod. Figure 8-7 illustrates how this is accomplished. The imaging, performed by two lenses and a concave mirror, is done so that the transverse coordinates remain unaltered while the coaxial coordinates are flipped around the center of the rods, see arrows inside rods in Figure 8-7. This arrangement not only performs part of the polarization control but also compensates for small residual asymmetries in the pumping of the rods. The relay cell is inserted in the beam path to avoid laser induced air breakdown at the focal points formed by the imaging system.

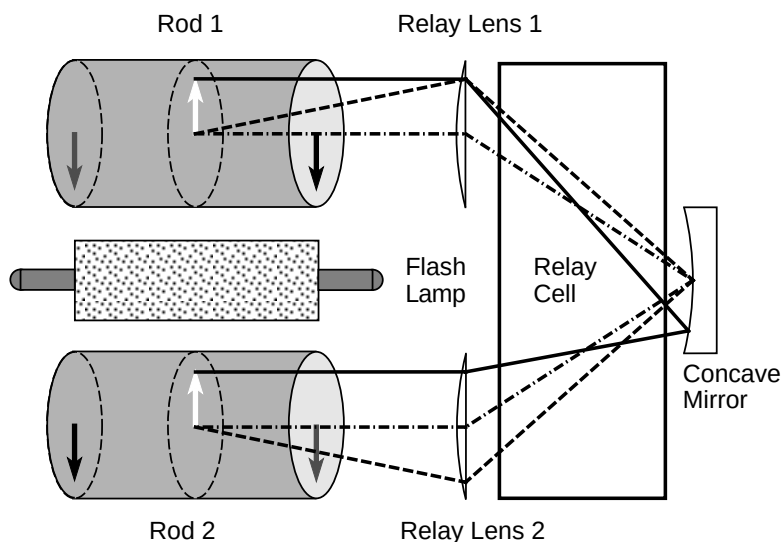


Figure 8-7. Principle of Relay Imaging.

Stimulated Brillouin Scattering

When light waves are scattered by density waves in a medium, a backwards traveling and frequency shifted component can be created in a process called Brillouin scattering. The density waves can be excited thermally or by the laser field itself through electrostriction (i.e., the macroscopic strain the electric field gives rise to). In the latter case the phenomenon is called Stimulated Brillouin Scattering or SBS. As is typical for processes that are self amplifying (stimulated), a minimum power density, a threshold, must be exceeded for it to occur. As long as the fluency is below this threshold value the process does not take place at all. When the fluency exceeds the threshold value the signal grows exponentially until limited by the onset of saturation. When the leading edge of the laser pulse returns through the rods it will decrease the gain. The effect of this can be seen in the trailing end of the laser pulse where there is a sudden drop in intensity.

A detailed theoretical analysis of the situation where a high power laser beam is focused into, for example CFC 113, shows that the field generated in the backwards direction by SBS is also the phase conjugate of the incident field. An illustration of how the reflectance of an incoming wave is different in a Phase Conjugate Mirror compared to an ordinary one is shown in Figure 8-8.

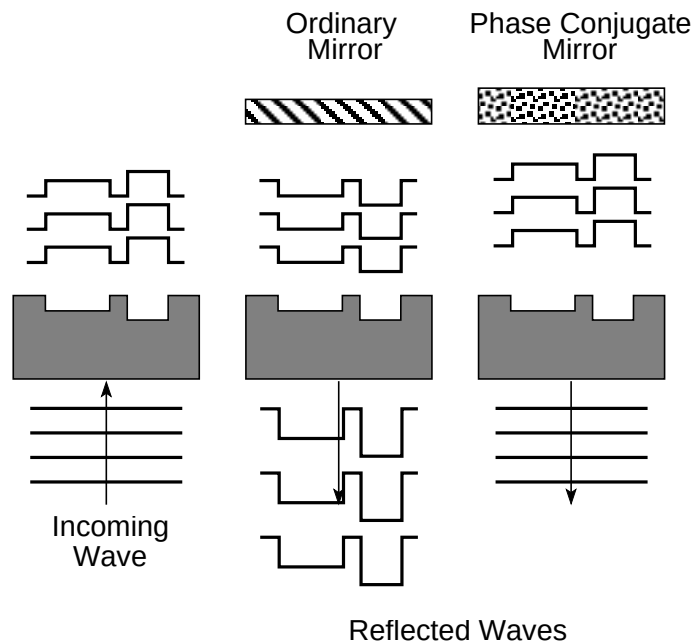


Figure 8-8. Illustration of the Difference Between Reflection in a Regular and a Phase Conjugate Mirror.

The pulse that undergoes reflection in the ordinary mirror, will after having traversed the distorting media a second time suffer from twice the distortion it had after its first pass while the pulse reflected in the phase conjugate mirror will be restored to its original shape.

In the Infinity laser this phenomena is used to correct for wave front distortions created by thermal and stress induced effects in the Nd:YAG rods. The distortions imposed on the pulse during its first pass through the amplifier are reversed on its second pass. Since this process is self correcting it is insensitive to changes in the degree and art of distortion. As a result the Infinity system can produce a laser pulse of superior quality that is independent of repetition rate and output power. There is, however, no polarization information in the phase conjugate process. This means that processes resulting in changes in the polarization state of the beam are not reversed or corrected. Instead the polarization state of the laser pulse is determined by the oscillator and the design of the beam path.

Harmonic Generation

The primary radiation from the Infinity laser is emitted at 1064 nm. For most applications, an output wavelength in the visible is, however, desired. This can be achieved by doubling the frequency of the Nd:YAG fundamental in a non-linear optical process called Harmonic Generation. A brief description of this process is given below.

Applying an electrical field to a linear dielectric medium induces a polarization in it according to the relationship:

$$P = \epsilon \chi E, \quad (1)$$

where E is the electric field vector, ϵ the permittivity of free space and χ is a dimensionless constant (electrical susceptibility) characteristic for the polarized material. A medium which can be described by this relationship is said to be linear. For a long time it was believed that all materials were linear and this assumption was one of the cornerstones of optics. Consequences of this assumption are that:

- optical properties are independent of light intensity
- the frequency of light does not change and
- light does not interact with light.

With the advent of the laser it has, however, become possible to create E-fields large enough to saturate the polarization. As a consequence the induced polarization will no longer be related to the electrical field strength in a linear way. For such cases a more general expression must be used. It is convenient to write this as a series expansion in the electrical field according to:

$$P = \epsilon (\chi_1 E + \chi_2 E^2 + \chi_3 E^3 + \dots), \quad (2)$$

where χ_1 is the regular linear susceptibility and χ_2, χ_3 are the second and third order non-linear susceptibilities respectively. Normally $\chi_1 \gg \chi_2 \gg \chi_3$ and hence the non-linear effects will be negligible unless the E-field is very large. If an applied field can be described by

$$E = E_0 \sin(\omega t) \quad (3)$$

the induced polarization, according to Eq. 2, takes the form:

$$P = \epsilon \chi_1 E \sin(\omega t) + \epsilon \chi_2 E^2 \sin^2(\omega t) + \dots \quad (4)$$

which can be rewritten, using standard trigonometric relationships, as:

$$P = \epsilon \chi_1 E \sin(\omega t) + 1/2 \epsilon \chi_2 E^2 \{1 - \cos(2\omega t)\} + \dots \quad (5)$$

In the last expression, the presence of the second term shows that a wave having twice the frequency of the fundamental can be formed in a non-linear medium. This phenomena is called frequency doubling or Second Harmonic Generation (SHG).

Since the refractive index is wavelength dependent it will usually be different for the fundamental and the frequency doubled light. This means that over a distance of several wavelengths the net frequency doubled power will be zero. In order to generate any substantial amount of frequency doubled light, the two refractive indexes needs to be the equal so that the phase difference between the waves remain constant. This situation is called phase matching and it can be achieved in materials that are birefringent (materials which have different refractive indexes for different polarization states). Figure 8-9 shows a cross section of what is called a phase-matching ellipsoid. The circles and ellipses show how the refractive indices, for two different polarization states and wavelengths, varies as a function of angle to the optical axis, which in the figure coincides with the Z-axis. As can be seen there is an angle, Θ_m , for which the refractive index for the fundamental beam, n_o^ω , is equal to the refractive index of the frequency doubled beam $n_e^{2\omega}$. When the crystal is tilted so that the incoming radiation propagates along this line it is said to be phase matched and frequency doubling will take place with the highest possible efficiency in the non-linear material.

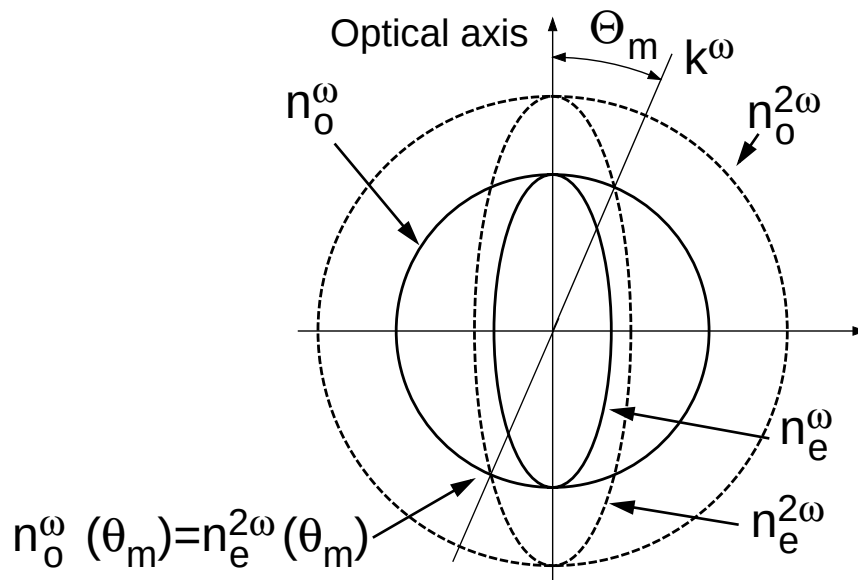


Figure 8-9. Illustration of Phase Matching in a Birefringent Media.

Since n_o and n_e have different temperature derivatives, phase matching can also be achieved by tuning the crystal temperature. In the Infinity system the crystals are, however, angle tuned.

After the crystal the two wavelengths can be separated using either a prism or, which is more common, a dichroic mirror that transmits one wavelength and reflects the other.

There are two different polarization arrangements that can be used for SHG, called type 1 and type 2 doubling. In the Infinity laser system type 1 doubling is used because it is the more efficient process of the two in BBO.

For some applications such as pumping of Optical Parametric Oscillators and of dye lasers running on Coumarin dyes, shorter wavelengths than 532 nm are desirable. Through a non-linear optical process similar to the one described above, radiation at 355 nm and at 266 nm can be produced starting with the Nd:YAG fundamental at 1064 nm. Third Harmonic Generation (THG) of 355 nm radiation is accomplished by frequency mixing, in a second crystal, the fundamental 1064 nm radiation of the Infinity laser and the frequency doubled radiation. Since energy is conserved, the resulting wavelength is given by $1 / (1/1064 + 1/532)$ which is equal to 355 nm. Radiation at 266 nm is produced by removing the left over 1064 nm after second harmonic generation, and then frequency doubling the 532 nm beam.

Second Harmonic Generation

After the pulse has been reflected out of the amplifying system by the thin film polarizer, it is sent through an angle tuned BBO crystal (assuming that this option has been installed). The crystal is mounted in a cell enclosed by two antireflection coated windows. The entrance and exit surfaces of the crystal have also been antireflection coated. In the crystal, the infrared radiation undergoes type I doubling which means that the polarization of the fundamental remains unchanged while the polarization of the second harmonic at 532 nm is perpendicular to the incoming fundamental, as shown in Figure 8-10. A dichroic mirror is used to separate the fundamental wavelength from the second harmonic.

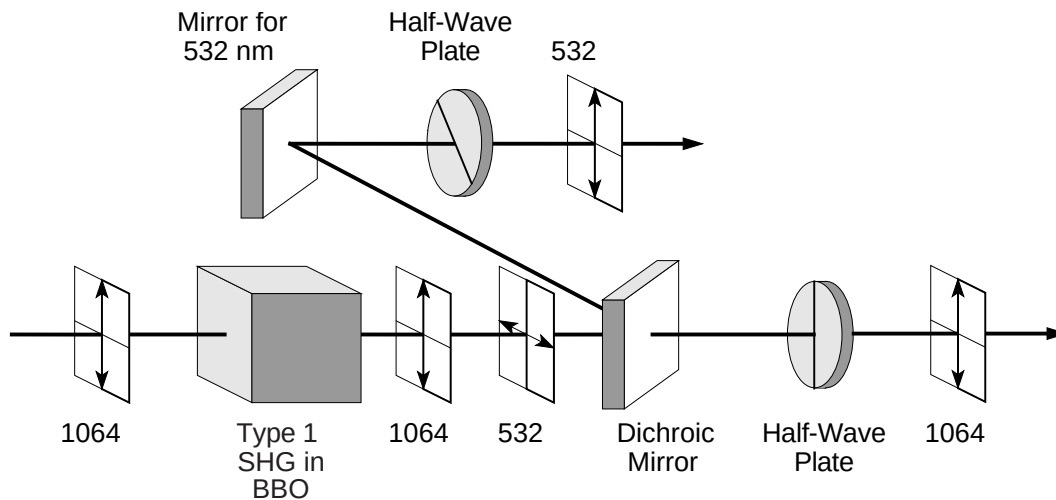


Figure 8-10. Polarization States Before and After Second Harmonic Generation.
(The Half-Wave Plates are Optional in the SHG Options Package).

Third Harmonic Generation

If the optional Third Harmonic Generation package has been installed, the beams (1064 nm and 532 nm) exiting the SHG crystal go through a wave plate, acting as a full wave plate for the second harmonic and as a half wave plate for the fundamental wavelength. The beams are then sent through a second angle tuned BBO crystal where they undergo Type I frequency mixing. The radiation produced at 355 nm is thereafter separated from the infrared and doubled radiation in a dichroic mirror, Figure 2-11.

Provisions are also made for installing an additional (and optional) dichroic mirror in the beam path containing the 532 and 1064 nm radiation. This dichroic (sometimes referred to as a stripper) directs the 1064 nm radiation into a beam dump while the 532 nm radiation is transmitted.

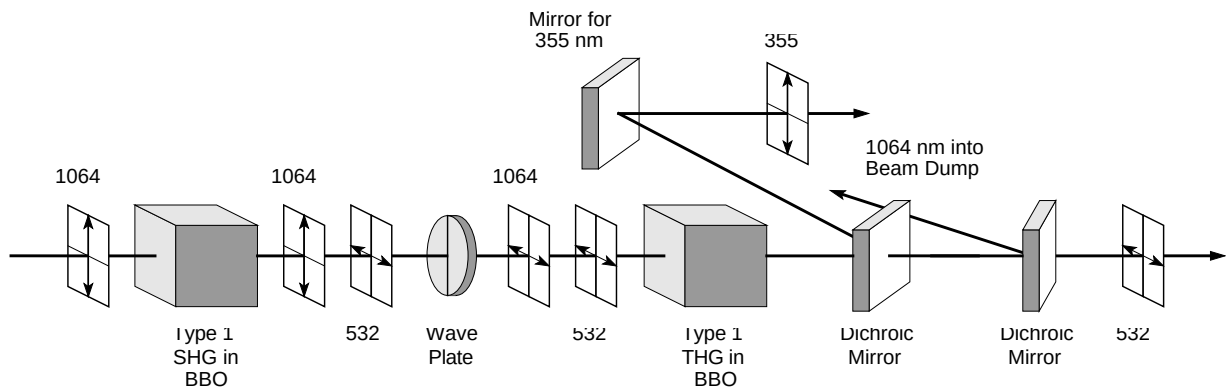


Figure 8-11. Polarization States Before, During and After Third Harmonic Generation.
(Dichroic Separating 1064 nm from 532 nm is Optional in THG Package Option)

Fourth Harmonic Generation

Radiation at 266 nm can be produced by frequency doubling the light at 532 nm, produced in the SHG process. This is done by adding one more BBO crystal to the set-up used for SHG. Since the IR fundamental is not used again after the SHG generation it is removed by a dichroic mirror, see Figure 2-12 (this dichroic is sometimes referred to as a stripper). After fourth harmonic generation the 532 nm and the 266 nm beams are separated in a dichroic mirror.

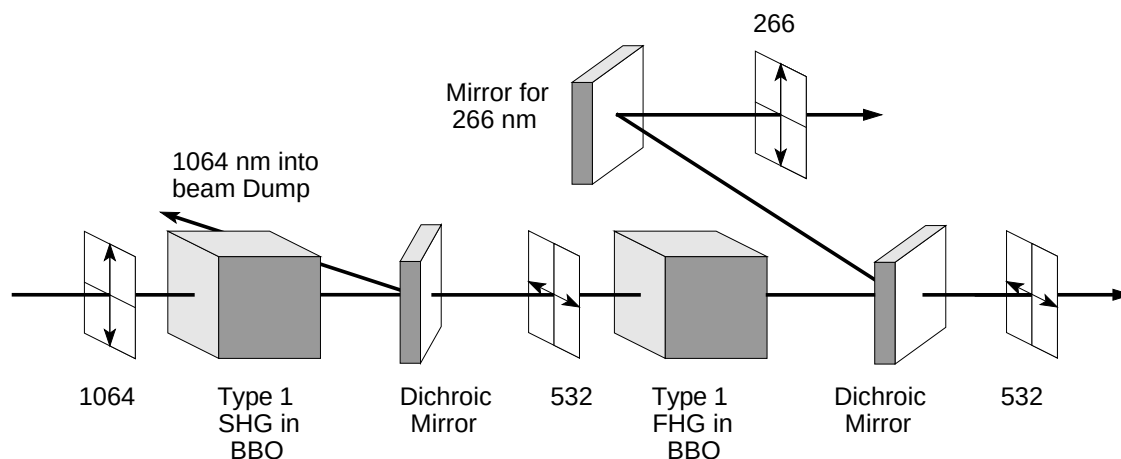


Figure 8-12. Polarization States Before, During and After Fourth Harmonic Generation.

OPERATOR'S MANUAL

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APPENDIX A

WARRANTY

Pulsed Nd:YAG Lasers Warranty

Coherent warrants to the original purchaser (the “buyer”) only that the Infinity Laser (a) conforms to Coherent’s published specifications, and (b) is free from defects in materials and workmanship.

The duration of the warranty is thirteen (13) months from date of shipment, or twelve (12) months from date of installation, whichever occurs first. Travel time and travel costs for on-site service are included during the first ninety (90) days of the warranty period. Crystals, lamps, and optics are not covered under the standard warranty or Productivity Plus Agreements. The warranty shall be void in the case of apparent misuse, incorrect operation or unauthorized repair.

Optics

Optics are warranted for a period of 90 days, subject to Coherent's operating procedures and specifications as outlined in the Operator's Manual.

Non-linear Crystals

The BBO crystals should be considered consumable items. Localized deterioration or failure may occur during normal use, and is not covered by warranty or Productivity Plus Agreements.

Flash Lamps

Lamps are consumable items and are not covered by any warranty or Productivity Plus Service Agreement. One spare lamp is included with all new Infinity systems. The expected lamp lifetime is greater than 50 million shots for normal system operation. Infinity has a resettable shot counter for the end-user's convenience.

Productivity Plus™ Service Agreements

Agreements are available to cover all Coherent Infinity Laser Systems. Productivity Plus Service Agreement coverage includes labor, zone-travel charges, and replacement parts, including the Nd:YAG rod. Coverage does not include consumable items such as, but not limited to, lamps, crystals, or optics. Productivity Plus agreements may be purchased at any time provided the system is currently under its original warranty or covered under an existing Productivity Plus agreement.

OPERATOR'S MANUAL

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APPENDIX B

PART NUMBERS

Part Numbers

The following parts can be ordered by contacting Coherent Customer Service at 1-800-367-7890 inside the U.S. Outside the U.S., Coherent Customer Service can be contacted by calling 408-764-4557.

Table B-1. Part Numbers for Optics in the Laser Head

DESCRIPTION	PART NUMBER
Flash lamp replacement kit	0167-381-00
Relay vacuum cell assembly	0167-310-00
SBS cell assembly	0165-938-00
Tilt assembly HR, Power detector	0168-088-00
Output window for laser head cover, IR/SHG	0167-311-00

Table B-2. Part Numbers for Power Supply, Water Side

DESCRIPTION	PART NUMBER
Particle filter	2603-0144
Deionization cartridge	2603-0141

Table B-3. Part Numbers for System Computer

DESCRIPTION	PART NUMBER
Infinity Computer Option	0166-879-00
Computer Only	2000-0089
Monitor Only	2000-0075

Table B-4. Part Numbers for Harmonic Generation

DESCRIPTION	PART NUMBER
SHG option, incl. crystal, crystal cell and mount, 2 dichroic mirrors with mounts.	0166-875-00
THG option, incl. crystal, crystal cell and mount, 2 dichroic mirrors with mounts, waveplate with mount.	0166-876-00
FHG option, incl. crystal, crystal cell and mount, 2 dichroic mirrors with mounts.	0168-599-00
Tested SHG crystal.	0168-079-00
Tested THG crystal.	0168-081-00
FHG crystal.	0169-277-00
5HG crystal	0170-309-00
SHG turning dichroic	0165-157-00
THG turning dichroic	0165-158-00
FHG turning dichroic	0167-025-00
Waveplate for THG (between SHG and THG)	0167-312-00
Crystalline Quartz prism (266 or 213 nm)	0168-684-00

Table B-5. Part Numbers for Optics Accessories

DESCRIPTION	PART NUMBER
One pair of 45° turning mirrors for 1064 nm incl. one left handed and one right handed Infinity mirror mount	0167-851-00
One pair of 45° turning mirrors for 1064 nm incl. one left handed and one right handed Infinity mirror mount. These mirrors transmit 532 nm.	0167-854-00
One pair of 45° turning mirrors for 532 nm incl. one left handed and one right handed Infinity mirror mount. These mirrors transmit 1064 nm.	0167-852-00
One pair of 45° turning mirrors for 355 nm incl. one left handed and one right handed Infinity mirror mount. These mirrors transmit 532 nm.	0167-853-00
A pair of Infinity mirror mounts without optics. One left handed and one right handed.	0167-855-00
Quarter-wave plate for 1064 nm	0167-856-00
Half-wave plate for 1064 nm incl. mount	0167-857-00
Half-wave plate for 532 nm incl. mount	0167-858-00
Half-wave plate for 355 nm incl. mount	0167-859-00
Beam expanding lens, f=-10 cm, dual AR coated for 1064/532 nm	0167-803-00
Beam expanding lens, f=-10 cm, dual AR coated for 532/355 nm	0167-810-00
Telescope (1.8x), two lenses + vacuum cell, dual AR coated for 1064/532 nm	0167-861-00
Telescope (1.8x), two lenses + vacuum cell, dual AR coated for 532/355 nm	0167-860-00
Telescope (2.0x), two lenses + vacuum cell, dual AR coated for 532/355 nm	0169-832-00
Telescope vacuum cell assembly, 1064/532 AR coated	0168-474-00
Telescope vacuum cell assembly, 532/355 AR coated	0168-475-00

OPERATOR'S MANUAL

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APPENDIX C

PACKING PROCEDURE

Introduction

The following is the factory recommended packing procedure for the Infinity laser system. This procedure should be followed if the Infinity system is to be shipped to another location after initial installation.

The Infinity laser system requires two shipping crates, one primarily for the laser head (P/N: 0166-949-00) and the second primarily for the power supply (P/N: 0166-950-00). Table C-1 gives a complete listing of the contents of each crate, in addition to the laser head and power supply, when the system was originally shipped from Coherent.

Table C-1. Contents of Infinity Laser Head and Power Supply Shipping Crates.

Laser Head Crate
1. PC and Laser Documentation
2. Spare Lamp
3. Computer Monitor
4. Deionization Cartridge
5. Computer Keyboard
6. Maintenance Kit
7. Water Hoses

Power Supply Crate
1. Computer (CPU)

Laser Head

The Infinity head crate consists of two compartments, see Figure C-1. The compartment for the Infinity head is foam filled (left side in Figure C-1), it is therefore very important to line the compartment with anti-static material before placing the laser head in the crate, see Figure C-1 and the left hand side of Figure C-2.

To prevent damage to laser head optical components the two POT assembly interlock pins (one on each side of the POT assembly covering the POT clips) should be removed and placed in the maintenance kit. The POT cover over the lamp leads should be secured using tape. Also, tape should be placed over the DPSS output port, and on both ends of the Faraday Isolator. Lens tissue should be taped over the two output ports in the Infinity head cover.

To prevent the umbilical connectors from becoming damaged the end should be enclosed in packing material (e.g., bubble wrap), see the right hand side of Figure C-2.



- 1. Infinity Laser Head
- 2. Anti-Static Wrap

- 3. Second Compartment for Infinity Components

Figure C-1. Overview of Infinity Head Crate



Figure C-2. Infinity Laser Head Properly Packed for Shipment

All other components of the Infinity laser system, with the exception of the computer, should be placed in the second compartment of the Infinity head crate (see Figure C-1, C-3, and Table C-1).



Figure C-3. Packing of Infinity Components, see Table C-1 for Part Descriptions

Power Supply

The Infinity power supply crate also consists of two compartments, see Figure C-4. Like the Infinity head crate, the compartment for the Infinity power supply is foam lined. It is therefore very important to wrap anti-static material around the power supply before placing it in the shipping crate. The front cover of the Power Supply crate should be utilized as a ramp for getting the power supply into the crate, see Figure C-4. The Infinity computer should be placed in the upper compartment of the Power Supply shipping crate. The key for the power supply should be placed in the Infinity Maintenance kit.

Note that the packing foam on the inside face of the Power Supply crate front is removable, see Figure C-5. Removing the piece furthest from the crate, when the cover is installed as a ramp, will allow the Power Supply to be inserted into the crate much more easily.

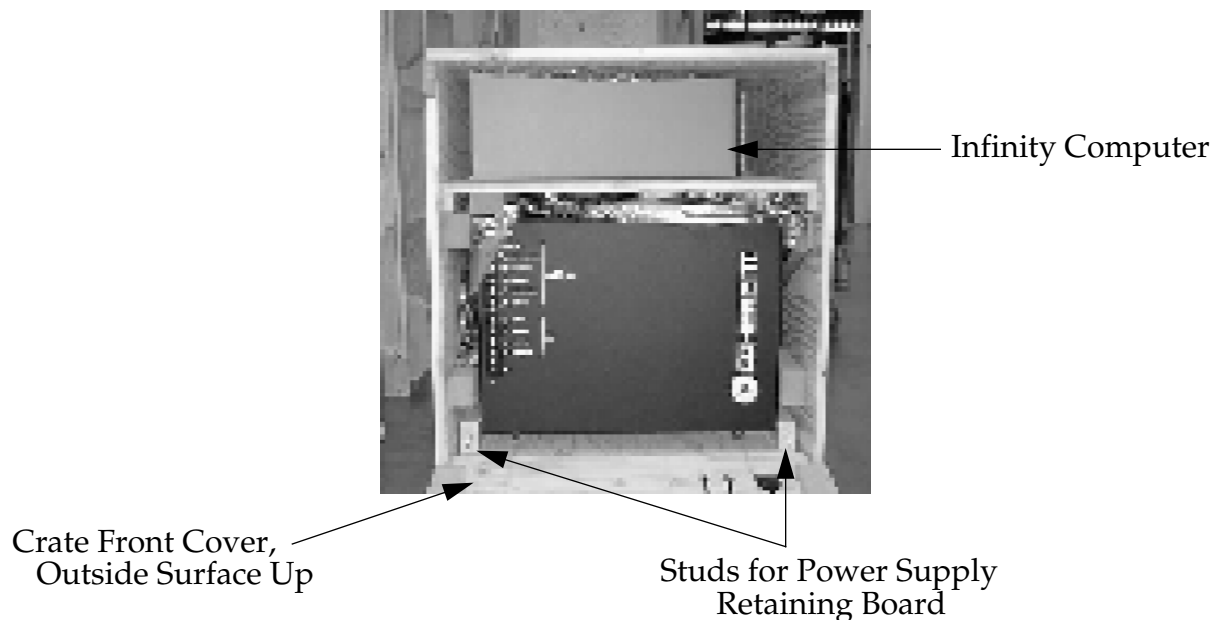


Figure C-4. Infinity Laser Head Properly Packed for Shipment

Once positioned in the crate the power supply is secured using the retaining board and wing nuts as illustrated in Figure C-5.

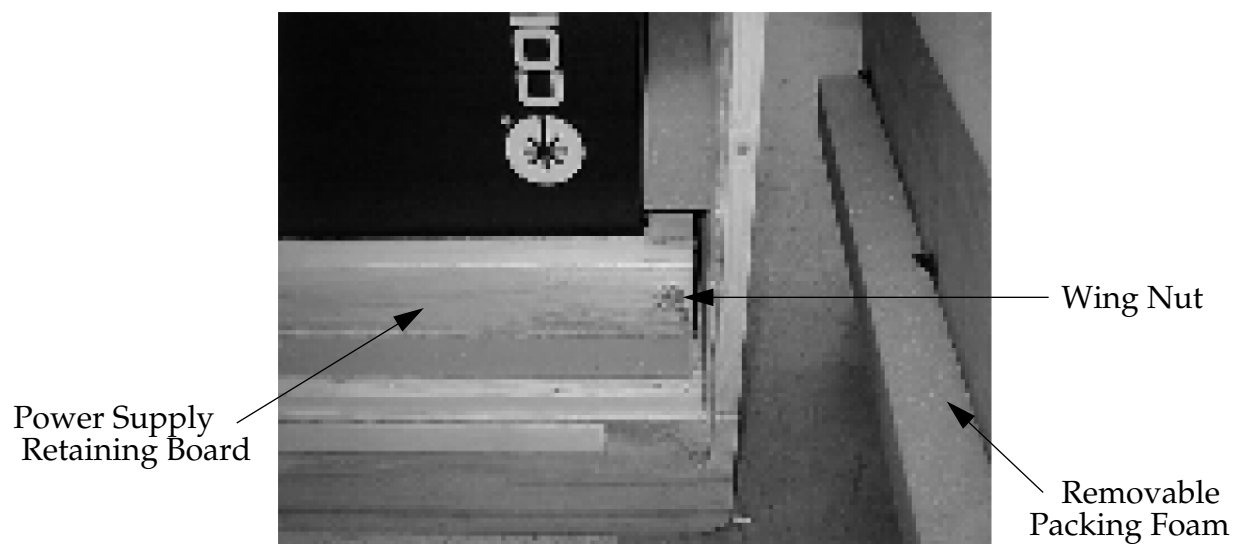


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